A Survey on Vision-Language-Action Models for Embodied AI

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***Abstract*—Embodied AI is widely recognized as a cornerstone of artificial general intelligence because it involves controlling embodied agents to perform tasks in the physical world. Building on the success of large language models and vision-language models, a new category of multimodal models—referred to as vision-language-action models (VLAs)—has emerged to address language-conditioned robotic tasks in embodied AI by leveraging their distinct ability to generate actions. The recent proliferation of VLAs necessitates a comprehensive survey to capture the rapidly evolving landscape. To this end, we present the first survey on VLAs for embodied AI. This work provides a detailed taxonomy of VLAs, organized into three major lines of research. The first line focuses on individual components of VLAs. The second line is dedicated to developing VLA-based control policies adept at predicting low-level actions. The third line comprises high-level task planners capable of decomposing long-horizon tasks into a sequence of subtasks, thereby guiding VLAs to follow more general user instructions. Furthermore, we provide an extensive summary of relevant resources, including datasets, simulators, and benchmarks. Finally, we discuss the challenges facing VLAs and outline promising future directions in embodied AI. A curated repository associated with this survey is available at: https://github.com/yueen-ma/Awesome-VLA.**

***Index Terms*—Vision-Language-Action Model, Embodied AI, Robotics, Multimodality, Large Language Model, World Model**

I. Introduction

Vision-language-action models (VLAs) are a class of multimodal models within the field of embodied AI, designed to process information from three modalities: vision, language, and action. Unlike conversational AI, embodied AI requires controlling physical embodiments that interact with the environment, and robotics is the most prominent domain of embodied AI. In language-conditioned robotic tasks, the policy must possess the capability to understand language instructions, visually perceive the environment, and generate appropriate actions, necessitating the multimodal capabilities of VLAs. The term was recently coined by RT-2 [1]. Compared to earlier deep reinforcement learning approaches, VLAs offer superior versatility, dexterity, and generalizability in complex environments. As a result, they are well-suited not only for controlled settings like factories but also for everyday tasks in household environments.

Early developments in deep learning primarily consist of unimodal models. In computer vision (CV), models like AlexNet [2] showcased the potential of artificial neural networks (ANNs). Recurrent neural networks laid the groundwork for numerous natural language processing (NLP) models, but have seen a transition in recent years, with Transformers [3] taking precedence. The Deep Q-network [4] demonstrated that ANNs can successfully tackle reinforcement learning (RL) problems. Leveraging advancements of unimodal models across diverse machine learning fields, multimodal models now address a wide range of tasks, such as visual question answering, image captioning, and text-to-video generation.

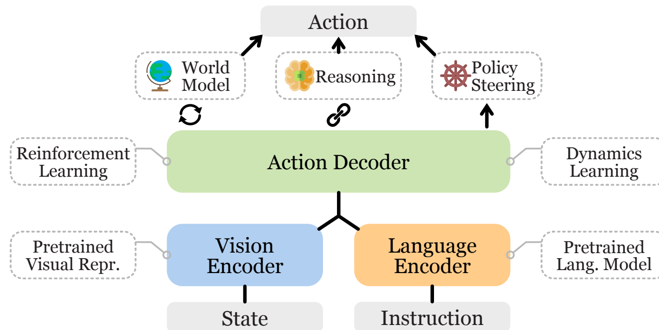

Figure 1: The general architecture of vision-language-action models. Three representative methods for action prediction are shown. Related components are presented in dashed boxes.

Conventional robot policies based on reinforcement learning have largely focused on addressing a limited set of tasks within controlled environments, such as item grasping [5]. However, there is a growing need for more versatile multi-task policies, akin to recent trends in large language models (LLMs) and vision-language models (VLMs). Developing a multi-task policy is challenging, as it requires learning a broader set of skills and adapting to diverse environments. Furthermore, specifying tasks via language instructions offers a more intuitive user-robot interface, which necessitates the development of language-conditioned robot policies.

Built upon the success of large VLMs, vision-language-action models have demonstrated their potential in addressing these challenges, as illustrated in Figure 1. Similar to VLMs, VLAs utilize vision foundation models as vision encoders to obtain pretrained visual representations of the current environmental state, such as object class, pose, and geometry. VLAs encode instructions using the token embeddings of their LLMs and employ various strategies to align vision and language embeddings, including approaches like BLIP-2 [6] and LLaVA [7]. By finetuning on robot data, the LLM can function as a decoder to predict actions and perform language-conditioned robotic tasks. These cross-disciplinary innovations drive the

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具身人工智能中视觉-语言-动作模型研究综述

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***摘要*——具身人工智能被广泛认为是通用人工智能的基石，因为它涉及通过控制具身智能体在物理世界中执行任务。基于大型语言模型和视觉-语言模型的成功，一种新的多模态模型——即视觉-语言-动作模型（VLAs）——应运而生，它利用这些模型独特的动作生成能力来解决具身人工智能中的语言驱动机器人任务。近年来VLAs的迅速普及要求我们对其进行全面综述，以把握其快速发展的现状。为此，我们首次针对具身人工智能领域的VLAs进行了系统性综述。本研究详细构建了VLAs的分类体系，将其划分为三大研究方向：第一方向聚焦于VLAs的各个独立组件；第二方向致力于开发能够有效预测低层级动作的基于VLAs的控制策略；第三方向则包含能够将长期任务分解为一系列子任务的高层任务规划器，从而引导VLAs遵循更通用的用户指令。此外，我们还对相关资源进行了全面梳理，涵盖数据集、仿真器及基准测试工具。最后，我们探讨了虚拟现实增强现实（VLAs）面临的挑战，并概述了具身人工智能领域未来的发展方向。本综述配套的精选资源库已发布于：https://github.com/yueen-ma/Awesome-VLA。**

***索引术语*— 视觉-语言-动作模型、具身人工智能、机器人技术、多模态技术、大型语言模型、世界模型**

I. 引言

视觉-语言-动作模型（VLAs）是具身人工智能领域中的一类多模态模型，专门用于处理来自视觉、语言和动作三种模态的信息。与对话式人工智能不同，具身人工智能需要控制与环境交互的物理实体，而机器人技术正是该领域最突出的应用方向。在语言引导型机器人任务中，智能体必须具备理解语言指令、视觉感知环境以及生成恰当动作的能力，这正是VLAs多模态能力的核心需求。该术语由RT-2研究团队最新提出[1]。相较于早期的深度强化学习方法，VLAs在复杂环境中的通用性、灵活性和泛化能力更胜一筹。因此，它们不仅适用于工厂等受控场景，更能轻松应对家庭环境中的日常任务。

深度学习的早期发展主要以单模态模型为主。在计算机视觉领域，AlexNet等模型[2]展现了人工神经网络

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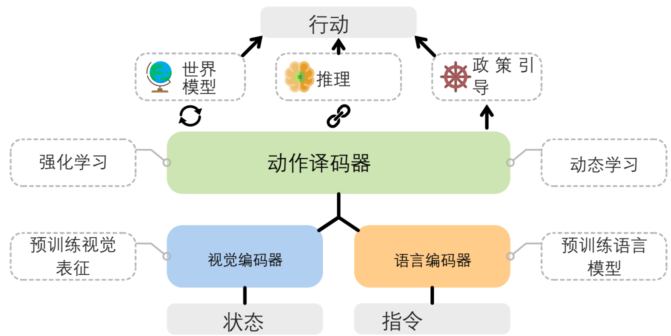
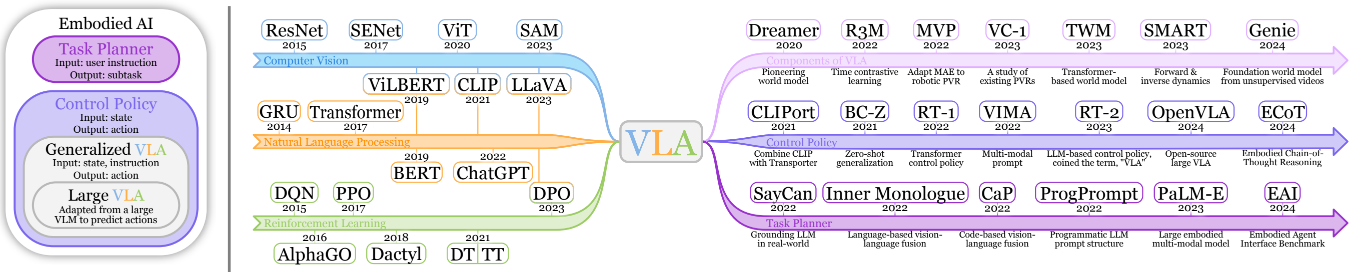


图1：视觉-语言-动作模型的总体架构。图中展示了三种代表性动作预测方法，相关组件以虚线框标示。

（ANN）的潜力。循环神经网络为众多自然语言处理（NLP）模型奠定了基础，但近年来已发生转变，Transformer架构[3]逐渐占据主导地位。深度Q网络[4]证明了ANN能够有效解决强化学习（RL）问题。随着单模态模型在机器学习各领域的持续进步，多模态模型现已能应对视觉问答、图像描述生成、文本转视频等多种任务。

基于强化学习的传统机器人策略主要聚焦于受控环境中的有限任务场景，例如物品抓取[5]。然而，随着大型语言模型（LLMs）和视觉语言模型（VLMs）等技术的最新发展，业界对更具通用性的多任务策略需求日益增长。开发多任务策略具有挑战性，因为这需要学习更广泛的技能并适应多样化环境。此外，通过语言指令指定任务能提供更直观的用户-机器人交互界面，这就要求开发基于语言条件的机器人策略。

基于大型视觉语言模型（VLM）的成功经验，视觉-语言-动作模型（VLAs）在应对这些挑战方面展现出显著潜力，如图1所示。与VLM类似，VLAs采用视觉基础模型作为视觉编码器，获取当前环境状态的预训练视觉表征（如物体类别、姿态和几何结构）。VLAs通过其大语言模型（LLM）的标记嵌入进行指令编码，并运用多种策略实现视觉与语言嵌入的对齐，包括BLIP-2[6]和LLaVA[7]等方法。通过在机器人数据上进行微调，LLM可作为解码器预测动作并执行语言条件下的机器人任务。这些跨学科创新推动了



(a) Concepts

(b) Timelines

Figure 2: (a) A Venn diagram that outlines the main concepts in embodied AI discussed in this paper. (b) Timelines that trace the evolution from unimodal models to vision-language-action models.

advancement of VLAs in embodied AI—a critical building block for artificial general intelligence (AGI).

VLAs are closely related to three lines of work, as depicted by the timelines in Figure 2b and the taxonomy in Figure 3. Some approaches focus on individual components of VLAs (§III-A), such as pretrained visual representations, dynamics learning, world models, and reasoning. Meanwhile, a substantial body of research is dedicated to low-level control policies (§III-B). In this category, language instructions and visual perceptions are fed into the control policy, which then generates low-level actions—such as translation and rotation—thereby rendering VLAs an ideal choice for control policies. In contrast, another category of models serves as high-level task planners responsible for task decomposition (§IV). These models break down long-horizon tasks into a sequence of subtasks that, in turn, guide VLAs toward the overall goal, as illustrated in Figure 4. Most current robotic systems adopt such a hierarchical framework [8]–[10] because the high-level task planner can leverage models with high capacity while the low-level control policy can focus on speed and precision, similar to hierarchical reinforcement learning.

To provide a more comprehensive overview of current progress in embodied AI, we propose a generalized definition of "VLA," as illustrated by the Venn diagram in Figure 2a. We define a VLA as any model capable of processing multimodal inputs from vision and language to produce robot actions that accomplish embodied tasks, typically following the architecture in Figure 1. The original concept of VLA referred to a model that adapts VLMs to robotic tasks [1]. Analogous to the distinction between LLMs and more general language models, we designate the original VLAs as "Large VLAs" (LVLAs) because they are based on LLMs or large VLMs.

Related Work. To the best of our knowledge, this survey is the first to review the recent progress of VLA models, a rapidly emerging research area. Previous surveys have investigated other facets of embodied AI. [11] comprehensively summarized foundation models in robotics up to 2023, while [12] focused on LLMs in robotics. [13] examined more recent vision, language, and robotic foundation models for general-purpose robots. [14] concentrates on real-world robot applications. In contrast, our work emphasizes VLA models, thereby complementing and extending the existing literature on embodied AI.

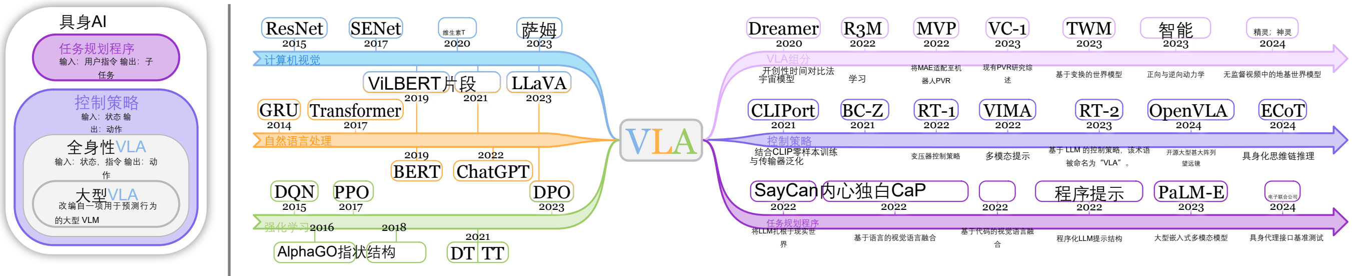
Contributions. To the best of our knowledge, this paper is the first comprehensive survey of emerging vision-language-action (VLA) models in the field of embodied AI.

- **Comprehensive Review.** We present a thorough review of emerging VLA models in embodied AI, covering various aspects including components, architectures, training objectives, robotic tasks, etc.
- **Taxonomy.** We introduce a taxonomy of VLAs based on the hierarchical framework of current robotic systems, comprising two hierarchies: a low-level control policy and a high-level task planner. Control policies execute low-level actions based on specified language commands and the perceived environment. Task planners provide guidance to control policies by decomposing long-horizon tasks into subtasks. We also discuss various essential components of VLAs.
- **Resources.** We summarize the necessary resources for training and evaluating VLAs, including recent datasets and benchmarks in real-world or simulated environments. We also discuss various approaches to address current challenges, such as data scarcity and inconsistency.
- **Future Directions.** We outline current challenges and promising future opportunities in the field, such as safety, foundation models, and real-world deployment.

II. Background

Embodied AI is a unique form of artificial intelligence that actively interacts with the physical environment. This sets it apart from other AI models, such as conversational AI, which primarily handles textual conversations, or generative AI models that focus on tasks like text-to-video generation. Embodied AI encompasses a broad spectrum of embodiments, including smart appliances, smart glasses, autonomous vehicles, and more. Among them all, robots stand out as one of the most prominent embodiments.

Robot learning is also usually framed as a reinforcement learning problem, represented as a Markov Decision Process (MDP) consisting of states (s), actions (a), and rewards (r). A MDP trajectory is denoted as $\tau = (s_1, a_1, r_1, \ldots, s_T, a_T, r_T)$. In certain scenarios, robotic tasks may also be viewed as Partially Observable Markov Decision Processes (POMDPs) due to incomplete observations. The primary objective of reinforcement learning is to train a policy capable of generating optimal actions for the current state $\pi(a_t|s_{\leq t}, a_{<t})$. Various methods, such as TD learning, policy gradients, etc.,



(a) 概念　　(b) 时间线

图2：(a) 维恩图，概述了本文讨论的具身人工智能（embodied AI）主要概念。(b) 时间线，追踪从单模态模型到视觉-语言-动作模型的演进过程。

具身人工智能中VLAs（虚拟语言架构）的发展——这是实现通用人工智能（AGI）的关键基础组件。

VLAs与三大研究方向密切相关，如图2b的时间线和图3的分类体系所示。部分研究方法聚焦于VLAs的各个组成部分（§III-A），例如预训练视觉表征、动态学习、世界模型及推理机制；而大量研究则致力于低级控制策略（§ III-B）——这类模型将语言指令与视觉感知输入控制策略，进而生成翻译、旋转等低级动作，因此使VLAs成为控制策略的理想选择。相比之下，另一类模型则充当高级任务规划器，负责任务分解（§IV）：这些模型将长期任务分解为一系列子任务，从而引导VLAs实现整体目标，如图4所示。当前大多数机器人系统采用这种分层框架[8]–[10]，因为高层任务规划器能够利用高容量模型，而低层控制策略则可专注于速度与精度，这类似于分层强化学习。

为了更全面地概述具身人工智能领域的最新进展，我们提出了“视觉语言模型（VLA）”的广义定义，如图2a中的维恩图所示。我们将VLA定义为能够处理视觉与语言等多模态输入、并生成实现具身任务所需机器人动作的模型，其架构通常遵循图1所示结构。原始VLA概念指的是将视觉语言模型（VLM）适配至机器人任务的模型[1]。类似于大语言模型（LLM）与通用语言模型的区分，我们将原始VLA称为“大型视觉语言模型（LVLAs）”，因其基于LLM或大型VLM架构。

相关研究。据我们所知，本综述是首个系统梳理VLA模型最新进展的研究，该领域属于快速发展的新兴研究方向。此前的综述主要探讨了具身人工智能的其他方面。[11] 全面总结了截至2023年机器人领域的基础模型，同时[12]重点关注了机器人领域中的大语言模型（LLMs）。[13] 针对通用机器人领域，对近期开发的视觉、语言及机器人基础模型进行了系统评估。[14] 聚焦于现实世界机器人应用。相比之下，我们的研究侧重于VLA模型，从而补充并扩展了现有具身人工智能领域的文献。

贡献。据我们所知，本文是关于具身人工智能领域中新兴视觉-语言-行动（VLA）模型的首篇综合性综述。

- **综合性综述。**本文对具身人工智能领域新兴的VLA模型进行了全面综述，涵盖组件、架构、训练目标、机器人任务等多个方面。
- **分类体系。**我们基于当前机器人系统的层级架构，提出了一种基于虚拟现实辅助系统（VLAs）的分类体系，该体系包含两个层级结构：低级控制策略与高级任务规划器。控制策略根据预设的语言指令及感知环境执行底层操作，而任务规划器则通过将长期任务分解为子任务来为控制策略提供指导。本文还探讨了虚拟现实辅助系统的核心组成部分。
- **资源。**我们总结了训练和评估虚拟实验室自动化（VLA）所需的必要资源，包括真实世界或模拟环境中的最新数据集和基准测试。同时，我们探讨了应对当前挑战（如数据稀缺性和不一致性）的多种方法。
- **未来发展方向。**我们概述了该领域当前面临的挑战及具有前景的未来机遇，例如安全性、基础模型以及实际部署应用。

II.背景

具身人工智能是一种独特的智能形态，能够主动与物理环境进行交互。这使其区别于其他人工智能模型——例如主要处理文本对话的对话式AI，或是专注于文本转视频生成等任务的生成式AI模型。具身人工智能涵盖多种应用场景，包括智能家电、智能眼镜、自动驾驶汽车等。其中，机器人堪称最具代表性的具身形态之一。

机器人学习通常也被框架为强化学习问题，表现为由状态（s）、动作（a）和奖励（r）组成的马尔可夫决策过程（MDP）。 MDP 轨迹表示为$\tau=(s_1, a_1, r_1, \ldots, s_T, a_T, r_T)$。在某些场景中，由于观测数据不完整，机器人任务也可视为部分可观测马尔可夫决策过程（POMDPs）。强化学习的主要目标是训练能够为当前状态π（$a_t|s_{\le t}$，$a_{<t}$）生成最优动作的策略。多种方法如时差学习、策略梯度等，

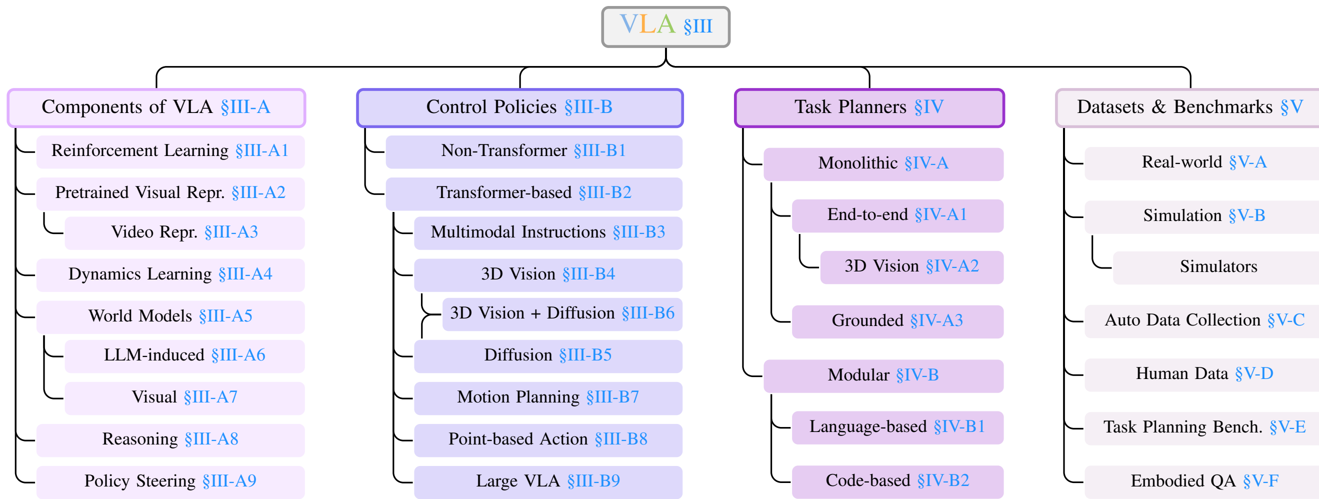


Figure 3: The taxonomy of VLA models. The organization of this survey follows this taxonomy.

can be employed to achieve this. However, in cases where defining the reward function proves challenging, imitation learning is utilized to directly model the action distribution within trajectories devoid of rewards $\tau = (s_1, a_1, \ldots, s_T, a_T)$. Furthermore, many multi-task robot models employ language as instructions p to determine which task or skill to execute, leading to the development of language-conditioned robot policies $\pi(a_t|p, s_{\le t}, a_{<t})$.

III. Vision-Language-Action Models

A. Components of VLA

A body of work focuses on individual components of VLA models, drawing on successes in CV, NLP, and RL. We introduce them through the lens of embodied AI.

1) Reinforcement Learning: Reinforcement learning (RL) laid the foundation for embodied AI and continues to help advance VLAs. Deep Q-Network (DQN) first demonstrated learning policies directly from high-dimensional pixel inputs, underscoring the need for greater model capacity in end-to-end RL. RL trajectories—sequences of states, actions, and rewards—naturally align with sequence modeling problems, making them well-suited to transformer architectures. Pioneering efforts in this direction include Decision Transformer (DT) [15] and Trajectory Transformer (TT) [16]. Gato [17] further extended this paradigm to a multimodal, multitask, multi-embodiment setting. $\pi^*_{0.6}$ [18] employs RL for VLAs to learn from experience.

Furthermore, a synergy has emerged between RL and LLMs, benefiting embodied AI in multiple ways. Reinforcement learning from human feedback (RLHF) aligns LLMs with human preferences and has also been applied to robot learning. SEED [19] utilizes RLHF alongside skill-based RL to address the sparse reward issue and to improve safety via human evaluation. Conversely, LLMs also enable novel RL methods. Reflexion [20] proposes a novel verbal reinforcement learning framework that replaces weight updates in RL models with linguistic feedback, which is naturally applicable to embodied decision-making, where planning also occurs in language form. Eureka [21] shows that LLMs can design reward functions for embodied agents that outperform expert human-engineered ones. As LLMs evolve, this synergy accelerates progress in RL and embodied AI.

2) Pretrained Visual Representations: The effectiveness of the vision encoder directly influences the performance of VLAs, as it provides crucial information regarding the current state s_t, such as object categories, positions, and affordance. Consequently, numerous methods are devoted to improving the quality of pretrained visual representations (PVRs). Their technical details are compared in Table I.

CLIP [22] has gained widespread adoption as a vision encoder in robotic models. The training objective of CLIP is to identify the correct text-image pair among all possible combinations in a given batch. CLIP undergoes training on the WIT dataset, comprising 400 million image-text pairs. This large-scale training allows CLIP to develop a rich understanding of the relationship between visual and textual information.

R3M [23] proposes two main pretraining objectives: time contrastive learning and video-language alignment. The objective of time contrastive learning is to minimize the distance between temporally proximal video frames while simultaneously increasing the separation between temporally distant ones. This objective aims to create PVRs that capture the temporal relationship within the video sequence. On the other hand, video-language alignment is to learn whether a video corresponds to a language instruction. This objective enriches the semantic relevance embedded in the PVRs. VIP [24] also capitalizes on the video temporal relationship.

MVP [25] adopts masked autoencoder (MAE) from computer vision to robotic datasets. The MAE involves masking out a portion of input patches to a ViT model and training it to reconstruct the corrupted patches. This approach closely resembles the masked language modeling technique used in BERT [26] and falls within the purview of self-supervised training. RPT [27] undergoes pretraining with a focus not only on reconstructing visual inputs but also on robotic actions and proprioceptive states.

Voltron [32] introduces a pretraining objective by incorporating language conditioning and language generation into the MAE objective. Employing an encoder-decoder Trans-

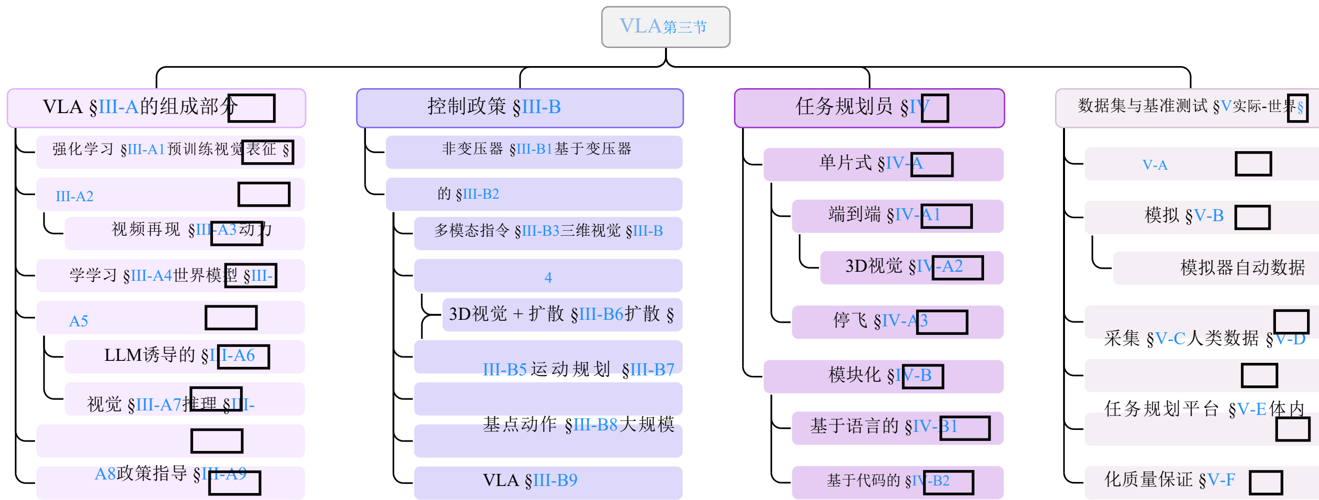


图3：VLA模型的分类体系。本综述的组织结构遵循该分类体系。

可以用来实现这一目标。然而，在定义奖励函数具有挑战性的案例中，会采用模仿学习直接建模缺乏奖励的轨迹中的动作分布$\tau=(s_1,a_1,\ldots,s_T,a_T)$。此外，许多多任务机器人模型利用语言作为指令$p$来确定执行哪个任务或技能，从而发展出语言条件化的机器人策略$\pi(a_t|p, s_{\le t}, a_{<t})$。

III. 愿景-语言-行动模型

A. VLA组分

大量研究聚焦于VLA模型的各个组成部分，借鉴了卷积神经网络（CV）、自然语言处理（NLP）和强化学习（RL）领域的成功经验。我们通过具身人工智能的视角对这些模型进行介绍。

*1) 强化学习：*强化学习（RL）为具身人工智能奠定了基础，并持续推动虚拟现实增强技术（VLAs）的发展。深度Q网络（DQN）首次实现了从高维像素输入中直接学习策略，凸显了端到端强化学习中对更大模型容量的需求。强化学习轨迹——由状态、动作和奖励组成的序列——与序列建模问题天然契合，使其特别适合Transformer架构。该领域的开创性研究包括决策Transformer（DT）[15]和轨迹Transformer（TT）[16]。Gato[17]进一步将这一范式拓展至多模态、多任务、多具身场景。$\pi^*_{0.6}$[18]采用强化学习技术使虚拟现实增强系统能够从经验中学习。

此外，强化学习（RL）与大语言模型（LLMs）之间已形成协同效应，从多维度推动具身人工智能的发展。基于人类反馈的强化学习（RLHF）使LLMs与人类偏好保持一致，该技术亦被应用于机器人学习领域。SEED[19]通过结合 RLHF与基于技能的强化学习，有效解决了奖励稀疏问题，并借助人工评估提升系统安全性。反之，LLMs也为新型强化学习方法提供了可能。Reflexion[20]提出了一种创新性语言强化学习框架，用语言反馈替代强化学习模型中的权重更新机制，该框架天然适用于具身决策场景——这类场景中规划过程同样以语言形式呈现。Eureka[21]实验证明，LLMs能为具身智能体设计出超越人类专家设计的奖励函数。随着LLMs技术的持续演进，这种协同效应正加速推动强化学习与具身人工智能领域的进步。

*2) 预训练视觉表征：*视觉编码器的有效性直接影响视觉语言表征（VLA）的性能，因为它提供了关于当前状态s_t的关键信息，例如物体类别、位置和可用性。因此，大量研究致力于提升预训练视觉表征（PVR）的质量。其技术细节对比见表I。

CLIP[22]作为视觉编码器已在机器人模型中得到广泛应用。其训练目标是在给定批次中从所有可能的组合中识别出正确的文本-图像对。CLIP基于包含4亿图像-文本对的WIT数据集进行训练，这种大规模训练使模型能够深入理解视觉信息与文本信息之间的关联关系。

R3M[23]提出了两个主要的预训练目标：时间对比学习与视频-语言对齐。时间对比学习的目标是在最小化时间上相邻视频帧之间距离的同时，增加时间上相距较远帧之间的分离度；这一目标旨在生成能够捕捉视频序列中时间关系的PVR。另一方面，视频-语言对齐旨在判断视频是否对应于语言指令，该目标能增强PVR中蕴含的语义相关性。VIP[24]同样充分利用了视频的时间关系。

MVP[25]将计算机视觉领域的掩码自编码器（MAE）应用于机器人数据集。MAE的核心思想是向ViT模型输入部分图像块并进行掩码处理，随后训练模型重建被损坏的图像块。该方法与BERT[26]中使用的掩码语言建模技术高度相似，属于自监督训练范畴。而RPT[27]在预训练阶段不仅专注于视觉输入的重建，还同时关注机器人动作及本体感觉状态的建模。

Voltron[32]通过将语言条件化和语言生成整合到MAE目标中，引入了预训练目标。该方法采用编码器-解码器Trans-

Table I: Pretrained visual representations. V: Vision. L: Language. CL: contrastive learning. TFM: Transformer. **Sim/Real**: simulated/real-world. **Mani/Navi**: manipulation/navigation. For simplicity, we only show the main part of the objective equation, omitting elements such as temperature, auxiliary loss, etc. $\mathcal{S}(\cdot)$ denotes a similarity measure.

Model	Network	Type	Objective	Notations	Robotic Tasks
CLIP [22]	ViT-B	VL-CL	$\sum_{i=1}^{N} -\log \frac{\exp(\mathcal{S}(x_i,y_i))}{\sum_{j=1}^{N}\exp(\mathcal{S}(x_i,y_j))}$	(x_i, y_i): image-text pair	(Used by CLIPort [28], EmbCLIP [29], and CoW [30])
R3M [23]	ResNet-50	Time-CL	$\sum_{i=1}^{T} -\log \frac{\exp(\mathcal{S}(x_i,x_j))}{\exp(\mathcal{S}(x_i,x_j))+\exp(\mathcal{S}(x_i,x_k))}$	x_i: i-th video frame, $i < j < k$	**Sim-Mani**: Meta-World, Franka Kitchen, Adroit
MVP [25]	ViT-B/L	MAE	$\sum_{i=1}^{N} -\log P\left(x_i \mid x_{\neq i}\right)$	x_i: image patch	**Real-Mani** (xArm7): pick, reach block, push cube, close fridge
VIP [24]	ResNet-50	Time-CL	$\sum_{i=1}^{T} -\log \frac{\exp(\mathcal{S}(x_0,x_j))}{\exp(\mathcal{S}(x_0,x_j))+\exp(\mathcal{S}(x_i,x_j))}$	x_i: i-th video frame, $0 < i < j$	**Real-Mani** (Franka): pick and place, close drawer, push bottle, fold towel
VC-1 [31]	ViT-L	MAE, CL	(A combination of previous works, including CLIP, R3M, MVP, and VIP.)	–	**Sim-Mani**: Meta-World, Adroit, DMC, TriFinger, Habitat 2.0; **Sim-Navi**: Habitat
Voltron [32]	TFM	MAE, Lang-Gen	$\sum_{i=1}^{N} -\log P\left(x_i \mid x_{\neq i}, y\right)$; $\sum_{i=1}^{N'} -\log P\left(y_i \mid x, y_{<i}\right)$	x_i: image patch; y: language instruction	**Sim-Mani**: Franka Kitchen; **Real-Mani** (Franka): custom study desk
RPT [27]	ViT	MAE	$\sum_{i=1}^{N} -\log P\left(x_i \mid x_{\neq i}, y, z, \dots\right)$	x, y, z: three distinct modalities	**Real-Mani** (Franka): stack, pick, pick from bin
DINOv2 [33]	ResNet, ViT	Self-distillation	$\sum_{x}\sum_{x' \neq x} H(P_t(x), P_x(x'))$	x, x': image views; $H()$: cross-entropy; P_t, P_s: teacher, student	(Used by OpenVLA [34], ReKep [35])
I-JEPA [36]	ViT	JEPA	$\frac{1}{M}\sum_{i=1}^{M}\sum_{j}\lVert x_j(i) - y_j(i)\rVert_2^2$	$x_j(i)$: j-th masked patch embedding of block i; $y_j(i)$: unmasked patch embedding	–
Theia [37]	ViT-T/S/B	Distillation	(Distillation of vision foundation models: ViT, CLIP, SAM, DINOv2, Depth-Anything.)	–	**Sim-Mani&Navi**: CortexBench (VC-1); **Real-Mani**: pick, place, open door/drawer

former structure, the pretraining alternates between language-conditioned masked image reconstruction and language generation from masked images. This enhances the alignment between language and vision modalities.

VC-1 [31] conducts an in-depth examination of prior PVRs and introduces an enhanced PVR model by systematically exploring optimal ViT configurations across diverse datasets. Additionally, they perform a comprehensive comparative analysis of their model against previous methods on various manipulation and navigation datasets, shedding light on the critical factors that contribute to the improvement of PVR. Another study [38] also compares previous PVRs obtained under supervised learning or self-supervised learning.

DINOv2 [33] proposes a new self-supervised training paradigm for PVRs that achieves performance beyond that of MAE. It employs a self-distillation framework in which the teacher and student networks receive different views of the same image and match their encoded representations. The student network is updated using SGD, while the teacher network is maintained as the EMA of the student network.

I-JEPA [36] is motivated by the joint-embedding predictive architectures proposed by [39]. It constructs a "primitive" internal world model by comparing the embeddings of patches. Unlike DINO, which uses cropped images, I-JEPA employs masked patches. Moreover, it differs from MAE because it is a non-generative approach.

Theia [37] proposes distilling various vision foundation models into a single model. By fusing their information from segmentation, depth, semantics, etc., it outperforms previous PVRs while requiring less data and a smaller model size.

3) Video Representations: Videos are simple sequences of images and can be represented by concatenating the usual PVRs of each frame. However, their multi-view nature enables a variety of unique representation techniques beyond those mentioned above, such as time contrastive learning and MAE. NeRF can be extracted from videos and contains rich 3D information for robot learning, as exemplified by F3RM [40] and 3D-LLM [41]. The recent 3D Gaussian Splatting (3D-GS) [42] surpasses NeRF in visual quality and rendering speed. Additionally, many videos contain audio, which can provide important cues for robot policies [43].

4) Dynamics Learning: Dynamics learning encompasses objectives aimed at endowing the model $f(\cdot)$ with an understanding of forward or inverse dynamics. Forward dynamics involves predicting the subsequent state resulting from a given action, whereas inverse dynamics entails determining the action required to transition from a previous state to a known subsequent state:

$$\begin{aligned} \text{Forward dynamics: } \hat{s}_{t+1} &\leftarrow f_{\text{fwd}}(s_t, a_t), \\ \text{Inverse dynamics: } \hat{a}_t &\leftarrow f_{\text{inv}}(s_t, s_{t+1}). \end{aligned} \tag{1}$$

Some approaches also frame these objectives as reordering problems for shuffled state sequences. Some forward dynamics methods closely resemble the image or video prediction pretraining used in PVRs. We compare them in Table II.

Vi-PRoM [44] presents three distinct pretraining objectives. The first involves a contrastive self-supervised learning objective designed to distinguish between different videos. The remaining two objectives are centered around supervised learning tasks: temporal dynamics learning, aimed at recovering shuffled video frames, and image classification employing pseudo labels. Through a comprehensive comparison with preceding pretraining methods, Vi-PRoM demonstrates its effectiveness for behavior cloning and PPO.

MIDAS [46] introduces an inverse dynamics prediction task as part of its pretraining. The objective is to train the model to predict actions from observations, formulated as a motion-following task. This approach enhances the model's understanding of the transition dynamics of the environment.

SMART [47] presents a pretraining scheme encompassing three distinct objectives: forward dynamics prediction, inverse dynamics prediction, and randomly masked hindsight control. The forward dynamics prediction task involves predicting the next latent state, while the inverse dynamics prediction task entails predicting the last action. In the case of hindsight control, the entire control sequence is provided as input, with

表I：预训练视觉表征。V：视觉。L：语言。CL：对比学习。 TFM ：Transformer。**Sim/Real**：模拟/真实世界。**Mani/Navi**：操作/导航。为简化说明，我们仅展示目标方程的主要部分，省略了温度、辅助损失等元素。S（·）表示相似度度量。

模型	网络	类型	目标	符号说明	机器人任务
CLIP [22]	ViT-B	VL-CL	$\sum_{i=1}^{N}$ _日志$\frac{\exp(\mathcal{S}(x_i,y_i))}{\sum_{j=1}^{N}\exp(\mathcal{S}(x_i,y_j))}$	(xi, yi): 图像-文本对	（由 CLIPort[28]、EmbCLIP[29]和 CoW[30]使用）
R3M [23]	ResNet-50	时间-CL	$\sum_{i=1}^{T}-$ 对数$\frac{指数(S(xi, xj))}{}$指数（$S(xi, xj)$)+指数（$S(xi, x k)$)	xi:i-第 i 帧视频，$i < j < k$	**Sim-Mani**：Meta-World、Franka Kitchen、Adroit
MVP [25]	ViT-B/L	MAE	$\sum_{i=1}^{N}$ 对数P (xi $x_{\neq i}$)	xi：图像块	**Real-Mani**(xArm7)：拾取、伸手阻挡、推动立方体、关闭冰箱
VIP [24]	ResNet-50	时间-CL	$\sum_{i=1}^{T}$ _日志$\frac{\exp(\mathcal{S}(x_0,x_j))}{\exp(\mathcal{S}(x_0,x_j))+\exp(\mathcal{S}(x_i,x_j))}$	xi:i-第 i 帧视频，$0 < i < j$	**Real-Mani**（Franka）：挑选并放置物品、关闭抽屉、推瓶、折叠毛巾
VC-1 [31]	ViT-L	MAE, CL	（整合了既往研究工作，包括CLIP、R3M、MVP及VIP等研究）	—	**Sim-Mani**：Meta-World、Adroit、DMC、TriFinger、Habitat 2.0；**Sim-Navi**：Habitat
伏特龙[32]	TFM	MAE，朗格-根	$\sum_{i=1}^{N}$ 对数P (xi $x_{\neq i}$, y）；$\Sigma_{i=1}^{N'}$_log P (yi | x, $y_{<i}$)	xi：图像块； y：语言教学	**Sim-Mani**：弗兰卡厨房； **Real-Mani**（Franka）：定制学习桌
RPT [27]	维生素T	MAE	$\sum_{i=1}^{N}$ 对数P (xi $x_{\neq i}$, y, z, …)	x, y, z：三种不同的模态	**Real-Mani**(Franka)：堆叠、拾取、从箱中拾取
DINOv2 [33]	ResNet、ViT	自己蒸馏	$\sum_{x}\sum_{x'\neq x}H(P_t(x),P_x(x'))$	x, x'：图像视图；H（）：交叉熵；Pt, Ps：教师，学生	（由OpenVLA[34]、ReKep[35]使用）
I-JEPA [36]	维生素T	JEPA	$\frac{1}{M}\sum_{i=1}^{M}\sum_{j}\lVert x_j(i)-y_j(i)\rVert_2^2$	$xj(i)$: 块i的第j个掩码补丁嵌入；$yj(i)$: 未掩码补丁嵌入	—
忒亚[37]	ViT-T/S/B	蒸馏	视觉基础模型的蒸馏方法：ViT、CLIP、SAM、DINOv2、Depth-Anything	—	**Sim-Mani&Navi**：CortexBench（VC-1）； **Real-Mani**：抓取、放置、开门/抽屉

在原有结构中，预训练过程交替进行语言条件下的掩蔽图像重建与基于掩蔽图像的语言生成任务。该机制增强了语言模态与视觉模态之间的对齐性。

VC-1[31]对先前的PVR研究进行了深入分析，并通过系统探索不同数据集中的最优ViT配置，提出了一种改进型PVR模型。此外，该研究还通过在多种操作与导航数据集上与既往方法进行全面对比分析，揭示了提升PVR效果的关键影响因素。另一项研究[38]则对比了基于监督学习与自监督学习方法获得的PVR结果。

DINOv2[33]提出了一种针对伪向量回归器（PVRs）的新型自监督训练范式，其性能超越了平均绝对误差（MAE）。该方法采用自蒸馏框架，使教师网络与学生网络能够接收同一图像的不同视角，并匹配其编码表征。学生网络通过随机梯度下降（SGD）进行更新，而教师网络则作为学生网络的弹性移动平均（EMA）进行维护。

I- JEPA[36]的灵感来源于[39]提出的联合嵌入预测架构。该方法通过比较图像块的嵌入向量来构建“原始”内部世界模型。与使用裁剪图像的DINO不同，I- JEPA 采用掩码图像块。此外，它与MAE的区别在于采用非生成式方法。

Theia[37]提出将多种视觉基础模型整合为单一模型。通过融合分割、深度、语义等信息，该模型在数据需求和模型体积方面均优于传统PVRs，同时展现出更优的性能表现。

*3) 视频表征：*视频是由图像组成的简单序列，可通过拼接各帧的常规PVRs进行表征。但其多视角特性使得除上述方法外，还可采用时间对比学习和MAE等独特表征技术。NeRF可从视频中提取并包含丰富的三维信息，适用于机器人学习领域，F3RM[40]和3D-LLM[41]即为典型范例。近期提出的三维高斯点阵技术（3D-GS）[42]在视觉质量和渲染速度方面超越了NeRF。此外，多数视频包含音频信息，可为机器人策略提供重要线索[43]。

*4) 动力学学习：*动力学学习的目标在于使模型f（·）具备对正向或逆向动力学的理解能力。正向动力学涉及预测给定动作导致的后续状态，而逆向动力学则需要确定从先前状态过渡到已知后续状态所需的动作：

$$\begin{aligned} &\text{正向动力学：} \hat{s}_{t+1} \leftarrow f_{\text{fwd}}(s_t, a_t), \\ &\text{逆动力学：} \hat{a}_t \leftarrow f_{\text{inv}}(s_t, s_{t+1})\text{。} \end{aligned} \tag{1}$$

部分研究方法也将这些目标表述为对随机化状态序列进行重排序的问题。某些前向动力学方法与物理视频再现器（PVRs）中使用的图像或视频预测预训练方法高度相似。我们在表II中对这些方法进行了比较。

Vi-PRoM[44]提出了三个明确的预训练目标。第一个目标采用对比式自监督学习方法，旨在实现不同视频间的区分识别。其余两个目标则聚焦于监督学习任务：时间动态学习（用于恢复随机打乱的视频帧）和伪标签图像分类。通过与现有预训练方法的全面对比，Vi-PRoM在行为克隆和PPO任务中展现出显著优势。

midas[46]在其预训练过程中引入了逆动力学预测任务。该任务旨在训练模型根据观测数据预测动作，具体表现为一项运动跟随任务。这种方法能增强模型对环境状态转移动态的理解。SMART[47]提出了一种包含三个不同目标的预训练方案：前向动力学预测、逆动力学预测以及随机掩码的后视控制。前向动力学预测任务涉及预测下一个潜在状态，而逆动力学预测任务则需要预测最后一个动作。在后视控制任务中，整个控制序列被作为输入提供。

Table II: Dynamics learning methods for VLAs. $f(\cdot)$ is the dynamic model. Fwd, inv: forward & inverse dynamics.

Model	Vision Encoder	Type	Objective	Notations	Robotic Tasks
Vi-PRoM [44]	ResNet	Temporal dynamics	$\sum_{i=1}^{T} \mathrm{CE}(i, f(x_i \mid x'))$	$f(\cdot)$: predicts frame index in raw seq. x given shuffled seq. x'	**Sim-Mani**: Meta-World, Franka Kitchen; **Real-Mani**: open & close drawer/door
MaskDP [45]	ViT (MAE)	MAE (implicit fwd & inv dynamics)	$\sum_{i=1}^{T} -\log P\left(x_i \mid x_{\neq i}, y\right)$	x_i: state or action token; y: the other modality	**Sim**: DeepMind Control Suite
MIDAS [46]	ViT, Mask R-CNN	Inverse dynamics	$\sum_{i=1}^{T} \mathrm{MSE}\left(a_t, f_{\text{inv}}(s_t, s_{t+1})\right)$	s_t, a_t: state, action	**Sim-Mani**: VIMA-Bench
SMART [47]	CNN	Forward & inverse dynamics	$\sum_{i=1}^{T} \big(\mathrm{MSE}\left(s_{t+1}, f_{\text{fwd}}(s_t, a_t)\right) + \mathrm{MSE}\left(a_t, f_{\text{inv}}(s_t, s_{t+1})\right) \big)$	s_t, a_t: state, action	**Sim**: DeepMind Control Suite
PACT [48]	ResNet-18, PointNet	Forward dynamics	$\sum_{i=1}^{T} \mathrm{MSE}\left(s_{t+1}, f_{\text{fwd}}(s_t, a_t)\right)$	s_t, a_t: state, action	**Sim-Navi**: Habitat; **Real-Navi** (MuSHR vehicle)
VPT [49]	ResNet	Inverse dynamics	$\sum_{i=1}^{T} \mathrm{MSE}\left(a_t, f_{\text{inv}}(s_t, s_{t+1})\right)$	s_t, a_t: state, action	**Sim**: Minecraft
GR-1 [50]	ViT (MAE)	Forward dynamics	$\sum_{i=1}^{T} \mathrm{MSE}\left(s_{t+1}, f_{\text{fwd}}(s_t, a_t)\right)$	s_t, a_t: state, action	**Sim-Mani**: CALVIN

some actions masked, and the model is trained to recover these masked actions. The incorporation of the first two dynamics prediction tasks facilitates capturing local and short-term dynamics, while the third task is designed to capture global and long-term temporal dependencies.

MaskDP [45] features the masked decision prediction task, wherein both state and action tokens are masked for reconstruction. This masked modeling task is specifically crafted to equip the model with an understanding of both forward and inverse dynamics. Notably, in contrast to preceding masked modeling approaches like BERT or MAE, MaskDP is applied directly to downstream tasks

PACT [48] introduces a pretraining objective aimed at modeling state-action transitions. It receives sequences of states and actions as input, and predicts each state and action token autoregressively. This pretrained model serves as a dynamics model, which can then be finetuned for various downstream tasks such as localization, mapping, and navigation.

VPT [49] proposes a video pretraining method that harnesses unlabeled internet data to pretrain a foundational model for the game of Minecraft. The approach begins by training an inverse dynamics model using a limited amount of labeled data, which is then utilized to label internet videos. Subsequently, this newly auto-labeled data is employed to train the VPT foundation model through behavior cloning. This methodology follows semi-supervised imitation learning. As a result of this process, the model demonstrates human-level performance across a multitude of tasks.

GR-1 [50] introduces video prediction pretraining tailored for a GPT-style model. The ability to anticipate future frames aligns with forward dynamics learning, contributing to more accurate action prediction.

5) World Models: A world model $P(\cdot)$ encodes commonsense knowledge about the world and predicts the future state for a given action [39]:

$$\hat{s}_{t+1} \sim P(\hat{s}_{t+1} | s_t, a_t). \tag{2}$$

It enables model-based control and planning for embodied agents, as they can search for an optimal action sequence in imaginary space before executing any real actions. Although forward dynamics learning also attempts to predict the next state, it is usually treated as a pretraining task or an auxiliary loss to enhance the RL-Transformer-based action decoder for the primary robotics tasks, rather than serving as a standalone module. Additionally, new embodied experiences can be sampled from visual world models that explicitly generate images or videos of future states.

Dreamer [51] employs three primary modules to construct a latent dynamics model: a representation model, responsible for encoding images into latent states; a transition model, which captures transitions between latent states; and a reward model, predicting the reward associated with a given state. Under the actor-critic framework, Dreamer utilizes an action model and a value model to learn behavior through imagination by propagating analytic gradients through the learned dynamics. Building upon this foundation, DreamerV2 [52] introduces a discrete latent state space along with an improved objective. DreamerV3 [53] extends its focus to a broader range of domains with fixed hyperparameters. DayDreamer [54] applies this method to physical robots performing real-world tasks.

IRIS [55] employs a GPT-like autoregressive Transformer as the foundation of its world model, with a VQ-VAE serving as the vision encoder. Subsequently, a policy is trained using imagined trajectories, which are unrolled from a real observation by the world model. TWM [56] also investigates the application of Transformers in building world models.

6) LLM-induced World Models: LLMs encompass a wealth of commonsense knowledge about the world, prompting many approaches to leverage that knowledge for improving VLAs.

DECKARD [57] prompts LLM to generate abstract world models (AWMs) represented as directed acyclic graphs [58], [59], specifically tailored for the task of item crafting in Minecraft. DECKARD iterates between two phases: in the Dream phase, it samples a subgoal guided by the AWM; in the Wake phase, DECKARD executes the subgoal and updates the AWM through interactions with the game. This guided approach enables DECKARD to achieve markedly faster item crafting compared to baselines lacking such guidance.

LLM-DM [60] uses an LLM to construct world models in planning domain definition language (PDDL)—a capability that LLM+P [61] did not achieve, as its PDDL world model was hand-crafted. The LLM also acts as an interface, mediating between the generated PDDL model and corrective feedback from syntactic validators and human experts. Finally, the PDDL world model serves as a symbolic simulator, assisting the LLM planner in generating plans.

RAP [62] repurposes an LLM to act as both a policy that predicts actions and a world model that provides the state transition distribution. Unlike previous chain-of-thought prompting methods, RAP incorporates Monte Carlo Tree Search (MCTS) to enable structured planning, allowing the LLM to

表II：VLAs的动力学学习方法。f（·）为动态模型。Fwd、inv：前向动力学与逆向动力学。

模型	视觉编码器	类型	目标	符号说明	机器人任务
Vi-PRoM [44]	ResNet	时间动态性	$\sum_{i=1}^{T}$ CE (i, $f(xi \mid x,)$)	f（·）: 在给定随机化序列x的原始序列x中预测帧索引 ′	**Sim-Mani**：元世界，弗兰卡厨房；**Real-Mani**：抽屉/门的开合功能
MaskDP [45]	ViT (MAE)	MAE（隐式正向与逆向动力学）	$\sum_{i=1}^{T}$ 对数$P(xi \mid x_{\neq}i,\ y)$	xi：状态或动作标记；y：另一种模式	**Sim**：DeepMind控制套件
迈达斯[46]	ViT，Mask R-CNN	逆动力学	$\sum_{i=1}^{T}$ MSE (at, $finv(st,\ st_{+1})$)	st，at：状态，动作	**Sim-Mani**： VIMA -Bench
SMART [47]	中心体	正向与逆向动力学	$\sum_{i=1}^{T}$ (MSE (st_{+1}, $ffwd(st,\ at)$) + MSE (at, $finv(st,\ st_{+1})$))	st，at：状态，动作	**Sim**：DeepMind控制套件
PACT [48]	ResNet-18、PointNet	正向动力学	$\sum_{i=1}^{T}$ MSE (st_{+1}, $ffwd(st,\ at)$)	st，at：状态，动作	**Sim-Navi**： 栖 息 地 ；**Real-Navi**（MuSHR车辆）
VPT [49]	ResNet	逆动力学	$\sum_{i=1}^{T}$ MSE (at, $finv(st,\ st_{+1})$)	st，at：状态，动作	**Sim**：《我的世界》
GR-1 [50]	ViT (MAE)	正向动力学	$\sum_{i=1}^{T}$ MSE (st_{+1}, $ffwd(st,\ at)$)	st，at：状态，动作	**Sim-Mani**：卡尔文

部分动作被遮蔽，模型需训练以恢复这些被遮蔽的动作。前两项动力学预测任务的引入有助于捕捉局部及短期动态特征，而第三项任务则旨在捕捉全局及长期时间依赖性。

MaskDP[45]采用掩码决策预测任务设计，该任务通过屏蔽状态标记和动作标记进行数据重构。这种掩码建模方法专门用于帮助模型理解正向与逆向动态关系。值得注意的是，与BERT或MAE等传统掩码建模方法不同，MaskDP可直接应用于下游任务场景。

PACT[48]引入了一个用于建模状态-动作转换的预训练目标。该模型以状态序列和动作序列作为输入，通过自回归机制对每个状态和动作标记进行预测。这个预训练模型作为动态模型使用，随后可针对定位、地图构建和导航等下游任务进行微调优化。

VPT[49]提出了一种视频预训练方法，该方法利用未标注的互联网数据对《我的世界》游戏的基础模型进行预训练。该方法首先使用少量标注数据训练逆向动力学模型，随后利用该模型对互联网视频进行标注。接着，通过行为克隆技术将这些新生成的自标注数据用于 VPT 基础模型的训练。该方法遵循半监督模仿学习原理。经过这一过程，模型在多项任务中展现出与人类相当的性能表现。

GR-1[50]引入了专为GPT风格模型定制的视频预测预训练方法。该模型具备预测未来帧的能力，与前向动力学学习机制相契合，从而提升动作预测的准确性。

5) 世界模型: 世界模型P（·）编码关于世界的常识性知识，并对给定动作的未来状态进行预测[39]：

$$\hat{s}_{t+1} \sim P(\hat{s}_{t+1} \mid s_t,\ at)\text{。} \tag{2}$$

该技术支持基于模型的控制与规划，使具身智能体能够在执行任何实际动作之前，于虚拟空间中搜索最优动作序列。尽管前向动力学学习同样旨在预测下一状态，但通常仅被视为预训练任务或辅助损失函数，用于增强基于RL-Transformer的动作解码器以应对主要机器人任务，而非作为独立模块使用。此外，新型具身体验可从视觉世界模型中采样获得——这些模型能显式生成未来状态的图像或视频。

Dreamer[51]采用三大核心模块构建潜在动力学模型：表征模型负责将图像编码为潜在状态；转移模型捕捉潜在状态间的转换关系；奖励模型则用于预测特定状态对应的奖励值。在演员-评论家框架下，Dreamer通过动作模型与价值模型，借助解析梯度在学习动态过程中的传播机制实现行为学习。在此基础上，DreamerV2[52]引入离散潜在状态空间并优化目标函数，DreamerV3[53]则将研究范围扩展至更多领域并固定超参数。DayDreamer[54]将该方法成功应用于执行真实场景任务的物理机器人系统。

IRIS[55]采用类GPT自回归Transformer作为其世界模型的基础架构，其中 VQ VAE 充当视觉编码器。随后通过从真实观测数据中解卷积生成的想象轨迹来训练策略模型。 TWM[56]还研究了Transformer在构建世界模型中的应用。

6) LLM诱导的世界模型: LLM包含大量关于世界的常识性知识，这促使许多研究方法利用这些知识来改进虚拟现实应用（VLAs）。

deckard[57]会引导大语言模型生成以有向无环图形式呈现的抽象世界模型（AWMs）[58]、[59]，这些模型专为《我的世界》中的物品制作任务量身定制。该系统采用两阶段迭代机制：在梦境阶段，系统根据AWM生成子目标进行采样；在清醒阶段，执行子目标并通过游戏交互实时更新AWM。这种智能引导机制使得deckard的物品制作效率较缺乏此类指导的基准模型显著提升。

LLM-DM[60]采用大语言模型（LLM）通过规划领域定义语言（PDDL）构建世界模型——这是LLM+P[61]未能实现的功能，因其 PDDL 世界模型是人工构建的。该LLM同时充当接口角色，在生成的 PDDL 模型与句法验证器及人类专家提供的修正反馈之间进行协调。最终，PDDL 世界模型作为符号模拟器，协助LLM规划器生成执行方案。

RAP[62]将大语言模型（LLM）重新定位为双重角色：既作为预测行为的策略模型，又作为提供状态转移分布的世界模型。与传统思维链提示方法不同，RAP采用蒙特卡洛树搜索（MCTS）技术实现结构化规划，使LLM能够

build a reasoning tree incrementally. This reasoning strategy helps RAP find a high-reward path that balances exploration and exploitation. Tree-Planner [63] improves efficiency by prompting the LLM only once to generate diverse paths within an action tree.

LLM-MCTS [64] builds upon RAP but extends the problem setting to POMDPs. As a world model, the LLM generates the initial belief of the current state; as a policy, it serves as a heuristic to guide action selection. By leveraging its commonsense knowledge, the LLM reduces the search space of MCTS, thereby improving search efficiency.

7) Visual World Models: Unlike LLM-induced world models, which are in textual form, visual world models generate images, videos, or 3D scenes of future states—aligning more closely with the physical world. They can further be utilized to generate new trajectories. This direction has been gaining increasing attention since Sora demonstrated world-simulation capabilities [65], as investigated by a dedicated survey [66].

Genie [67] introduces a new class of generative models, termed Generative Interactive Environments. It consists of three main components: a spatiotemporal video tokenizer, an autoregressive dynamics model, and a latent action model. After being trained on unlabeled videos in an unsupervised manner, Genie allows users to interact with the generative environment on a frame-by-frame basis. Consequently, Genie establishes a foundation world model.

3D-VLA [68] proposes a 3D world model capable of goal generation. It processes visual inputs, such as images, depth maps, and point clouds, and then generates a goal state—either as an image or a point cloud—using diffusion models in response to the user's query. The generated goal state can subsequently be used to guide robot control.

UniSim [69] builds a generative model based on real-world interaction videos. It is capable of simulating visual outcomes for both high-level and low-level actions, which can then be leveraged as new experiences for training embodied agents. E2WM [70] even treats existing simulators as world models to collect embodied experiences via MCTS.

8) Reasoning: Reasoning has become a key capability of LLMs, as demonstrated by chain-of-thought (CoT) methods [71], [72]. In embodied AI, researchers are exploring how to leverage CoT reasoning to refine the decision-making process.

Reasoning is naturally compatible with high-level task planning, as both often occur in the language domain. ThinkBot [73] applies CoT to recover missing action descriptions in sparse human instructions, thereby enhancing instruction coherence and boosting success rates in difficult tasks. ReAct [74] interleaves reasoning traces and actions, where CoT can help create action plans, inject commonsense knowledge, and handle exceptions, ultimately improving embodied decision-making. RAT [75] integrates CoT with retrieval-augmented generation (RAG) to mitigate hallucinations and thus improve embodied planning. Tree-Planner [63] employs a tree-of-thoughts approach for task planning.

Another paradigm involves equipping low-level control policies with reasoning capabilities, as pioneered by ECoT [76]. This method trains OpenVLA [34] to conduct embodied CoT reasoning about plans, sub-tasks, motions, and visual features, before predicting low-level actions. By relying on this multi-step reasoning rather than the "muscle memory" of VLAs, it improves success rates on challenging generalization tasks without requiring additional robot data. CoT-VLA [77] introduces visual CoT reasoning for VLAs.

9) Policy Steering: Policy steering can enhance VLA performance at test-time without the need for expensive retraining. V-GPS [78] re-ranks generated actions based on a learned value function, while RoboMonkey [79] employs a VLM-based verifier to select the optimal action from a sampled set.

Strengths and Limitations.

a) PVRs: Although time contrastive learning with videos and text-guided pretraining methods, such as CLIP, provide image-level information, they lack the pixel-level details offered by MAE-based self-supervised learning methods. Pixel-level information contains rich details—including segmentation masks, object positions, and depth estimation—that are generally more useful for robot manipulation tasks requiring high precision, as demonstrated by VC-1's comparison [31]. DINOv2 learns both pixel- and image-level features by combining masked image modeling with a momentum encoder and multi-crop augmentation, with benefits on downstream robotic tasks evidenced by OpenVLA [34]. I-JEPA focuses on patches in the representation space and, as a result, captures low-level image features more effectively than view-invariance methods such as DINO. Theia distills various off-the-shelf vision foundation models into a single model that surpasses isolated individual models, as demonstrated by comprehensive evaluations against most existing PVRs in robot learning [37].

b) Forward & inverse dynamics: Forward dynamics learning is generally more challenging than inverse dynamics learning because predicting future states is more complex than predicting past actions. Consequently, the difficulty of forward dynamics often leads to greater performance improvements, as demonstrated in SMART. However, inverse dynamics models can be used to generate action labels for datasets that contain only states, such as raw robot manipulation videos.

c) World Models & Reasoning: Although both world models and reasoning methods can be applied to low-level control policies and high-level task planners, current approaches remain distinct. World models are predominantly used to interact with control policies because they excel at generating the immediate next state given low-level actions. In contrast, CoT-based reasoning methods focus on task planning since they express thought chains in text, making them well-suited for refining text-based task plans.

B. Low-level Control Policies

Through the integration of an action decoder with perception modules, such as vision encoders and language encoders, a VLA model π_θ with parameters θ is formed as a control policy to execute language instructions p:

$$\hat{a}_t \sim \pi_\theta(\hat{a}_t | p, s_{\leq t}, a_{<t}). \tag{3}$$

It can also be referred to as a low-level policy, low-level controller, or action primitive. The diversity among VLAs

采用增量式推理树构建方法。该推理策略能帮助RAP找到平衡探索与利用的高回报路径。Tree-Planner[63]通过仅需一次引导LLM在动作树中生成多样化路径，从而显著提升效率。

LLM- MCTS[64]在RAP基础上进行扩展，将问题设定延伸至POMDPs。作为世界模型，LLM生成当前状态的初始信念；作为策略，它作为启发式方法指导行动选择。通过利用常识知识，LLM缩小了 MCTS 的搜索空间，从而提升搜索效率。

*7) 视觉世界模型：*与基于LLM生成的文本形式世界模型不同，视觉世界模型能够生成未来状态的图像、视频或3D场景——其表现更贴近物理世界。这些模型还可用于生成新的运动轨迹。自Sora展现出世界模拟能力以来[65]，这一研究方向备受关注（详见专项综述[66]）。Genie[67]提出了一类新型生成模型，称为生成式交互环境模型，其包含三个主要组件：时空视频分词器、自回归动力学模型和潜在动作模型。
在通过无监督方式对未标注视频进行训练后，Genie允许用户以逐帧方式与生成环境进行交互。因此，Genie建立了基础世界模型。

3D-VLA[68]提出了一种具备目标生成功能的三维世界模型。该模型可处理图像、深度图及点云等视觉输入数据，随后通过扩散模型响应用户查询，生成目标状态（可为图像或点云形式）。生成的目标状态可进一步用于指导机器人控制。

UniSim[69]基于真实世界交互视频构建了一个生成模型，能够模拟高层级和低层级动作的视觉效果，这些效果可作为训练具身智能体的新经验来源。E2WM[70]甚至将现有模拟器视为世界模型，通过 MCTS 技术收集具身经验。

*8) 推理能力：*推理能力已成为大语言模型（LLMs）的关键能力，这一点可通过思维链（CoT）方法[71]得到验证。[72]在具身人工智能领域，研究人员正在探索如何利用概念理论推理（CoT reasoning）来优化决策过程。

推理能力与高层次任务规划天然兼容，因为两者常在语言领域中同时发生。ThinkBot[73]通过应用概念理论（CoT）来补全人类指令中的缺失动作描述，从而提升指令连贯性并提高复杂任务成功率。ReAct[74]采用推理轨迹与动作的交互式设计，其中概念理论可辅助制定行动计划、注入常识知识并处理异常情况，最终优化具身化决策能力。RAT[75]将概念理论与检索增强生成（RAG）技术相结合，有效缓解幻觉现象从而提升具身化规划效果。Tree-Planner[63]则采用思维树结构方法进行任务规划。

另一种范式是为低级控制策略赋予推理能力，这一方法由ECoT[76]率先提出。该方法通过训练OpenVLA[34]对计划、子任务、运动及视觉特征进行具身化的CoT推理，再据此预测低级动作。相较于传统VLAs依赖的“肌肉记忆”，这种多步骤推理机制能在无需额外机器人数据的情况下，显著提升复杂泛化任务的成功率。CoT-VLA[77]则为VLAs引入了视觉CoT推理技术。

*9) 策略引导：*策略引导可在测试阶段提升VLA性能，且无需昂贵的重新训练。V-GPS[78]基于学习到的价值函数对生成的动作进行重新排序，而RoboMonkey[79]则采用基于VLM的验证器从采样集合中选择最优动作。

优势与局限性

*a) PVRs：*尽管基于视频的时间对比学习及文本引导的预训练方法（如CLIP）能提供图像级信息，但它们缺乏基于MAE的自监督学习方法所提供的像素级细节。像素级信息包含丰富的细节——包括分割掩码、物体位置和深度估计——这些细节对于需要高精度的机器人操作任务通常更为有用，VC-1的对比实验已证明了这一点[31]。DINOv2通过将掩码图像建模与动量编码器及多裁剪增强技术相结合，同时学习像素级和图像级特征，并在OpenVLA[34]所示的下游机器人任务中展现出优势。I- JEPA 专注于表征空间中的图像块，因此相较于DINO等视图不变性方法，能更有效地捕捉低级图像特征。Theia将多种现成的视觉基础模型整合为单一模型，其性能超越了单独使用的各个模型——这一点已在针对大多数现有PVRs的机器人学习全面评估中得到验证[37]。

*b) 正向与逆向动力学：*正向动力学学习通常比逆向动力学学习更具挑战性，因为预测未来状态比预测过去动作更为复杂。正向动力学的高难度往往能带来更显著的性能提升，SMART实验结果即印证了这一点。不过逆向动力学模型也可用于为仅包含状态数据的数据集（如原始机器人操作视频）生成动作标签。

*c) 世界模型与推理方法：*尽管世界模型和推理方法均可应用于低级控制策略与高级任务规划系统，但现有研究路径仍存在显著差异。世界模型主要应用于控制策略交互场景，因其擅长根据低级动作生成即时的下一状态。相比之下，基于概念图（CoT）的推理方法侧重于任务规划领域，由于其通过文本形式呈现思维链路，因此特别适用于优化基于文本的任务规划方案。

B. 低级控制策略

通过将动作解码器与感知模块（如视觉编码器和语言编码器）集成，形成了一个参数为θ的VLA模型π_θ，该模型作为执行语言指令p的控制策略：

$$a\wedge_t \sim \pi_\theta(a\wedge_t|p,\ s_{\leq t},\ a_{<t})。 \quad (3)$$

它也可称为低级策略、低级控制器或动作原语。VLAs之间的多样性

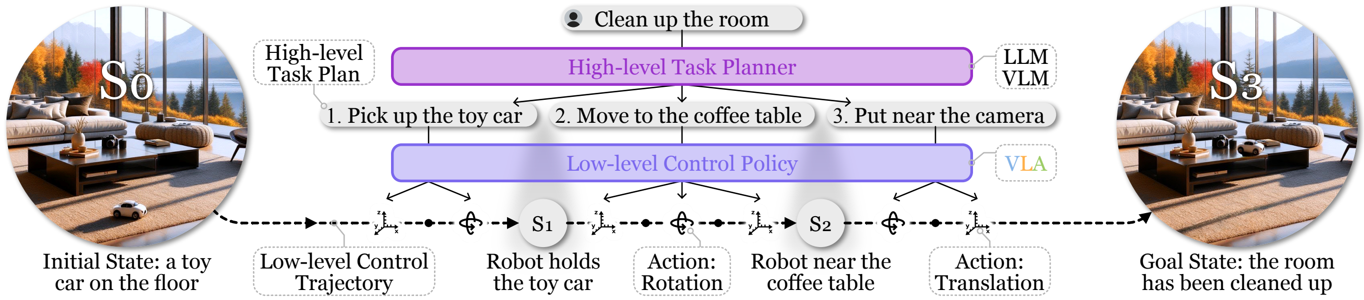


Figure 4: Illustration of a hierarchical robot policy. The high-level task planner decomposes the user instruction into subtasks, which are then executed step by step by the low-level control policy.

arises from individual modules and overall architectures. The general architecture is shown in Figure 1. This section explores various approaches to designing low-level control policies. Table III summarizes their technical details.

1) Non-Transformer Control Policies: Prior to the adoption of Transformer models, early control policies for language-conditioned robotic tasks varied significantly in architecture.

CLIPort [28] integrates CLIP with the Transporter network, creating a two-stream architecture. The CLIP vision encoder extracts "semantic" information from the RGB image, while the Transporter network extracts "spatial" information from the RGB-D image. The CLIP sentence encoder encodes the language instruction and guides the output $SE(2)$ action, consisting of paired pick and place end-effector poses. It represents an early demonstration of language-conditioned pick-and-place capabilities.

BC-Z [80] processes two types of task instructions: a language instruction or a human demonstration video. The environment is presented to the model in the form of an RGB image. Then the instruction embedding and the image embedding are combined through the FiLM layer, culminating in the generation of actions. This conditional policy is asserted to exhibit zero-shot task generalization to unseen tasks.

MCIL [81] represents a pioneering robot policy that integrates free-form natural language conditioning. This is in contrast to earlier approaches that typically rely on conditions in the form of task IDs or goal images. MCIL introduces the capability to leverage unlabeled and unstructured demonstration data. This is achieved by training the policy to follow either image or language goals, with a small fraction of the training dataset consisting of paired image and language goals.

HULC [82] introduces several techniques aimed at enhancing robot learning architectures. These include a hierarchical decomposition of robot learning, a multimodal Transformer, and discrete latent plans. The Transformer learns high-level behaviors, hierarchically dividing low-level local policies and the global plan. Additionally, HULC incorporates a visuo-lingual semantic alignment loss based on contrastive learning to align VL modalities. HULC++ [83] further integrates a self-supervised affordance model. This model guides HULC to the actionable region specified by a language instruction, enabling it to fulfill tasks within this designated area.

UniPi [84] treats the decision-making problem as a text-conditioned video generation problem. To predict actions, UniPi generates a video based on a given text instruction, and actions are extracted from the frames of the video through inverse dynamics. This innovative policy-as-video formulation offers several advantages, including enhanced generalization across diverse robot tasks and the potential for knowledge transfer from internet videos to real robots.

2) Transformer-based Control Policies: Since the introduction of Transformers, control policies have converged to similar Transformer-based architectures.

Interactive Language [86] presents a robotic system wherein the low-level control policy can be guided in real-time by human instructions conveyed through language, enabling the completion of long-horizon rearrangement tasks. The efficacy of such language-based guidance is primarily attributed to the utilization of a meticulously collected dataset containing diverse language instructions, which surpasses previous datasets by an order of magnitude in scale.

Hiveformer [87] places significant emphasis on leveraging multi-view scene observations and maintaining the full observation history for a language-conditioned policy. This approach represents an advancement over previous systems, such as CLIPort and BC-Z, that only use the current observation. Notably, Hiveformer stands out as one of the early adopters of Transformer architecture as its policy backbone.

Gato [17] proposes a model that can play Atari games, caption images, and stack blocks, all with a single set of model parameters. This achievement is facilitated by a unified tokenization scheme, harmonizing the input and output across diverse tasks and domains. Consequently, Gato enables the simultaneous training of different tasks. Astra [126] optimizes this architecture via trajectory attention.

RoboCat [93] proposes a self-improvement process designed to enable an agent to rapidly adapt to new tasks with as few as 100 demonstrations. This self-improvement process iteratively finetunes the model and self-generates new data with the finetuned model. Built upon the Gato model, RoboCat incorporates the VQ-GAN image encoder. During training, RoboCat predicts not only the next action but also future observations. The effectiveness of the self-improvement process is demonstrated through comprehensive experiments conducted in both simulated and real-world environments under multi-task, multi-embodiment settings.

RT-1 [95], developed by the same team as BC-Z, shares similarities with BC-Z but introduces some key distinctions. Notably, RT-1 employs a vision encoder based on the more efficient EfficientNet, departing from BC-Z's use of ResNet.

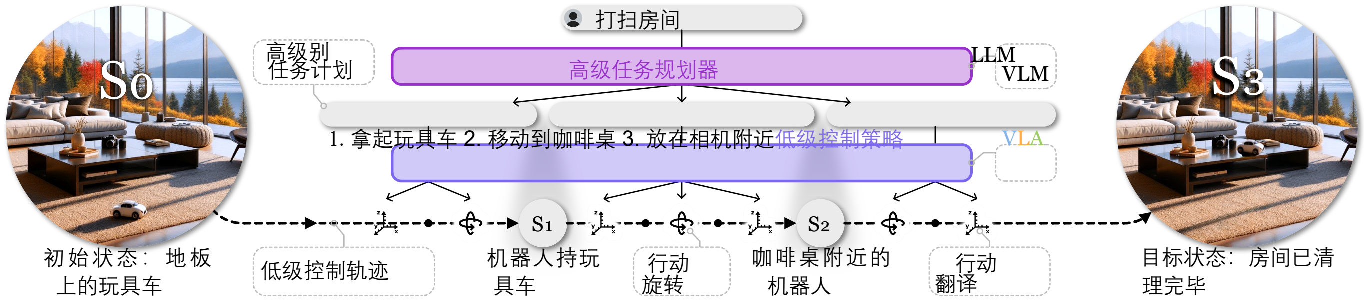


图4：分层机器人策略示意图。高层任务规划器将用户指令分解为子任务，这些子任务随后由低层控制策略逐步执行。

该架构由各个模块及整体架构共同构成。通用架构如图1所示。本节探讨了设计低层控制策略的多种方法。表III汇总了这些方法的技术细节。

*1) 非Transformer控制策略：*在Transformer模型应用之前，针对语言条件化机器人任务的早期控制策略在架构设计上存在显著差异。

CLIPort[28]将CLIP与Transporter网络进行整合，构建了双流架构。CLIP视觉编码器从RGB图像中提取“语义”信息，而Transporter网络则从RGB-D图像中提取“空间”信息。CLIP句子编码器对语言指令进行编码，并指导输出$SE(2)$动作——该动作由成对的抓取与放置末端执行器位姿组成。这标志着语言条件化抓取-放置能力的早期验证成果。

BC-Z[80]系统可处理两种类型的任务指令：语言指令或人类演示视频。模型接收的环境信息以RGB图像形式输入，随后通过FiLM层将指令嵌入与图像嵌入进行融合，最终生成动作指令。该条件策略被证实具有零样本任务泛化能力，能够有效应对未见过的任务场景。

MCIL[81]代表了一种开创性的机器人策略，其整合了自由形式自然语言条件反射机制。这与早期通常依赖任务ID或目标图像形式条件的方法形成鲜明对比。 MCIL 引入了利用未标注且非结构化演示数据的能力，通过训练策略跟随图像或语言目标实现这一目标，其中训练数据集中仅有少量包含配对图像与语言目标的数据。

HULC[82]提出了一系列旨在优化机器人学习架构的技术方案，包括机器人学习的层次化分解方法、多模态Transformer模型以及离散潜在规划机制。该Transformer模型通过分层结构学习高层次行为模式，将低层级局部策略与全局规划进行有机整合。此外， HULC 引入基于对比学习的视觉-语言语义对齐损失函数，有效实现视觉模态间的语义对齐。 HULC ++[83]进一步集成自监督功能位模型，该模型能根据语言指令指定的可操作区域引导 HULC 执行任务，确保其在预设区域内精准完成操作。

UniPi[84]将决策问题转化为文本驱动的视频生成任务。该系统通过生成基于文本指令的视频来预测动作，并利用逆向动力学方法从视频帧中提取动作信息。这种创新的“策略即视频”框架具有多重优势：不仅能显著提升机器人在多样化任务中的泛化能力，还可实现互联网视频知识向真实机器人系统的迁移应用。

*2) 基于Transformer的控制策略：*自Transformer技术问世以来，控制策略已逐渐趋同于基于Transformer的架构体系。

交互式语言系统[86]构建了一个机器人系统，该系统能够通过语言指令实时引导底层控制策略，从而实现长期规划重组任务的高效完成。这种基于语言的引导机制之所以高效，主要得益于其采用了经过精心收集的多样化语言指令数据集，其规模远超以往同类数据集，达到数量级提升。

Hiveformer[87]在语言条件策略中特别强调利用多视角场景观测数据，并完整保留观测历史记录。相较于仅使用当前观测数据的CLIPort和BC-Z等早期系统，该方法实现了显著改进。值得注意的是，Hiveformer作为Transformer架构作为策略骨干网络的早期采用者，在技术实现上具有开创性意义。

Gato[17]提出了一种模型，能够通过单一参数集完成Atari游戏模拟、图像字幕生成以及积木堆叠等任务。该成果得益于统一的分词方案，实现了跨任务与跨领域的输入输出协调统一。因此，Gato支持不同任务的同步训练。Astra[126]通过轨迹注意力机制对该架构进行了优化。

RoboCat[93]提出了一种自我改进机制，旨在使智能体仅需100次示范即可快速适应新任务。该机制通过迭代优化模型，并利用优化后的模型自动生成新数据。基于Gato模型，RoboCat整合了 VQ -GAN图像编码器。在训练过程中，RoboCat不仅能预测下一步动作，还能预测后续观测结果。通过在模拟环境和真实世界环境中进行的多任务、多实体场景下的全面实验，验证了该自我改进机制的有效性。

RT-1[95]由与BC-Z相同的团队开发，与BC-Z具有相似性但引入了一些关键差异。值得注意的是，RT-1采用基于更高效EfficientNet的视觉编码器，而BC-Z则使用ResNet。

Table III: VLAs that serve as low-level control policies. ♢ indicates large VLAs. Closely related (non-VLA models) are included in brackets. The remaining are generalized VLAs. BC: behavior cloning (cont/disc: continuous/discrete action). Xattn: cross-attention. Concat: concatenation. Quant.: quantization. p/s: prompt/state vision encoder. [SC]: self-collect data.

Model	Vision Encoder	Language Encoder / VLM	Action Decoder/Head	Architecture	Training Objectives	Training Datasets	Environments, Embodiments, Tasks, and Skills
CLIPort [28]	CLIP-ResNet50, Transporter-ResNet	CLIP-GPT	LingUNet	Hadamard	BC ($SE(2)$)	[SC]	**Sim**: Ravens
MCIL [81]	Custom-CNN	USE	RNN	LMP	MCIL	Play-LMP, [SC]	**Sim**: 3D Playroom environment
HULC [82], [83]	MCIL CNN	Sentence-BERT	RNN	LMP	MCIL	CALVIN data	**Sim**: CALVIN
Language costs [85]	CLIP-ViT, UNet	CLIP-GPT	STORM	Concat	MLE (cost map)	[SC]	**Sim** & **Real** (Franka): pick, place
Interactive Language [86]	ResNet	CLIP-GPT	TFM	Xattn	BC (cont)	[SC: Language-Table]	**Sim** & **Real** (xArm6): rearrangement
Hiveformer [87]	UNet	CLIP-GPT	TFM, UNet	Concat	BC (cont)	RLBench data	**Sim**: RLBench
PerAct [88]	3D CNN	CLIP-GPT	PerceiverIO	Concat	3D affordance	RLBench data, [SC]	**Sim**: RLBench; **Real** (Franka): pick, place, stack, open drawer, sweep, insert peg, etc.
Act3D [89]	CLIP-ResNet50	CLIP-GPT	TFM	Xattn	3D affordance (voxel)	RLBench data, [SC]	**Sim**: RLBench; **Real** (Franka): reach target, wipe, stack, etc.
RVT [90], RVT-2 [91]	CLIP-ResNet50	CLIP-GPT	TFM	Concat	2D affordance (project to 3D)	RLBench data, [SC]	**Sim**: RLBench; **Real** (Franka): stack, press, place
RoboPoint [92]	ViT-L/14	Vicuna-V1.5 13B	–	Concat	2D affordance (project to 3D)	[SC]	**Real** (Franka): pick, place
Gato [17]	ViT	Sent.Piece	TFM	Concat	BC (cont & disc)	[SC]	**Sim** & **Real** (Sawyer): RGB-stacking
(RoboCat) [93]	VQ-GAN (p, s)	–	TFM	Quant.	BC, observation prediction	Self-improvement	**Sim** & **Real** (Sawyer, Franka, KUKA): stacking, building, lifting, insertion, removal
VIMA [94]	ViT, Mask R-CNN	T5	TFM	Xattn	BC ($SE(2)$)	[SC:VIMA-Data]	**Sim** (Ravens): VIMA-Bench
BC-Z [80]	ResNet18 (p, s)	USE	MLP	FiLM	BC (cont)	[SC]	**Real** (EDR): pick-place/wipe/drag, grasp, push
RT-1 [95]	EfficientNet	USE	TFM	FiLM	BC (disc)	[SC: Fractal]	**Real** (EDR): pick-place, move, knock
MOO [94]	OWL-ViT (p), EfficientNet (s)	USE	TFM	FiLM	BC (disc)	[SC]	**Real** (EDR): pick, move near, knock, place upright, place into
Q-Transformer [96]	EfficientNet	USE	TFM	FiLM	TD error	Fractal, Auto-collect	**Sim**: pick; **Real** (EDR): pick, place, open/close drawer, move near
(RT-Trajectory) [97]	EfficientNet	–	TFM	–	BC (disc)	[SC]	**Real** (EDR): pick, place, fold towel, swivel chair, etc.
(ACT) [98]	ResNet18	–	CVAE-TFM	–	BC (cont, action chunking)	[SC] with ALOHA	**Sim**: transfer cube, bimanual insertion; **Real** (ViperX, WidowX): slot battery, open cup, etc
RoboAgent / MT-ACT [99]	CNN	–	CVAE-TFM	FiLM	BC (cont, action chunking)	RoboSet	**Real** (Franka): pick, place, open/close drawers, pour, push, drag, etc.
RoboFlamingo [100]	CLIP-ViT-L/14	LLaMA, MPT, GPT-NeoX	LSTM, TFM	Xattn	BC (cont)	CALVIN data	**Sim**: CALVIN
RoboUniView [101]	UVFormer	(Model design follows RoboFlamingo)			BC (cont)	CALVIN data	**Sim**: CALVIN
DeeR-VLA [102]	(Model design follows RoboFlamingo++)				BC (cont)	CALVIN data	**Sim**: CALVIN
Instruct2Act [103]	CLIP, SAM	ChatGPT	Robot APIs	Tool-Use	–	–	**Sim** (Ravens): VIMA-Bench
VoxPoser [104]	ViLD, MDETR, OWL-ViT, SAM	GPT-4	MPC	Tool-Use	–	–	**Sim**: Sapien; **Real** (Franka): move&avoid, set table, close drawer, open bottle, sweep trash
UniPi [84]	Imagen video	T5-XXL	CNN	Xattn	Inverse dynamics	[SC:PF], Ravens, BridgeV1	**Sim**: Painting Factory (PF), Ravens
(Diffusion Policy) [105]	ResNet18	–	UNet, TFM	–	DDPM	[SC]	**Sim**: Robomimic, Franka Kitchen, etc; **Real** (UR5, Franka): push-T, flip mug, pour sauce
(DP3) [106]	DP3 Encoder	(Model design follows Diffusion Policy)			DDIM	[SC]	**Sim**: (72 tasks); **Real** (Franka, Allegro hand): roll-up, dumpling, drill, pour
SUDD [107]	ResNet18	CLIP-GPT	UNet, TFM	Concat	DDPM	Lang-guided data generation	**Sim**: MuJoCo; **Real** (UR5e): pick, place
Octo [108]	CNN	T5-base	TFM	Concat	DDPM	OXE	**Real**: BridgeV2, CMU Baking, Stanford Coffee, Berkeley Peg Insert, etc.
3D Diffuser Actor [109]	(Model design follows Act3D)				DDPM	RLBench, CALVIN, [SC]	**Sim**: RLBench, CALVIN; **Real** (Franka): put, open, close, stack, etc.
MDT [110]	CLIP-ViT-B/16 (p), Voltron/ResNet18 (s)	CLIP-GPT	DiT	Concat	DDIM	CALVIN, LIBERO, [SC]	**Sim**: CALVIN, LIBERO; **Real** (Franka): pick, push, open, close, etc.
RDT-1B [111]	SigLIP	T5-XXL	DiT	Xattn	DDPM	(Aggregated Datasets)	**Real** (ALOHA dual-arm): wash, pour, fold, etc.
RT-2 [1] ♢	ViT-4B, ViT-22B	PaLI-X, PaLM-E	Symbol-tuning	Concat	BC (disc), Co-fine-tuning	Fractal, VQA	**Sim**: Language-Table; **Real**: RT-1 evaluation tasks
RT-H [112] ♢	(Model design follows RT-2)				BC (disc)	Diverse+Kitchen	**Real**: Diverse+Kitchen eval tasks
RT-X [113] ♢	(Models from RT-1 and RT-2)				BC (disc)	[SC: OXE]	**Real**: BridgeV2, RT-1 evaluation tasks, etc.
OpenVLA [34] ♢	DINOv2, SigLIP	Prismatic-7B	Symbol-tuning	Concat	BC (disc)	OXE, DROID	**Real**: BridgeV2, RT-1 evaluation tasks, Franka-Tabletop, DROID, etc.
OpenVLA-OFT [114] ♢	(Improves OpenVLA with OFT recipe)				BC (cont, parallel decode w/ chunk.)	LIBERO, [SC]	**Sim**: LIBERO; **Real** (ALOHA setup): fold, scoop, put
TraceVLA [115] ♢	(Model design follows OpenVLA, adding visual trace prompting)				BC (disc)	BridgeV2, Fractal, [SC]	**Sim**: SimplerEnv; **Real** (WidowX): pick, push, fold, swipe
π_0 [116] ♢	SigLIP	PaliGemma	Action expert	Concat	Flow matching	OXE, [SC: π-cross-embod.]	**Real** (general robot control): sweep, open, pack, etc.
RoboMamba [117] ♢	CLIP-ViT	Mamba	MLP	Concat	BC (cont)	[SC]	**Sim**: Sapien; **Real** (Franka): open door/box, staple
SpatialVLA [118] ♢	SigLIP, Ego3D Position Encoding	PaliGemma 2	MLP	Concat	BC (Adaptive Action Grids)	OXE, BridgeV2, Fractal, RH20T	**Sim**: SimplerEnv, LIBERO; **Real** (WidowX): pick, place, close, push
TinyVLA [119] ♢	ViT	Pythia	Diffusion Policy	Concat	DDPM	OXE, [SC]	**Sim**: MetaWorld; **Real** (Franka, UR5): place, stack, flip mug, close drawer, open box
CogACT [120] ♢	DINOv2, SigLIP	LLaMA 2	DiT	Concat	DDIM	OXE, [SC]	**Sim**: SimplerEnv; **Real** (Realman, Franka): pick, place, stack, open/close oven
DexVLA [121] ♢	ViT, ResNet-50	Qwen2-VL 2B, DistilBERT,	ScaleDP	Concat	DDPM	[SC]	**Real** (Franka, UR5e, AgileX): pick, fold shirt, bus table, pour
HybridVLA [122] ♢	DINOv2, SigLIP, CLIP	LLaMA 2, Phi-2	MLP	Concat	DDIM	OXE, DROID	**Sim**: RLBench; **Real** (Franka, AgileX): pick-place, pour, open drawer, fold, etc.
LAPA [123] ♢	C-ViViT, VQ-GAN	LWM-Chat-1M	MLP	Concat	LAPA	OXE, BridgeV2, Something V2	**Sim**: Language-Table, SimplerEnv; **Real** (Franka): pick, cover with towel, knock over
WorldVLA [124] ♢	VQ-GAN	Chameleon	TFM	Quant.	BC (disc, chunk.), world model	LIBERO, OpenVLA data	**Sim**: LIBERO
UniVLA [125] ♢	MoVQ	Emu3	TFM	Quant.	BC (disc, chunk.), world model	(Sim benchmark data)	**Sim**: CALVIN, LIBERO, SimplerEnv

表III：作为低级控制策略的VLAs。◊表示大型VLAs；密切相关（非VLA模型）的模型列于括号内；其余均为通用型VLAs。BC：行为克隆（cont/disc：连续/离散动作）；Xattn：交叉
注意。连接：连接操作。量化：量化处理。p/s：提示/状态视觉编码器。[SC]：自收集数据。

模型	视觉编码器	语言编码器/ VLM	动作解码器/头部	建筑学	培训目标	训练数据集	环境、实施例、任务和技能
CLIPort [28]	CLIP-ResNet50、Transporter-ResNet	CLIP-GPT	LingUNet	哈达玛德	BC ($SE(2)$)	[SC]	**Sim**：乌鸦
MCIL [81]	自定义CNN	用	递归神经网络	LMP	MCIL	Play- LMP，[SC]	**Sim**：3D游戏室环境
HULC [82], [83]	MCIL CNN	句子BERT	递归神经网络	LMP	MCIL	卡尔文数据	**Sim**：卡尔文
语言成本[85]	CLIP-ViT，UNet	CLIP-GPT	风暴	粘合	MLE（成本图）	[SC]	**模拟与真实**（Franka）：选择、放置
交互式语言 [86]	ResNet	CLIP-GPT	TFM	Xattn	BC（续）	[SC: LanguageTable]	**模拟与真实**（xArm6）：重排
Hiveformer [87]	联网	CLIP-GPT	TFM 、UNet	粘合	BC（续）	RLBench 数据	**Sim**: RLBench
PerAct [88]	3D CNN	CLIP-GPT	PerceiverIO	粘合	三维可及性	RLBench数据[SC]	**Sim**：RLBench；**Real**（Franka）：选择、放置、堆叠、打开抽屉、清扫、插入钉子等。
Act3D [89]	CLIP-ResNet50	CLIP-GPT	TFM	Xattn	三维功能区（体素）	RLBench数据[SC]	**Sim**：RLBenchmark；**Real**（Franka）：达到目标、清除、堆叠等。
RVT [90], RVT-2 [91]	CLIP-ResNet50	CLIP-GPT	TFM	粘合	二维可及性（项目转3D）	RLBench数据[SC]	**Sim**: RLBench；**Real**(Franka)：堆栈、压入、放置
RoboPoint [92]	ViT-L/14	Vicuna-V1.5 13B	—	粘合	二维可及性（项目转3D）	[SC]	**实际**（Franka）：挑选、放置
猫[17]	维生素T	Sent. 片	TFM	粘合	BC（结肠与结肠系膜）	[SC]	**模拟与真实**（Sawyer）：RGB叠加
(RoboCat) [93]	VQ-GAN (p, s)	—	TFM	定量	观测预测	自己改善	**模拟与真实**（Sawyer，Franka，KUKA）：堆叠、构建、提升、插入、移除
VIMA [94]	ViT，Mask R-CNN	T5	TFM	Xattn	BC ($SE(2)$)	[SC: VIMA -数据]	**Sim**(Ravens)： VIMA -Bench
BC-Z [80]	ResNet18（p，s）	用	MLP	菲林	BC（续）	[SC]	**实际**(EDR)：选择位置/擦拭/拖动、抓取、推动
RT-1 [95]	高效网络	用	TFM	菲林	BC（光盘）	[SC：分形]	**实际**(EDR)：选位、移动、击打
MOO [94]	OWL-ViT（p），EfficientNet (s)	用	TFM	菲林	BC（光盘）	[SC]	**实际**（EDR）：拾取、靠近、敲击、竖立、放入
Q-Transformer [96]	高效网	用	TFM	电影；影片	TD错误	分形，自收集	**Sim**：选择；**Real**（EDR）：选择、放置、打开/关闭抽屉、靠近
（RT轨迹）[97]	高效网	—	TFM	—	BC（光盘）	[SC]	**实际**(EDR)：挑选、放置、折叠毛巾、旋转椅子等。
（ACT）[98]	ResNet18	—	CVAE-TFM	—	BC（内容、动作分块）	伴发[SC] 夏威夷人问候语	**Sim**：转移立方体，双手插入式；**Real**（ViperX、Widox）：插槽式电池、开放式杯等
RoboAgent / MTACT [99]	中心体	—	CVAE-TFM	电影；影片	BC（内容、动作分块）	RoboSet	**实际**（弗兰卡）：挑选、放置、开关抽屉、倾倒、推动、拖拽等动作。
RoboFlamingo [100]	CLIP-ViT-L/14	LLaMA、 MPT 、GPT-NeoX	激光扫描跟踪法 TFM	Xattn	BC（续）	卡尔文数据	**Sim**：卡尔文
RoboUniView [101]	UV成型机	（模型设计参照RoboFlamingo）			BC（续）	卡尔文数据	**Sim**：卡尔文
DeeR-VLA [102]	（模型设计遵循RoboFlamingo++）				BC（续）	卡尔文数据	**Sim**：卡尔文
Instruct2Act [103]	CLIP，SAM	ChatGPT	机器人API	工具使用	—	—	**Sim**(Ravens)： VIMA -Bench
VoxPoser [104]	ViLD, MDETR, OWL-ViT、SAM	GPT-4	MPC	工具使用	—	—	**Sim**：Sapien；**Real**（Franka）：移动&躲避、摆桌子、关抽屉、开瓶子、扫垃圾
UniPi [84]	图片视频	T5-XXL	中心体	Xattn	逆动力学	[SC: PF]，乌鸦，BridgeV1	**Sim**：绘画工厂（PF），乌鸦
（扩散策略）[105]	ResNet18	—	UNet，TFM		分布式数据处理管理	[SC]	**模拟**：仿生机器人、Franka Kitchen等；**真实**（UR5、Franka）：推T、翻转杯子、倒酱料
(DP3) [106]	DP3编码器	（模型设计遵循扩散策略）			DDIM	[SC]	**Sim**：（72 项 任 务）；**Real**（Franka、Allegro 手部）：卷曲、饺子、钻孔、倾倒
SUDD [107]	ResNet18	CLIP-GPT	UNet，TFM	粘合	分布式数据处理管理	语言引导的数据生成	**Sim**：MuJoCo；**Real**（UR5e）：选择、放置
Octo [108]	中心体	T5碱基	TFM	粘合	分布式数据处理管理	OXE	**实际应用**：BridgeV2、CMU Baking、Stanford Coffee、Berkeley Peg Insert等。
3D扩散器演员 [109]	（模型设计遵循Act3D标准）				分布式数据处理管理	RLBench，calvin，[SC]	**Sim**: RLBench，calvin；**Real**(Franka): put，open，close，stack 等。
MDT [110]	CLIP-ViT-B/16（p），Voltron/ResNet18（s）	CLIP-GPT	DiT	粘合	DDIM	卡尔文，LIBERO，州[南卡罗来纳州]	**Sim**：calvin， LIBERO；**Real**（Franka）：pick、push、open、close等。
RDT-1B [111]	SigLIP	T5-XXL	DiT	Xattn	分布式数据处理管理	（聚合数据集）	**实际操作**（Aloha双臂设备）：清洗、倾倒、折叠等。
RT-2 [1] ◊	ViT-4B, ViT-22B	PaLI-X、PaLM-E符号	调谐	粘合	BC（盘式），Cofine调谐	分形， VQA	**Sim**：语言表；**Real**：RT-1评估任务
RT-H [112] ◊	（模型设计遵循RT-2标准）				BC（光盘）	Diverse+Kitchen	**Real**：多样化+厨房评估任务
RT-X [113] ◊	（来自RT-1和RT-2的模型）				BC（光盘）	[SC: OXE]	**实际应用**：BridgeV2、RT-1评估任务等。
OpenVLA [34] ◊	DINOv2，SigLIP	棱镜-7B	符号调谐	粘合	BC（光盘）	OXE，DROID	**实际应用**：BridgeV2、RT-1评估任务、Franka-Tabletop、droid等。
OpenVLA-OFT [114] ◊	（通过OFT配方改进了OpenVLA）				BC（连续、平行使用块进行解码。）	LIBERO，[SC]	**Sim**：LIBERO；**Real**（aloha设置）：折叠、scoops、放置
TraceVLA [115] ◊	（模型设计遵循OpenVLA标准，并增加可视化轨迹提示功能）				BC（光盘）	BridgeV2，Fractal，[SC]	**Sim**：SimplerEnv；**Real**（WidowX）：pick、push、fold、swipe
π_0 [116] ◊	SigLIP	巴利语宝石	行动专家	粘合	流匹配	OXE，[SC:π-交叉-实体]	**实际应用**（通用机器人控制）：清扫、开门、打包等。
RoboMamba [117] ◊	CLIP-ViT	曼巴蛇	MLP	粘合	BC（续）	[SC]	**Sim**：Sapien；**Real**（Franka）：开门/开箱，钉合
SpatialVLA [118] ◊	SigLIP，Ego3D 位置编码	PaliGemma 2	MLP	粘合	BC（自适应操作网格）	OXE, BridgeV2, 分形，RH20T	**Sim**：SimplerEnv，LIBERO；**Real**(WidowX): pick，place，close，push
TinyVLA [119] ◊	维生素T	皮提亚	扩散政策	粘合	分布式数据处理管理	OXE, [SC]	**Sim**：MetaWorld；**Real**（Franka，UR5）：放置、堆叠、翻转杯子、关闭抽屉、打开盒子
CogACT [120] ◊	DINOv2，SigLIP	LLaMA 2	DiT	粘合	DDIM	OXE, [SC]	**Sim**：SimplerEnv；**Real**(Realman，Franka)：选择、放置、堆叠、打开/关闭烤箱
DexVLA [121] ◊	ViT，ResNet-50	Qwen2-VL 2B, DistilBERT	规模DP	粘合	分布式数据处理管理	[SC]	**实际**（Franka、UR5e、AgileX）：挑选、折叠衬衫、摆设餐桌、倾倒
HybridVLA [122] ◊	DINOv2、SigLIP、CLIP	LLaMA 2，Phi-2	MLP	粘合	DDIM	OXE，DROID	**Sim**: RLBench；**Real**（Franka、AgileX）：取物、倾倒、打开抽屉、折叠等操作。
LAPA [123] ◊	C-ViViT，VQ -GAN	LWM-Chat-1M	MLP	粘合	自细胞碱性磷酸酶活性	OXE 、BridgeV2、Something V2	**Sim**：语言-表格，更简单的环境；**Real**（Franka）：选择，用毛巾覆盖，打翻
WorldVLA [124] ◊	VQ-GAN	变色蜥蜴	TFM	定量	BC（磁盘、数据块），世界模型	LIBERO, OpenVLA数据	**Sim**： LIBERO
UniVLA [125] ◊	MoVQ	Emu3	TFM	定量	BC（磁盘、数据块），世界模型	（模拟基准数据）	**Sim**：calvin、LIBERO、更简单的环境

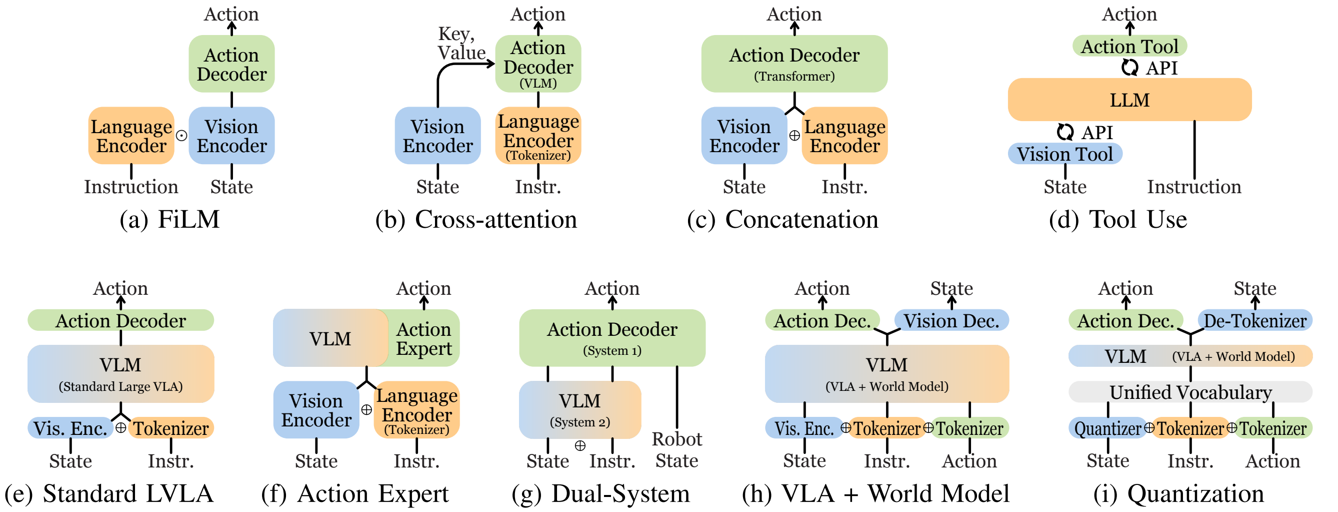


Figure 5: Representative architectures of VLA models. Some correspond to the "Architecture" column in Table III. ⊙: Hadamard product. ⊕: Concatenation.

However, RT-1 does not use video as a task instruction. Additionally, RT-1 replaces the MLP action decoder in BC-Z with a Transformer decoder, producing discretized actions. This modification enables RT-1 to attend to past images, enhancing its performance compared to BC-Z.

Q-Transformer [96] extends RT-1 by introducing autoregressive Q-functions. In contrast to RT-1, which learns expert trajectories through imitation learning, Q-Transformer adopts Q-learning methods. Alongside the TD error objective of Q-learning, a conservative regularizer is incorporated to ensure that the maximum value action remains in-distribution. This approach allows Q-Transformer to leverage not only successful demonstrations but also failed trajectories for learning.

RT-Trajectory [97] adopts trajectory sketches as policy conditions instead of relying on language conditions or goal conditions. These trajectory sketches consist of curves that delineate the intended trajectory for the robot end-effector to follow. They can be manually specified through a graphical user interface, extracted from human demonstration videos, or generated by foundation models. RT-Trajectory's policy is built upon the backbone of RT-1 and trained to control the robot arm to accurately follow the trajectory sketches. This approach facilitates generalization to novel objects, tasks, and skills, as trajectories from various tasks are transferable.

ACT [98] builds a conditional VAE policy with action chunking, requiring the policy to predict a sequence of actions rather than a single one. During inference, the action sequence is averaged using a method called temporal ensembling. RoboAgent [99] extends ACT via MT-ACT, demonstrating that action chunking improves temporal consistency. It also introduces an inpainting-based semantic augmentation method.

RoboFlamingo [100] adapts the existing Flamingo VLM to a robot policy by attaching an LSTM-based policy head. This demonstrates that pretrained VLMs can be effectively transferred to language-conditioned robotic manipulation tasks.

A recent trend in LLMs is equipping them with tool-use capabilities by generating code that calls tools via APIs [127]. Instruct2Act [103] follows this paradigm by integrating vision and action tools, enabling LLMs to perform robotic tasks.

3) Control Policies for Multimodal Instructions: Multi-modal instruction enables new ways to specify tasks, such as through demonstrations, by naming novel objects, or by pointing with a finger or mouse click.

VIMA [128] places a significant emphasis on multimodal prompts and the generalization capabilities of models. By incorporating multimodal prompts, more specific and intricate tasks can be formulated compared to pure text prompts. VIMA introduces four main types of tasks: object manipulation, visual goal reaching, novel concept grounding, one-shot video imitation, visual constraint satisfaction, and visual reasoning. These tasks are often challenging or even infeasible to express using only language prompts. VIMA-Bench has been developed to evaluate across four generalizability levels: placement, novel combinatorial, novel object, and novel task.

MOO [94] extends RT-1 to handle multimodal prompts. Leveraging the backbone of RT-1, it incorporates OWL-ViT to encode images within the prompt. By expanding the RT-1 dataset with new objects and additional prompt images, MOO enhances the generalization capabilities of RT-1. This extension also facilitates new methods of specifying target objects, such as pointing with a finger and clicking on graphical user interfaces.

4) Control Policies with 3D Vision: In our 3D world, it is reasonable to assume that 3D vision provides richer information than 2D images.

Point clouds are a common representation for 3D visual inputs due to their straightforward derivation from RGB-D inputs, as demonstrated by both DP3 [106] and 3D Diffuser Actor [109]. Voxels have likewise been extensively studied. VER [129] proposes voxelizing multi-view images into 3D cells in a coarse-to-fine manner, enhancing performance in vision-language navigation tasks. PerAct [88] facilitates efficient task learning with only a few demonstrations by leveraging voxel representations in both the observation and action spaces. The input comprises voxel maps reconstructed from multi-view RGB-D images, while the output corresponds to the best

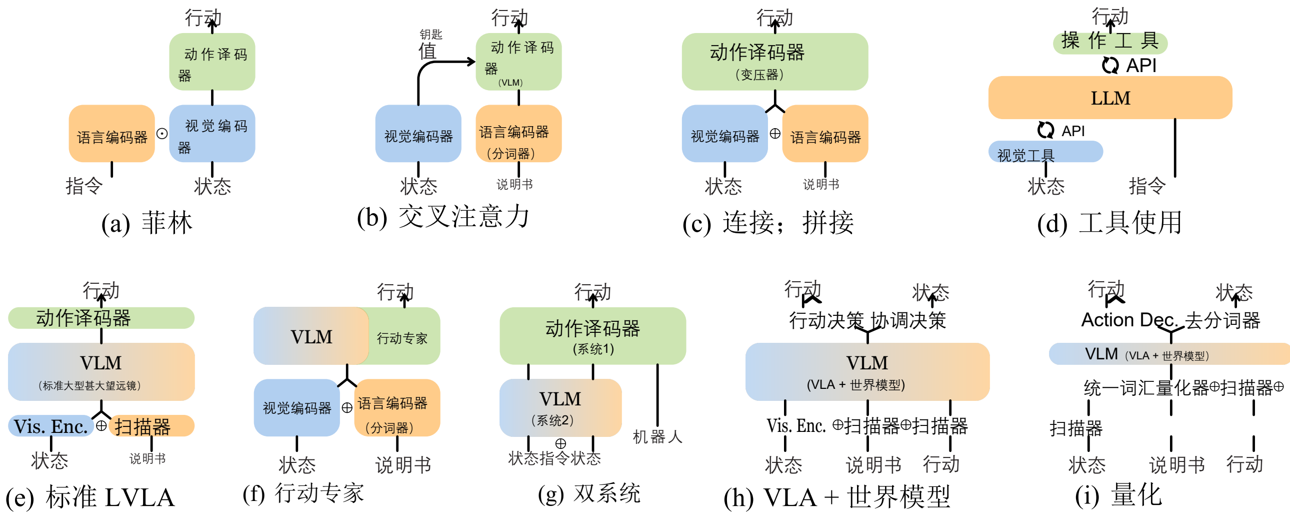


图5：VLA模型的典型架构。部分架构对应表III中的“架构”列。$\odot$：哈达玛积。$\oplus$：连接操作。

然而，RT-1并未将视频作为任务指令。此外，RT-1在BCZ 中用Transformer解码器替换了 MLP 动作解码器，生成离散化动作。这一改进使RT-1能够关注历史图像，相较于BC-Z显著提升了性能。

Q-Transformer[96]通过引入自回归Q函数对RT-1模型进行了扩展。与通过模仿学习获取专家轨迹的RT-1不同，Q-Transformer采用了Q学习方法。在Q学习的时差误差目标基础上，该模型还引入了保守正则化机制，以确保最大值动作始终处于分布范围内。这种设计使得Q-Transformer不仅能利用成功示范轨迹进行学习，还能从失败轨迹中汲取经验进行优化。

RT-Trajectory[97]采用轨迹草图作为策略条件，而非依赖语言条件或目标条件。这些轨迹草图由一系列曲线组成，用于描绘机器人末端执行器应遵循的预期运动路径。它们可通过图形用户界面手动指定、从人类演示视频中提取，或由基础模型自动生成。RT-Trajectory的策略基于RT-1的核心架构构建，并经过训练以精确控制机械臂沿轨迹草图运动。这种设计有助于实现对新物体、新任务及新技能的泛化能力，因为来自不同任务的轨迹具有可迁移性。

ACT[98]通过动作分块技术构建条件 VAE 策略，要求策略预测动作序列而非单一动作。在推理阶段，采用时序集成方法对动作序列进行平均处理。RoboAgent[99]通过MT-ACT对ACT进行扩展，证明动作分块技术能提升时间一致性，并引入基于图像修复的语义增强方法。

RoboFlamingo[100]通过附加基于 LSTM 的策略头部，将现有的Flamingo VLM 适配为机器人策略。这表明预训练的视觉语言模型（VLM）可有效迁移至语言条件下的机器人操作任务。

最近在大语言模型（LLMs）中的一个趋势是通过生成调用API工具的代码来赋予其工具使用能力[127]。Instruct2Act[103]遵循这一范式，通过整合视觉与动作工具，使LLMs能够执行机器人任务。

*3) 多模态指令的控制策略：*多模态指令能够通过演示、命名新物体、手指指向或鼠标点击等方式，为任务描述提供创新方法。

VIMA[128]特别强调多模态提示和模型的泛化能力。与纯文本提示相比，通过引入多模态提示可以构建更为具体和复杂的任务。 VIMA 定义了四大主要任务类型：物体操作、视觉目标达成、新颖概念定位、单次视频模仿、视觉约束满足以及视觉推理。这些任务通常仅使用语言提示难以或无法实现。 VIMA -Bench旨在评估四个泛化能力层级：位置感知、新颖组合任务、新颖物体识别和新颖任务执行。

MOO[94]对RT-1模型进行了扩展，使其能够处理多模态提示。该模型基于RT-1的主干架构，通过整合OWL-ViT技术实现提示内容中的图像编码。通过在RT-1数据集上新增物体样本和提示图像，MOO显著提升了模型的泛化能力。这一扩展还支持多种目标对象指定方式，例如通过手指指向和点击图形用户界面等交互操作。

*4) 基于三维视觉的控制策略：*在三维世界中，可以合理假设三维视觉比二维图像能提供更丰富的信息。

点云因其可直接从RGB-D输入中推导得出，成为三维视觉输入的常见表示形式，这一点在DP3[106]和3D Diffuser Actor[109]研究中均得到验证。体素同样被广泛研究：VER[129]提出将多视角图像以粗到细的方式体素化为三维单元，显著提升了视觉语言导航任务的性能；PerAct[88]通过在观察空间和动作空间中利用体素表征，仅需少量示范即可实现高效任务学习。其输入由多视角RGB-D图像重建的体素图构成，输出则对应最佳

voxel for guiding the gripper's movement. RoboUniView [101] improves performance by injecting 3D information from multi-perspective images through a novel UVFormer vision encoder, which is pretrained on a 3D occupancy task.

In contrast, Act3D [89] introduces a continuous resolution 3D feature field, with adaptive resolutions based on the current task, addressing the computational cost of voxelization. RVT, RVT-2 [90], [91] propose re-rendering images from virtual views of the scene point cloud and using these images as inputs, rather than directly relying on 3D inputs.

5) Diffusion-based Control Policies: Diffusion-based action generation leverages the success of diffusion models in the field of computer vision.

Diffusion Policy [105] formulates a robot policy as a DDPM. This approach incorporates a variety of techniques, including receding horizon control, visual conditioning, and the time-series diffusion Transformer. The effectiveness of this diffusion-based visuomotor policy is underscored by its proficiency in multimodal action distributions, high-dimensional action spaces, and training stability.

SUDD [107] presents a framework where an LLM guides data generation, and subsequently, the filtered dataset is distilled into a visuo-linguo-motor policy. This framework achieves language-guided data generation by composing an LLM with a suite of primitive robot utilities, such as grasp samplers and motion planners. It then extends Diffusion Policy by incorporating language-based conditioning for multi-task learning and facilitates the distillation of the filtered dataset.

Octo [108] introduces a Transformer-based diffusion policy characterized by a modular open-framework design, allowing for flexible connections from different task definition encoders, observation encoders, and action decoders to the Octo Transformer. Being among the first to utilize the OXE dataset [113], Octo demonstrates positive transfer and generalizability across diverse robots and tasks.

MDT [110] adapts the newly introduced DiT model from computer vision to the action prediction head. DiT, originally proposed as a Transformer-based diffusion model, replaces the classical U-Net architecture for video generation. Coupled with two auxiliary objectives—masked generative foresight and contrastive latent alignment—MDT demonstrates better performance than the U-Net-based diffusion model, SUDD.

RDT-1B [111] is a diffusion-based foundation model for bimanual manipulation, also built on DiT. It addresses data scarcity by introducing a unified action format across various robots, enabling pretraining on heterogeneous multi-robot datasets with over 6K trajectories. As a result, RDT scales up to 1.2B parameters and demonstrates zero-shot generalization.

6) Diffusion-based Control Policies with 3D Vision: Several works have proposed combining 3D Vision with diffusion-based policies. DP3 [106] introduces 3D inputs to a diffusion policy, resulting in improved performance. Similarly, 3D Diffuser Actor [109] shares the core idea of DP3 but differs in model architecture by combining Act3D with Diffusion Policy. 3D-MoE [130] explores an efficient mixture-of-experts architecture with DiT-based action diffusion using a rectified flow scheduler.

7) Control Policies for Motion Planning: Motion planning involves decomposing movement tasks into discrete waypoints while satisfying constraints such as obstacle avoidance and kinematic limits.

Language costs [85] presents a novel approach to robot correction using natural language for human-in-the-loop robot control systems. This method leverages predicted cost maps generated from human instructions, which are then utilized by the motion planner to compute the optimal action. This framework enables users to correct goals, specify preferences, or recover from errors through intuitive language commands.

VoxPoser [104] employs LLM and VLM to create two 3D voxel maps that represent affordance and constraint. It leverages the programming capability of LLMs and the perception capability of VLMs. LLM translates language instructions into executable code, invoking VLM to obtain object coordinates. Based on the composed affordance and constraint maps, VoxPoser employs model predictive control to generate a feasible trajectory for the robot arm's end-effector. Notably, VoxPoser does not require any training, as it directly connects LLM and VLM for motion planning.

RoboTAP [131] breaks down demonstrations into stages marked by the opening and closing of the gripper. In each stage, RoboTAP uses the TAPIR algorithm to detect active points that track the relevant object from the source to the target pose. The path can then be used by visual servoing to control the robot. A motion plan is created by stringing together these stages, enabling few-shot visual imitation.

8) Control Policies with Point-based Action: Recent research has explored leveraging the capabilities of VLMs to select or predict point-based actions—a cost-effective alternative to building full VLAs.

PIVOT [132] casts robotics tasks as visual question answering, leveraging VLMs to select the best robot action from a set of visual proposals. Visual proposals are annotated in the form of keypoints on images. The VLM is iteratively prompted to refine them until the best option is identified.

RoboPoint [92] finetunes VLM using the task of spatial affordance prediction, which is to point at where to act on the image. These affordance points on 2D images are subsequently projected into 3D space using depth maps, forming the predicted robot action.

ReKep [35] is a constraint function that maps 3D keypoints in the scene to a numerical cost. Robotics manipulation tasks can be represented as a sequence of ReKep constraints, which are produced with large vision models and VLMs. Consequently, robot actions can be obtained by solving the constrained optimization problem.

9) Large VLA: Large VLA is equivalent to the original VLA definition proposed by RT-2 [1], as illustrated in Figure 2a. This terminology is analogous to the distinction between LLMs and general language models, or between large VLMs and general VLMs.

RT-2 [1] endeavors to harness the capabilities of large multimodal models in robotics tasks, drawing inspiration from models like PaLI-X and PaLM-E (see the architecture in Fig. 5e). The approach introduces co-fine-tuning, aiming to fit the model to both Internet-scale visual question answering

用于引导夹具运动的体素。RoboUniView[101]通过采用基于三维空间占用任务预训练的新型UVFormer视觉编码器，将多视角图像中的三维信息注入系统，从而提升性能。

相比之下，Act3D[89]引入了连续分辨率的3D特征场，其分辨率会根据当前任务动态调整，从而有效降低了体素化的计算成本。 RVT 、 RVT -2[90][91] 建议从场景点云的虚拟视图中重新渲染图像，并将这些图像作为输入，而非直接依赖三维输入数据。

*5) 基于扩散的控制策略：*基于扩散的动作生成技术借鉴了计算机视觉领域中扩散模型的成功经验。

扩散策略[105]将机器人策略表述为一个 DDPM 。该方法融合了多种技术，包括后视地平线控制、视觉条件化以及时间序列扩散Transformer。这种基于扩散的视觉运动策略的有效性体现在其对多模态动作分布、高维动作空间以及训练稳定性的出色表现。

SUDD[107]提出了一种框架，其中大型语言模型（LLM）指导数据生成过程，随后将过滤后的数据集提炼为视觉-语言-运动策略。该框架通过将LLM与抓取采样器、运动规划器等基础机器人工具组合，实现了语言引导的数据生成。通过引入基于语言的条件机制来支持多任务学习，并优化扩散策略，从而有效促进了过滤数据集的提炼过程。

Octo[108]提出了一种基于Transformer的扩散策略，其采用模块化开放式框架设计，能够灵活连接不同的任务定义编码器、观察编码器和动作解码器至Octo Transformer。作为首批使用 OXE 数据集[113]的研究之一，Octo在多种机器人和任务中展现出良好的迁移性能与泛化能力。

MDT[110]将计算机视觉领域新提出的DiT模型适配至动作预测模块。DiT最初作为基于Transformer的扩散模型被提出，可替代传统U-Net架构用于视频生成。通过结合掩码生成预测和对比式潜在对齐两个辅助目标，MDT展现出优于基于U-Net的扩散模型SUDD的性能表现。

RDT -1B[111]是基于扩散机制的双臂操作基础模型，同样基于DiT架构构建。该模型通过引入跨机器人统一动作格式来解决数据稀缺问题，支持在包含超过6000条轨迹的异构多机器人数据集上进行预训练。因此， RDT 可扩展至12亿参数规模，并展现出零样本泛化能力。

*6) 基于扩散机制的三维视觉控制策略：*多项研究提出将三维视觉技术与扩散控制策略相结合。DP3[106]通过向扩散策略输入三维数据，显著提升了模型性能。类似地，三维扩散器Actor[109]与DP3共享核心思想，但通过整合Act3D与扩散策略的模型架构实现了创新。3D-MoE[130]则采用基于DiT的动作扩散机制，结合修正流调度器构建高效的专家混合架构。

*7) 运动规划中的控制策略：*运动规划的核心在于将运动任务分解为离散航点，同时满足避障和运动学限制等约束条件。

语言成本法[85]为基于自然语言的人机协同机器人控制系统提供了一种创新的机器人校正方案。该方法利用根据人类指令生成的预测成本图，由运动规划器据此计算出最优动作方案。这一框架使用户能够通过直观的语言指令来修正目标、指定偏好或纠正错误。VoxPoser[104]则运用大语言模型（LLM）和视觉语言模型（VLM）生成两个三维体素图，分别表示操作可能性与约束条件。该系统充分发挥了LLM的编程能力与VLM的感知能力：LLM将语言指令转化为可执行代码，并调用 VLM 获取物体坐标。基于组合的可用性与约束映射，VoxPoser采用模型预测控制技术为机械臂末端执行器生成可行运动轨迹。值得注意的是，VoxPoser无需任何训练过程，因其直接将大语言模型（LLM）与 VLM 相结合用于运动规划。

RoboTAP[131]将演示过程划分为多个阶段，每个阶段以夹爪的开启和关闭为标志。在每个阶段中，RoboTAP采用tapir算法检测追踪相关物体从源位姿到目标位姿的活动轨迹点。随后可通过视觉伺服控制技术利用该路径驱动机器人运动。通过将这些阶段串联起来，可生成运动规划，从而实现小样本视觉模仿。

*8) 基于点动作的控制策略：*近期研究探索了利用虚拟实验室模型（VLM）的能力来选择或预测基于点的动作——这是构建完整虚拟实验室架构（VLA）的一种成本效益更高的替代方案。

pivot[132]将机器人任务转化为视觉问答问题，通过利用视觉语言模型（VLMs）从视觉提案集中筛选最佳机器人动作。视觉提案以图像关键点形式标注，系统通过迭代提示 VLM 优化这些关键点，直至确定最优方案。

RoboPoint[92]通过空间可用性预测任务对 VLM 进行微调，该任务旨在指示图像上应执行操作的位置。这些二维图像上的可用性点随后通过深度图投影到三维空间中，从而生成预测的机器人动作。

ReKep[35]是一种约束函数，用于将场景中的三维关键点映射为数值代价。机器人操作任务可表示为一系列ReKep约束条件，这些约束条件由大型视觉模型和视觉语言模型（VLM）生成。因此，通过求解约束优化问题即可获得机器人动作。

*9) 大型VLA：*大型VLA等同于RT-2[1]提出的原始VLA定义，如图2a所示。该术语分类类似于大语言模型（LLMs）与通用语言模型之间的区别，或大型VLM与通用VLM之间的区分。

RT-2[1]致力于在机器人任务中充分利用大型多模态模型的能力，其设计灵感来源于PaLI-X和PaLM-E等模型（具体架构见图5e）。该方法引入了协同微调机制，旨在使模型同时适应互联网规模的视觉问答任务。

(VQA) data and robot data. This training scheme enhances generalizability and brings about emergent capabilities.

RT-H [112] introduces an action hierarchy that includes an intermediate prediction layer of language motions, situated between language instructions and low-level actions (translation and rotation). This additional layer facilitates improved data sharing across different tasks. For example, both the language instructions "pick" and "pour" may involve the language motion "move the arm up". Moreover, this action hierarchy enables users to specify corrections to recover from failures, which the model can then learn from.

RT-X [113] builds upon the previous RT-1 and RT-2 models. These models are retrained using the newly introduced open-source large dataset, named Open X-Embodiment (OXE), which is orders of magnitude larger than previous datasets. The resulting models, RT-1-X and RT-2-X, both outperform their original versions.

OpenVLA [34] was later developed as an open-source counterpart to RT-2-X. They additionally explored efficient fine-tuning methods, including LoRA and model quantization. OpenVLA-OFT [114] applies an Optimized Fine-Tuning (OFT) recipe for improved efficiency and performance. TraceVLA [115] finetunes OpenVLA to enable visual trace prompting, enhancing spatial-temporal awareness.

π_0 [116] proposes a flow-matching architecture for transforming VLMs into VLAs. By incorporating an additional action expert based on the mixture-of-experts framework (see the architecture in Fig. 5f), it effectively inherits the Internet-scale knowledge in the VLM while extending its capabilities to address robotic tasks.

RoboMamba [117] replaces the costly Transformer with a Mamba state space model that has linear inference complexity, achieving efficient robotic reasoning and action capabilities.

SpatialVLA [118] introduces Ego3D Position Encoding for injecting 3D information and represents actions with Adaptive Action Grids for enhanced generalizability and robustness.

LAPA [123] devises the first unsupervised pretraining method for VLAs based on latent actions [67]. This approach employs a three-stage process to learn from internet-scale unlabeled videos. First, a VQ-VAE framework is pretrained to extract quantized latent actions between image frames. Next, a VLA model is pretrained to predict these latent actions. Finally, the model is finetuned using only a small robotic dataset to map the latent actions to actual robot actions.

The integration of large VLAs with diffusion has also been gaining popularity. TinyVLA [119] leverages the Diffusion Policy, while CogACT [120] utilizes a DiT action diffusion module. DexVLA [121] proposes embodied curriculum learning to progressively train a diffusion action expert, incorporating sub-step reasoning for decomposing long-horizon tasks. HybridVLA [122] integrates diffusion with the autoregression paradigm to fully leverage VLMs' reasoning capabilities.

GR00T N1 [133] introduces a dual-system architecture to build a robot foundation model for humanoid robots (see the architecture in Fig. 5g). Its VLM-based System 2 processes image observations and language instructions at 10 Hz, while System 1 is a diffusion module that generates closed-loop motor actions in real time (120 Hz).

NORA-1.5 [134] unifies a VLA with a world model through reward-guided post-training. Genie Envisioner [135] is a world foundation platform that integrates a world model and a VLA within a single video-generative framework. See the general architecture in Fig. 5h.

Visual autoregressive modeling (VAR) [136] with quantized visual tokens demonstrates improved performance over diffusion models in image generation. This suggests that the three modalities of VLAs can be unified under the autoregressive paradigm [137]. WorldVLA [124] and UniVLA [125] advance this direction by integrating VLAs with world models (see the architecture in Fig. 5i). It quantizes multimodal data into discrete tokens, forming a shared vocabulary of quantized multimodal tokens. Consequently, all modalities can be modeled autoregressively, enabling not only action and text generation but also image generation, thereby constituting a world model.

Strengths and Limitations.

a) Architectures: Representative VLA architectures are illustrated in Figure 5. FiLM is used in RT-1, and thus its follow-up models inherit this mechanism. While cross-attention may offer superior performance with smaller model sizes, concatenation is simpler to implement and can achieve comparable results with larger models [128]. Quantization unifies multimodal tokens into a shared vocabulary, thus enabling integration with world models. The tool-use paradigm of LLMs can also be applied to robotic tasks.

b) Action types and their training objectives: Most low-level control policies predict actions for the end-effector pose while abstracting away the motion planning module that produces more fine-grained motions. While this abstraction facilitates better generalization to different embodiments, it also imposes limitations on dexterity.

The behavior cloning (BC) objective is used in imitation-learning, with different variants for different action types. The BC objective for continuous action can be written as:

$$\mathcal{L}_{\text{Cont}} = \sum_{t} \text{MSE}\left(a_t, \hat{a}_t\right), \tag{4}$$

where $\text{MSE}(\cdot)$ stands for mean squared error. a_t is the action annotation from expert demonstrations.

Discrete action is achieved by dividing the action value range into a fixed number of bins. Its BC objective is:

$$\mathcal{L}_{\text{Disc}} = \sum_{t} \text{CE}\left(a_t, \hat{a}_t\right), \tag{5}$$

where $\text{CE}(\cdot)$ stands for cross-entropy loss.

The BC objective of $SE(2)$ action is:

$$\mathcal{L}_{SE(2)} = \text{CE}(a_{\text{pick}}, \hat{a}_{\text{pick}}) + \text{CE}(a_{\text{place}}, \hat{a}_{\text{place}}). \tag{6}$$

The DDPM objective in diffusion-based control policies is:

$$\mathcal{L}_{\text{DDPM}} = \text{MSE}\left(\varepsilon^k, \varepsilon_\theta(a_t + \varepsilon^k, k)\right), \tag{7}$$

where ε^k is a random noise for iteration k; ε_θ is the noise prediction network, i.e., the VLA model.

While discrete action has demonstrated superior performance in RT-1 [95], Octo [108] argues that it leads to early grasping issues. $SE(2)$ actions require the model to predict

(VQA)数据与机器人数据。该训练方案可提升泛化能力并催生涌现能力。

RT-H[112]设计了一个动作层级结构，其中包含一个语言动作预测中间层，该层位于语言指令与低级动作（如平移和旋转）之间。这一额外层级有助于提升不同任务之间的数据共享效率。例如，语言指令“pick”和“pour”都可能涉及语言动作“将手臂向上移动”。此外，该动作层级结构允许用户指定故障后的修正方案，模型可据此进行学习。

RT-X[113]是在前代RT-1和RT-2模型基础上开发的。这些模型利用新推出的开源大型数据集——Open X-Embodiment（OXE）进行了重新训练，该数据集规模比以往任何数据集都大数个数量级。最终生成的RT-1-X和RT-2-X模型均优于其原始版本。

OpenVLA[34]后来被开发为RT-2-X的开源版本。研究团队还探索了包括LoRA和模型量化在内的高效微调方法。OpenVLA-OFT[114]采用优化微调（OFT）方案以提升效率和性能表现。TraceVLA[115]通过微调OpenVLA实现可视化轨迹提示功能，显著增强了时空感知能力。

π_0[116]提出了一种流匹配架构，用于将视觉语言模型（VLMs）转换为视觉语言应用（VLAs）。通过基于专家混合框架引入额外的动作专家（参见图5f所示架构），该方案在继承 VLM 中互联网规模知识的同时，扩展了其处理机器人任务的能力。

RoboMamba[117]采用具有线性推理复杂度的Mamba状态空间模型替代了成本高昂的Transformer模型，从而实现了高效的机器人推理与动作能力。

SpatialVLA[118]引入了 Ego3D 位置编码技术用于注入三维信息，并采用自适应动作网格（Adaptive Action Grids）来表示动作，从而提升模型的泛化能力和鲁棒性。

LAPA[123]首次提出基于潜在动作的视觉语言动作（VLA）无监督预训练方法[67]。该方法采用三阶段流程从互联网规模的未标注视频中进行学习：首先通过 VQ-VAE 框架预训练，提取图像帧间的量化潜在动作；接着预训练VLA模型以预测这些潜在动作；最后仅使用小型机器人数据集进行微调，将潜在动作映射为实际机器人动作。

将大型视觉语言模型（VLA）与扩散机制相结合的研究方案正日益受到关注。TinyVLA[119]采用了扩散策略，CogACT[120]则运用了扩散式动作扩散模块。DexVLA[121]提出具身化课程学习方法，通过逐步训练扩散动作专家模型，并结合子步骤推理机制来分解长时域任务。HybridVLA[122]将扩散机制与自回归范式相结合，充分挖掘了视觉语言模型的推理能力。

GR00T N1[133]采用双系统架构构建人形机器人基础模型（架构示意图见图5g）。其基于 VLM 的系统2以10Hz频率处理图像观测与语言指令，而系统1作为扩散模块可实时生成闭环运动指令（120Hz）。

NORA-1.5[134]通过奖励引导的后训练机制，将VLA与世界模型统一起来。Genie Envisioner[135]是一个世界基础平台，它在一个统一的视频生成框架内整合了世界模型和VLA。具体整体架构详见图5h。

基于量化视觉标记的视觉自回归建模（VAR）[136]在图像生成任务中展现出优于扩散模型的性能表现。这表明视觉语言模型（VLAs）的三种模态可通过自回归范式实现统一[137]。WorldVLA[124]与UniVLA[125]通过将VLAs与世界模型相结合（参见图5i所示架构）推动了这一方向。该方法将多模态数据量化为离散标记，形成共享的量化多模态标记词汇表。由此所有模态均可进行自回归建模，不仅支持动作与文本生成，还能实现图像生成，从而构成世界模型。

优势与局限性

*a) 架构设计：*代表性VLA架构如图5所示。RT-1采用了FiLM模型，因此后续模型继承了该机制。虽然交叉注意力机制可能在模型规模较小的情况下提供更优性能，但拼接结构实现更简单，且在大型模型中也能取得相近效果[128]。量化技术将多模态标记统一为共享词汇表，从而实现与世界模型的整合。大语言模型的工具使用范式同样适用于机器人任务场景。

*b) 行动类型及其训练目标：*大多数低级控制策略在预测末端执行器姿态时，会抽象化处理生成更精细运动的运动规划模块。这种抽象虽然有助于更好地泛化到不同应用场景，但也限制了操作的灵活性。

行为克隆（BC）目标被应用于模仿学习中，针对不同动作类型采用多种变体。连续动作的BC目标可表述为：

$$\mathcal{L}_{\text{Cont}} = \sum_t \text{MSE}\left(a_t, \hat{a}_t\right), \tag{4}$$

其中 MSE（·）表示均方误差。a_t为专家演示中的动作标注。

离散动作是通过将动作值范围划分为固定数量的区间来实现的。其边界条件（BC）目标函数为：

$$\mathcal{L}_{\text{Disc}} = \sum_t \text{CE}\left(a_t, \hat{a}_t\right), \tag{5}$$

其中 CE（·）表示交叉熵损失。$SE(2)$ 行

动的 BC 目标为：

$$L_{SE(2)} = \text{CE}(a_{pick}, \hat{a}_{pick}) + \text{CE}(a_{place}, \hat{a}_{place}). \tag{6}$$

基于扩散控制策略中的 DDPM 目标为：

$$\mathcal{L}_{\text{DDPM}} = \text{MSE}(\varepsilon^k, \ \varepsilon_\theta(a_t + \varepsilon^k, \ k)), \tag{7}$$

其中ε^k是第k次迭代的随机噪声；ε_θ为噪声预测网络，即VLA模型。

虽然离散动作在RT-1[95]中表现出更优的性能，但Octo[108]指出这会导致早期抓取问题。$SE(2)$动作要求模型进行预测

Table IV: High-level task planners.

Model	Vision Model	Language Model / VLM	Low-level Control Policy	Architecture	Require Training	Environments, Embodiments, Tasks, and Skills
PaLM-E [10]	ViT, OSRT	PaLM	Interactive Language, RT-1	Concat	✓	**Sim**: TAMP; **Sim&Real-Mani**: Language-Table; **Real**: SayCan setup
EmbodiedGPT [138]	EVA ViT-G/14	LLaMA-7B	MLP	Xattn	✓	**Sim-Mani**: Franka Kitchen, Meta-World
LEO [139]	ConvNeXt, PointNet++	Vicuna-7B	MLP	Concat	✓	**Sim-Mani**: CLIPort task subset; **Sim-Navi**: Habitat-web
3D-LLM [41]	CLIP, 3D-CLR, ConceptFusion	Flamingo, BLIP-2	DD-PPO	Concat	✓	**Sim-Navi**: Habitat
ShapeLLM [140]	ReCon++	LLaMA	–	Concat	✓	3D MM-Vet Benchmark
SayCan [9]	–	PaLM, FLAN	BC-Z, MT-Opt	–	✗	**Real-Navi&Mani** (EDR): office kitchen
Translated ⟨LM⟩ [141]	–	GPT-3, Codex	–	–	✗	**Sim**: VirtualHome
$(SL)^3$ [142]	–	T5-small	seq2seq	–	✓	**Sim-Navi**: ALFRED
Inner Monologue [8]	MDETR, CLIP	InstructGPT, PaLM	CLIPort, heuristics, SayCan policies	Language	✗	**Sim&Real-Mani**: Tabletop Rearrangement; **Real-Navi&Mani**: Kitchen Mobile Mani.
LLM-Planner [143]	HLSM object detector	GPT-3	HLSM	Language	✓	**Sim-Navi**: ALFRED
LID [144]	CNN	GPT-2	–	Lang., Embed	✓	**Sim**: VirtualHome
SMs [145]	CLIP, ViLD	GPT-3, RoBERTa	CLIPort	Lang., Code	✗	**Sim-Mani**: pick, place
ProgPrompt [146]	ViLD	GPT-3	API	Code	✗	**Sim**: VirtualHome
ChatGPT for Robotics [147]	YOLOv8	ChatGPT	API	Code	✗	**Real-Navi**: drone flight; **Sim-Navi**: AirSim, Habitat
CaP [148]	ViLD, MDETR	GPT-3, Codex	API	Code	✗	**Sim-Mani** & **Sim-Navi**: pick, place, etc.
DEPS [149]	CLIP	ChatGPT	MC-Controller, Steve-1	Code	✗	**Sim**: Minecraft
ConceptGraphs [150]	SAM, CLIP, LLaVA	GPT-4	API	Code (JSON)	✗	**Sim**: AI2-THOR; **Real** (Spot Arm): pick-place

only the pick and place end-effector poses, which is sufficient for many tabletop manipulation tasks. However, more complex tasks—such as "pouring water into a cup"—may require additional DoFs, thereby necessitating $SE(3)$ actions [35]. Although point-based actions can be coarse-grained, they are more easily obtained from VLMs in a zero-shot manner.

c) RT series: RT-1 [95] inspired a series of "Robotic Transformer" models. The Transformer backbone surpasses previous RNN backbones by harnessing the higher capacity of Transformers to absorb larger robot datasets. Preceding RT-1 was BC-Z, which solely utilized MLP layers for action prediction. Subsequent to RT-1, several works emerged, each introducing new capabilities. MOO adapted RT-1 to accommodate multimodal prompts. RT-Trajectory enabled RT-1 to process trajectory sketches as prompts. Q-Transformer utilized Q-learning to train RT-1. RT-2, based on ViT and LLM, introduced a completely different architecture from RT-1. RT-X retrained RT-1 and RT-2 with a significantly larger dataset, resulting in improved performance. Based on RT-2, RT-H [112] introduces action hierarchies for better data sharing.

d) LVLA vs generalized VLA: While LVLAs can greatly enhance instruction-following abilities because they can better parse user intentions, concerns arise regarding their training cost and deployment speed. Slow inference speed, in particular, can significantly impact performance in dynamic environments, as changes in the environment may occur during inference. Therefore, several methods have been proposed to improve efficiency. TinyVLA [119] focused on inference speed and data efficiency through a smaller VLM and diffusion head for robot action. DeeR-VLA [102] proposes only partially activating the model with dynamic inference with early-exit.

e) Scaling Law: Similar to LLMs, scaling laws have also been observed in robotics [151], [152], revealing the importance of model size, dataset size, and the diversity of environments and objects. Further research in this area can guide the development of VLAs with robust in-the-wild generalization capabilities.

IV. Task Planners

A high-level task planner π_ϕ aims to divide a complex task ℓ into a sequence of subtasks p_i (i.e., task plan), each serving as an instruction to low-level control policies π_θ:

$$[p_1, p_2, \ldots, p_N] \sim \pi_\phi(\ell, s_t). \tag{8}$$

This process is sometimes referred to as task or subgoal decomposition and is closely related to task and motion planning (TAMP) and embodied decision making. When equipped with task planners, VLAs can complete more complex, long-horizon tasks, as illustrated in Figure 4. Details are summarized in Table IV. Ideally, task plans should also incorporate optimal scheduling for these subtasks.

A. Monolithic Task Planners

A single LLM or multimodal LLM (MLLM) can typically generate task plans by employing a tailored framework or through finetuning on embodied datasets. We refer to these as *monolithic* models.

1) End-to-end Task Planners: Similar to LVLAs, task planners can be implemented as end-to-end multimodal LLMs, leveraging their Internet-scale knowledge for task planning.

PaLM-E [10] integrates ViT and PaLM to create a large embodied multimodal language model capable of performing high-level embodied reasoning tasks. Based on perceived images and high-level language instructions, PaLM-E generates a text plan that serves as instructions for low-level robotic policies. In a mobile manipulation environment, they map the generated plan to executable low-level instructions with SayCan [9]. As the low-level policy executes actions, PaLM-E can also replan based on changes in the environment. With PaLM as its backbone, PaLM-E can handle both normal visual question answering (VQA) tasks, along with the additional embodied VQA tasks.

EmbodiedGPT [138] introduces the embodied-former, which outputs task-relevant instance-level features. This is achieved by incorporating information from vision encoder embeddings and embodied planning information provided by LLM. The instance feature serves to inform the low-level policy about the immediate next action to take.

表IV：高级任务规划师

模型	视觉模型	语言模型 / VLM	低级控制策略	架构	培训要求	环境、实施例、任务和技能
PaLM-E [10]	ViT, OSRT	帕尔姆	交互式语言 RT-1	粘合		**Sim**: TAMP；**Sim&Real-Mani**: 语言表格；**真实**：SayCan设置
EmbodiedGPT [138]	EVA ViT-G/14	LLaMA-7B	MLP	Xattn	✓	**Sim-Mani**：弗兰卡·基钦，Meta-World
LEO [139]	ConvNeXt、PointNet++	Vicuna-7B	MLP	粘合		**Sim-Mani**：CLIPort任务子集；**Sim-Navi**：栖息地网
3D-LLM [41]	CLIP、3D-CLR、ConceptFusion	火烈鸟，BLIP-2	DD-PPO	粘合	✓	**Sim-Navi**：生境
ShapeLLM [140]	ReCon++	LLaMA	–	粘合	✓	三维MM-Vet基准测试
SayCan [9]	–	PaLM、FLAN	BC-Z，MT-Opt	–	✗	**Real-Navi&Mani**（EDR）：办公室厨房
翻译〈LM〉[141]	–	GPT-3，Codex	–	–	✗	**Sim**: VirtualHome
$(SL)^3$ [142]	–	T5-小	seq2seq	–	✓	**Sim-Navi**：alfred
内心独白[8]	MDETR，片段	InstructGPT、PaLM	CLIPort、启发式算法、SayCan策略	语言	‹	**Sim&Real-Mani**：桌面重组；**Real-Navi&Mani**：厨房移动Mani。
LLM-Planner [143]	HLSM 目标探测器	GPT-3	HLSM	语言	✓	**Sim-Navi**：alfred
LID [144]	中心体	GPT-2	–	长，嵌入	✓	**Sim**: VirtualHome
SMs [145]	CLIP，ViLD	GPT-3、RoBERTa	CLIPort	Lang.，代码	✗	**Sim-Mani**：采摘、放置
ProgPrompt [146]	病毒性心力衰竭	GPT-3	API	编码	✗	**Sim**: VirtualHome
ChatGPT在机器人领域的应用 [147]	YOLOv8	ChatGPT	API	编码	✗	**Real-Navi**：无人机飞行；**Sim-Navi**：AirSim、Habitat
CaP [148]	ViLD, MDETR	GPT-3，Codex	API	编码	✗	**Sim-Mani & Sim-Navi**：选择、放置等操作
DEPS [149]	剪辑	ChatGPT	MC控制器，Steve-1	编码	✗	**Sim**：《我的世界》
ConceptGraphs [150]	SAM、CLIP、LLaVA	GPT-4	API	代码（JSON）	✗	**Sim**: AI2-THOR；**Real**（Spot Arm）：拾取-放置

仅需选择放置末端执行器的姿态即可完成多数桌面操作任务。然而对于更复杂的任务（例如“将水倒入杯子”），可能需要增加自由度，这就需要采用$SE(3)$动作[35]。尽管基于点的动作可采用粗粒度处理方式，但通过零样本学习方法从视觉语言模型中获取这些动作更为便捷。

*c) RT系列：*RT-1[95]启发了一系列“机器人Transformer”模型。Transformer主干网络通过利用Transformer更强的容量来处理更大的机器人数据集，超越了以往的RNN 主干网络。RT-1之前是BC-Z，它仅使用 MLP 层进行动作预测；RT-1之后出现了多项改进工作，每项都引入了新功能：MOO对RT-1进行了适配以支持多模态提示；RT-Trajectory 使 RT-1 能够处理轨迹草图作为提示；Q-Transformer采用Q学习方法训练RT-1；基于ViT和LLM的RT-2采用了与RT-1完全不同的架构； RTX 利用更大规模的数据集对RT-1和RT-2进行重新训练，提升了性能；而基于RT-2的RT-H[112]则引入了动作层级结构以优化数据共享。

*d) LVLA 与广义VLA的对比：*虽然低维VLA（LVLAs）能通过更精准解析用户意图显著提升指令执行能力，但其训练成本与部署速度仍存在隐患。特别是在动态环境中，推理速度过慢可能严重影响性能表现——因为推理过程中环境参数可能持续变化。为此学界提出了多种提升效率的方法：TinyVLA[119]通过采用更小的 VLM 和扩散头来优化机器人动作推理速度与数据效率；而DeeR-VLA [102]则采用动态推理策略，通过提前退出机制实现模型部分激活。

*e) 缩放定律：*与大型语言模型（LLMs）类似，机器人技术领域也观察到了缩放定律[151]。[152]这揭示了模型规模、数据集规模以及环境与对象多样性的关键重要性。该领域的进一步研究将有助于开发具备强大实际场景泛化能力的VLAs（虚拟实验室环境）。

IV.任务规划者

高级任务规划器π_ϕ旨在将复杂任务ℓ划分为一系列子任务p_i（即任务计划），每个子任务均作为对低级控制策略π_θ的指令：

$$[p_1, p_2, \ldots, p_N] \sim \pi_\phi(\ell, s_t). \tag{8}$$

该过程有时被称为任务或子目标分解，与任务与运动规划（TAMP）及具身化决策制定密切相关。如图4所示，配备任务规划器的虚拟实验室助理（VLAs）能够完成更复杂、时间跨度更长的任务。具体细节详见表IV。理想情况下，任务计划还应包含对子任务的最优调度方案。

A. 单片式任务规划器

单个LLM或多模态LLM（MLLM）通常可通过采用定制框架或基于具身数据集的微调来生成任务计划。我们将这类模型称为*单体式*模型。

*1) 端到端任务规划器：*与LVLAs类似，任务规划器可采用端到端多模态大语言模型（LLM）实现，通过利用其互联网规模的知识库进行任务规划。

PaLM-E[10]通过整合视觉表征网络（ViT）与PaLM模型，构建了一个具备高级具身推理能力的大型多模态语言模型。该模型基于感知图像和高级语言指令生成文本计划，为低级机器人策略提供执行指令。在移动操作环境中，系统通过SayCan[9]将生成的计划转化为可执行的低级指令。当低级策略执行动作时，PaLME还能根据环境变化进行动态重构。依托PaLM作为核心架构，PaLM-E不仅能处理常规视觉问答（VQA）任务，还可承担额外的具身VQA 任务。

EmbodiedGPT[138]引入了具身形式生成器（embodied-former），该模块可输出与任务相关的实例级特征。其工作原理是通过整合视觉编码器嵌入信息以及大语言模型（LLM）提供的具身规划信息来实现。这些实例特征能够为低级策略提供关于当前应采取的即时下一步行动的指导信息。

2) End-to-end Task Planners with 3D Vision: Some task planners also explore the use of 3D vision. Because the majority of current MLLMs deal with images as visual inputs, they require architectural changes to incorporate 3D visual inputs, and therefore, they are usually end-to-end models.

LEO [139] uses a two-stage training strategy to integrate a point cloud encoder with an LLM: the first stage focuses on 3D vision-language alignment, followed by the second stage, which involves 3D vision-language-action instruction tuning. LEO performs well not only in 3D question-answering tasks but also in manipulation, navigation, and task planning.

3D-LLM [41] injects 3D information into LLMs and empowers them to perform 3D tasks, such as 3D-assisted dialog and navigation, using features from point cloud, gradSLAM, and neural voxel field. MultiPLY [153] is an object-centric embodied LLM that incorporates even more modalities, including audio, tactile, and thermal.

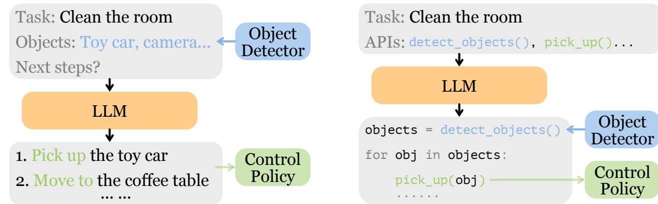
ShapeLLM [140] is built on the novel 3D vision encoder ReCon++, which distills knowledge from multi-view image and text teachers, along with a point cloud MAE. By integrating ReCon++ with LLaMA, ShapeLLM improves embodied interaction performance on their newly proposed 3D benchmark, 3D MM-Vet.

3) Grounded Task Planners: Grounded task planning involves generating high-level actions while considering whether they can be executed by low-level control policies.

SayCan [9] is a framework designed to integrate high-level LLM planners with low-level control policies. In this framework, the LLM planner accepts users' high-level instruction and "says" what the most probable next low-level skill is, a concept referred to as *task-grounding*. The low-level policy provides the value function as the affordance function, determining the possibility that the policy "can" complete the skill, known as *world-grounding*. By considering both LLM's plan and affordance, the framework selects the optimal skill for the current state.

Translated ⟨LM⟩ [141] employs a two-step process to translate high-level instructions into executable actions. Initially, a pretrained causal LLM is utilized for *plan generation*, breaking down the high-level instruction into the next action expressed in free-form language phrases. Then, as these phrases may not directly map to VirtualHome actions, a pretrained masked LLM is then employed for *action translation*. This step involves calculating the similarity between the generated action phrases and the VirtualHome action. The translated action is appended to the plan, and the updated plan is read by the LLM to generate the next action phrase. The two-step process is repeated until a complete plan is formed. A "Re-prompting" strategy [154] is further proposed to generate corrective actions when the agent encounters precondition errors.

$(SL)^3$ [142] is a learning algorithm that alternates between three steps: segmentation, labeling, and parameter update. In the segmentation step, high-level subtasks are aligned with low-level actions, subtask descriptions are then inferred in the labeling step, and finally, the network parameters are updated. This approach enables a hierarchical policy to discover reusable skills with sparse natural language annotations.



(a) Language-based (b) Code-based

Figure 6: Different approaches to connect LLM to multi-modal modules in modular task planners.

B. Modular Task Planners

Finetuning end-to-end models on embodied data can be expensive, and some approaches adopt a modular design by assembling off-the-shelf LLMs and VLMs into task planners. These approaches can also be viewed as following the tool-use architecture [127].

1) Language-based Task Planners: Language-based approaches use natural language descriptions as the medium for exchanging multimodal information, as shown in Figure 6a.

Inner Monologue [8] sits between high-level command and low-level policy to enable closed-loop control planning. It employs LLM to generate language instructions for low-level policies and dynamically updates these instructions based on feedback received from them. The feedback encompasses various sources: success feedback, object and scene feedback, and human feedback. As the feedback is communicated to the LLM in textual format, no additional training is required for the LLM. A similar approach is used in ReAct [74].

LLM-Planner [143] introduces a novel approach to constructing a hierarchical policy comprising a high-level planner and a low-level planner. The high-level planner harnesses the capabilities of LLM to generate natural language plans, while the low-level planner translates each subgoal within the plan into primitive actions. While sharing similarities with previous methods in its overall architecture, LLM-Planner distinguishes itself by incorporating a re-planning mechanism, aiding the robot to "get unstuck".

LID [144] introduces a novel data collection procedure termed Active Data Gathering (ADG). A key aspect of ADG is hindsight relabeling, which reassigns labels to unsuccessful trajectories, effectively maximizing the utilization of data irrespective of their success. By converting all environmental inputs into textual descriptions, their language-model-based policy demonstrates enhanced combinatorial generalization.

Socratic Models (SMs) [145] present a unique framework wherein diverse pretrained models are effectively composed without the need for finetuning. The framework is based on the key component, named multimodal-informed prompting, facilitating information exchange among models with varied multimodal capabilities. The idea is to utilize multimodal models to convert non-language inputs into language descriptions, effectively unifying different modalities within the language space. Beyond excelling in conventional multimodal tasks, SMs showcase their versatility in robot perception and planning. In addition to natural language plans, task plans can also be represented in the form of pseudocode.

*2) 基于三维视觉的端到端任务规划器：*部分任务规划器还探索了三维视觉的应用。由于当前大多数多语言大语言模型（MLLM）以图像作为视觉输入，因此需要对其架构进行修改以整合三维视觉输入，故这些模型通常属于端到端架构。

LEO[139]采用两阶段训练策略，将点云编码器与大语言模型（LLM）进行集成：第一阶段专注于三维视觉-语言对齐，第二阶段则涉及三维视觉-语言-动作指令调优。该模型不仅在三维问答任务中表现优异，同时在物体操控、导航及任务规划等场景中也展现出卓越性能。

3D-LLM[41]将三维信息注入大语言模型（LLM），使其能够利用点云、gradSLAM及神经体素场等特征执行三维任务，例如三维辅助对话与导航。MultiPLY[153]是一种以物体为中心的具身化大语言模型，其整合了更多模态信息，包括音频、触觉和热成像数据。

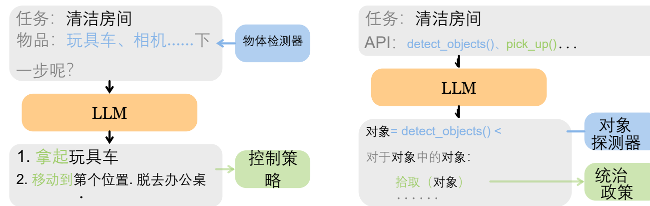
ShapeLLM[140]基于创新的3D视觉编码器ReCon++构建，该编码器能从多视角图像和文本教师模型中提炼知识，并结合点云MAE。通过将ReCon++与LLaMA集成，ShapeLLM在它们新提出的3D基准测试3D MM-Vet上显著提升了具身交互性能。

*3) 基于现实条件的任务规划器：*基于现实条件的任务规划涉及在生成高层级动作的同时，考虑这些动作是否能够通过低层级控制策略执行。

SayCan[9]是一个旨在将高级语言模型规划器与低级控制策略相结合的框架。该框架中，高级语言模型规划器接收用户的高层次指令，并“预测”最可能的下一低级技能动作——这一概念被称为*任务锚定*。低级策略则通过价值函数作为可操作性函数，评估该策略“能够”完成该技能动作的可能性，即*世界锚定*。通过综合考量高级语言模型的规划方案与可操作性特征，该框架可为当前状态选择最优技能动作。

〈LM [141]采用两步流程将高层指令转化为可执行动作。首先利用预训练因果型大语言模型进行计划*生成*，将高层指令分解为自由形式语言短语表示的后续动作。由于这些短语可能无法直接映射到VirtualHome动作，随后采用预训练掩码型大语言模型进行*动作翻译*。该步骤通过计算生成动作短语与VirtualHome动作之间的相似度来实现。将翻译后的动作添加至计划后，LLM会读取更新后的计划以生成新的动作短语。该两步流程持续迭代直至形成完整计划。研究还提出“重提示”策略[154]，用于在智能体遇到前置条件错误时生成修正动作。

$(SL)^3$[142]是一种学习算法，其流程包含三个交替执行的步骤：分割、标注和参数更新。在分割步骤中，高层子任务与底层动作进行对齐；随后在标注步骤中推断出子任务描述；最后更新网络参数。该方法使得分层策略能够在自然语言标注稀疏的情况下发现可复用技能。



(a) 基于语言的 (b) 基于代码的

图6：模块化任务规划器中将大语言模型（LLM）与多模态模块连接的不同方法。

B. 模块化任务规划器

在具身数据上对端到端模型进行微调可能成本高昂，部分方法采用模块化设计，通过将现成的大型语言模型（LLMs）和视觉语言模型（VLMs）组装成任务规划器。这些方法也可视为遵循工具使用架构[127]。

*1) 基于语言的任务规划器：*基于语言的方法使用自然语言描述作为交换多模态信息的媒介，如图6a所示。

内部独白系统[8]位于高层指挥与底层策略之间，旨在实现闭环控制规划。该系统采用大语言模型（LLM）生成底层策略的语言指令，并根据策略反馈动态更新指令内容。反馈来源包括成功反馈、目标与场景反馈以及人工反馈等多种渠道。由于反馈信息以文本形式直接传递给LLM，因此无需额外训练即可运行。类似方法在ReAct系统[74]中也得到了应用。

LLM-Planner[143]提出了一种构建分层策略的新方法，该策略由高层规划器和低层规划器组成。高层规划器利用大语言模型（LLM）的能力生成自然语言计划，而低层规划器则将计划中的每个子目标转化为基础动作。尽管整体架构与先前方法存在相似性，LLM-Planner通过引入重规划机制实现了差异化设计，该机制能帮助机器人“摆脱困境”。

LID[144]提出了一种名为主动数据收集（ADG）的新型数据收集方法。 ADG 的核心在于事后重标注技术，该技术可为未成功轨迹重新分配标签，从而有效提升数据利用率（无论其是否成功）。通过将所有环境输入转化为文本描述，基于语言模型的策略展现出更强的组合泛化能力。

苏格拉底模型（SMs）[145]提出了一种独特的框架，能够无需微调即可有效整合多种预训练模型。该框架的核心组件名为“多模态信息提示”，能够促进具备不同多模态能力的模型之间的信息交流。其核心思想是利用多模态模型将非语言输入转化为语言描述，从而在语言空间内有效整合不同模态。除了在传统多模态任务中表现优异外，SMs 在机器人感知与规划领域也展现出强大的灵活性。除自然语言计划外，任务计划还可采用伪代码形式进行表示。

2) Code-based Task Planners: Code-based task planners leverage the coding ability of LLMs to generate task plans in the form of a program. Object detectors, VLMs, and control policies can be invoked via APIs, as shown in Figure 6b.

ProgPrompt [146] introduces a novel task-planning approach by prompting LLMs with program-like specifications detailing available actions and objects. This enables LLMs to generate high-level plans for household tasks in a few-shot manner. Environmental feedback can be incorporated through assertions within the program.

ChatGPT for Robotics [147] takes advantage of the programming ability of ChatGPT to facilitate "user on the loop" control, a departure from the conventional "engineer in the loop" methodology. The procedure includes several steps: firstly, a list of APIs is defined, such as an object-detection API, a grasp API, a move API; secondly, a prompt is then constructed for ChatGPT, specifying the environment, API functionality, task goal, etc.; thirdly, iteratively prompting ChatGPT to write code with the defined APIs that can execute the task, provided the access to simulation and user feedback for evaluating the code quality and safety; finally, executing the ChatGPT generated code. In this procedure, ChatGPT serves as a high-level task planner, and actions are generated through function calls to corresponding low-level APIs.

Code as policies (CaP) [148] also leverages the code-writing capability of LLMs. It employs GPT-3 or Codex to generate policy code, which, in turn, invokes perception modules and control APIs. CaP exhibits proficiency in spatial-geometric reasoning, generalization to new instructions, and parameterization for low-level control primitives. By leveraging the multimodal capabilities of GPT-4V, COME-robot [155] eliminates the need for perception APIs in CaP. This also opens up possibilities for open-ended reasoning and adaptive planning within a closed-loop framework, enabling capabilities such as failure recovery and free-form instruction following.

DEPS [149] stands for "Describe, Explain, Plan, and Select". This approach employs an LLM to generate plans and explain failures based on feedback descriptions collected from the environment—a process referred to as "self-explanation", aiding in re-planning. Additionally, DEPS introduces a trainable goal selector to choose among parallel candidate sub-goals based on how easily they can be achieved, a crucial aspect often overlooked by other high-level task planners.

ConceptGraphs [150] introduces a method to convert observation sequences into open-vocabulary 3D scene graphs. Objects are extracted from RGB images using 2D segmentation models, and VLMs are employed to caption objects and establish inter-object relations, resulting in the formation of the 3D scene graph. This graph can then be translated into a text description (JSON), offering rich semantic and spatial relationships between entities to LLMs for task planning.

Strengths and Limitations.

Monolithic task planners that utilize grounded task planning focus on generating executable plans. End-to-end models share an architecture similar to most LVLAs and can be finetuned on specialized embodied data to achieve better performance. However, the training costs of such large models can be a concern. In contrast, *modular* task planners are more readily deployable because they leverage off-the-shelf LLMs and VLMs. Language-based task planners offer the advantage of seamless integration of LLMs and VLMs, as they are designed to operate in the natural language space. However, they often require extra steps to align the generated task plans with language instructions that are admissible to low-level control policies. Conversely, while code-based task planners may require manually wrapping VLMs and control policies in APIs and preparing clear documentation in advance, they enable code debugging and provide greater controllability. Nevertheless, their performance can be constrained by the programming capabilities of existing models.

Table V: Embodied Datasets. Per RT-X, skills correspond to verbs, and tasks are different combinations of verbs and objects. ∗ Dataset collected in simulation rather than the real world. Some datasets are continually updated, and we include only the original version. Table adapted from [156], [157].

Dataset	Skills	Tasks	Scenes	Episodes	Collection	Obs.	Instruction	Robots
MIME [158]	12	20	1	8.3K	Human teleop.	RGBD	Demo	Baxter
∗RoboTurk [159]	2	–	1	2.1K	Human teleop.	RGB	–	Sawyer
RoboNet [160]	–	–	10	162K	Script	RGB	Goal image	(7 robots)
MT-Opt [161]	2	12	1	800K	Script	RGB	Lang	(7 robots)
BC-Z [80]	3	100	1	25.9K	Human teleop.	RGB	Lang, demo	EDR
Fractal [95]	12	700+	2	130K	Human teleop.	RGB	Lang	EDR
MOO [94]	5	–	–	59.1K	Human teleop.	RGB	Multimodal	EDR
∗VIMA [128]	17	–	1	650K	Script	RGB	Multimodal	UR5
RoboSet [162]	12	38	11	98.5K	Human, script	RGBD	Lang	Franka
BridgeV2 [163]	13	–	24	60.1K	Human, script	RGBD	Lang	WidowX
RH20T [156]	42	147	7	110K+	Human teleop.	RGBD	Lang	(4 robots)
DROID [157]	86	–	564	76K	Human teleop.	RGBD	Lang	Franka
OXE [113]	527	160K	311	1M+	Aggregate data	RGBD	Lang	(22 robots)

V. Datasets and Benchmarks

A. Real-world Robot Datasets & Benchmarks

Embodied AI faces significant data scarcity issues because real-world robot data is not as readily available as language data. Collecting real-world robot datasets poses multiple challenges. Firstly, it is impeded by the cost and time required to procure robotic equipment, set up environments, and gather expert data through dedicated policies or human teleoperation. Secondly, the diverse types and configurations of robots introduce inconsistencies in sensory data, control modes, gripper types, etc. Lastly, accurately capturing object 6D poses and reproducing setups remains elusive. We summarize recent robot datasets in Table V. In addition, real-world benchmarks are further complicated by the need for human evaluation.

B. Simulators, Simulated Robot Datasets & Benchmarks

Many researchers resort to simulated environments to circumvent real-world obstacles and scale the data collection process. We compare simulators and simulated benchmarks in Table VI. Nevertheless, this strategy presents its own challenges, chief among which is the sim-to-real gap. This discrepancy arises when models trained on simulated data exhibit poor performance during real-world deployment. The causes of this gap are multifaceted, encompassing unrealistic rendering quality, inaccuracies in physics simulations, and domain shifts characterized by object properties and robot motion planners. For instance, simulating non-rigid objects such as deformable objects or liquids presents significant difficulties. Moreover, importing new objects into simulators

*2) 基于代码的任务规划器：*基于代码的任务规划器利用大语言模型（LLM）的编码能力，以程序形式生成任务计划。如图6b所示，对象检测器、虚拟语言模型（VLM）及控制策略均可通过API调用。

ProgPrompt[146]提出了一种创新的任务规划方法，通过向大语言模型（LLM）输入包含可用动作和对象的程序化规范进行提示。该方法使LLM能够以少量样本数据生成家庭任务的高层次规划方案。程序中的断言语句可整合环境反馈信息。

机器人领域的ChatGPT[147]通过利用ChatGPT的编程能力，实现了“用户参与控制”模式，这与传统的“工程师主导控制”方法论形成显著差异。该流程包含三个关键步骤：首先定义一系列API接口，包括目标检测API、抓取API和移动API；其次为ChatGPT构建提示词，明确环境参数、API功能及任务目标等要素；第三阶段通过迭代提示让ChatGPT基于预定义API编写执行任务的代码，并同步接入仿真环境和用户反馈以评估代码质量与安全性；最后执行ChatGPT生成的代码。在此过程中，ChatGPT充当高级任务规划器，通过调用对应底层API接口生成具体操作指令。

代码即策略（CaP）[148]同样充分利用了大语言模型的代码编写能力。该系统采用GPT-3或Codex生成策略代码，进而调用感知模块和控制API接口。CaP在空间几何推理、新指令泛化能力以及低级控制原语参数化方面表现出色。通过整合GPT-4V的多模态能力，COME机器人[155]成功实现了在CaP架构中省去感知API接口的需求。这一创新还为闭环框架下的开放式推理与自适应规划开辟了可能性，使得故障恢复和自由形式指令执行等核心功能得以实现。

DEPS[149]代表“描述、解释、规划与选择”。该方法通过大型语言模型生成计划方案，并基于从环境中收集的反馈描述进行故障分析——这一过程被称为“自我解释”，有助于优化后续规划。此外，DEPS引入了可训练的目标选择器，可根据子目标的可实现性进行并行候选子目标筛选，这是其他高级任务规划器常忽视的关键环节。

ConceptGraphs[150]提出了一种将观测序列转换为开放词汇表三维场景图的方法。该方法首先通过二维分割模型从RGB图像中提取物体，随后采用视觉语言模型（VLM）为物体添加描述性标签并建立物体间关系，最终构建出三维场景图。该图谱可转化为文本描述（JSON），为大语言模型（LLM）提供实体间丰富的语义与空间关系，从而支持任务规划。

优势与局限性

*单一模块化*任务规划器采用基于现实环境的任务规划方法，专注于生成可执行计划。端到端模型采用与大多数语言视觉学习算法（LVLAs）相似的架构，并可通过专用具身化数据进行微调以提升性能。然而，此类大型模型的训练成本可能如表V：具身化数据集所示。在RT-X框架中，技能对应动词，任务则是动词与对象的不同组合。 数据集来源于模拟环境而非真实世界。部分数据集持续更新，本文仅采用原始版本。表格改编自[156][157].

数据集	技能	任务	场景剧集		收集	观察	指令	机器人
MIME [158]	12	20	1	8.3K	人类远视。	RGBD	演示物	巴克斯特
∗RoboTurk [159]	2	—	1	2.1K	人类远视。	RGB	—	索耶
RoboNet [160]	—	—	10	162K	脚本	RGB	目标图像	（7台机器人）
MT-Opt [161]	2	12	1	800K	脚本	RGB	长的	（7台机器人）
BC-Z [80]	3	100	1	25.9K	人类远视。	RGB	朗，演示	EDR
分形[95]	12	700+	2	130K	人类远视。	RGB	长的	EDR
MOO [94]	5	—	—	59.1K	人类远视。	RGB	多峰值	EDR
∗VIMA [128]	17	—	1	650K	脚本	RGB	多峰值	UR5
RoboSet [162]	12	38	11	98.5K	人类，脚本	RGBD	长的	弗兰卡
BridgeV2 [163]	13	—	24	60.1K	人类，脚本	RGBD	长的	WidowX
RH20T [156]	42	147	7	110K+	人类远视。	RGBD	长的	（4台机器人）
Droid[157]	86	—	564	76K	人类远视。	RGBD	长的	弗兰卡
OXE [113]	527	160K	311	1M+	聚合数据	RGBD	长的	（22台机器人）

相比之下，*模块化*任务规划器更具部署便利性，因其基于现成的大型语言模型（LLMs）和视觉语言模型（VL-Ms）。基于语言的任务规划器具有无缝集成LLMs与VLMs的优势，因其设计初衷即在自然语言空间中运行。但这类方案通常需要额外步骤，才能使生成的任务计划与符合底层控制策略的语言指令保持一致。而基于代码的任务规划器虽需手动将VLMs和控制策略封装到API接口中，并预先准备清晰文档，但能实现代码调试功能并提供更强的可控性。不过其性能仍可能受限于现有模型的编程能力。

V. 数据集与基准测试

A. 真实世界机器人数据集与基准测试

实体化人工智能面临显著的数据稀缺性问题，因为现实世界中的机器人数据不像语言数据那样容易获取。收集真实机器人数据集存在多重挑战：首先，获取机器人设备、搭建实验环境以及通过专用策略或人工远程操作收集专家数据所需的成本和时间会形成阻碍；其次，机器人类型与配置的多样性导致传感器数据、控制模式、抓取器类型等方面存在不一致性；最后，准确捕捉物体六维位姿并复现实验场景仍难以实现。表V汇总了近期主流机器人数据集。此外，现实世界基准测试还因需要人工评估而面临更多复杂性。

B. 模拟器、模拟机器人数据集及基准测试

许多研究者采用模拟环境来规避现实世界中的障碍并扩大数据收集规模。我们在表VI中对比了模拟器与模拟基准测试结果。然而，这种策略也存在固有挑战，其中最突出的是模拟与现实之间的差距。当基于模拟数据训练的模型在实际部署中表现不佳时，就会产生这种差异。造成这种差距的原因具有多维性，包括渲染质量不真实、物理模拟存在误差，以及以物体属性和机器人运动规划为特征的领域偏移现象。例如，模拟非刚性物体（如可变形物体或液体）会面临重大技术难题。此外，将新物体导入模拟器时也存在诸多挑战。

Table VI: Simulators and simulated benchmarks. Control: continuous control tasks. D, S, A, N: depth, segmentation, audio, normal. Force: simulated contact force between end-effector and item. PD: pre-defined. Table adapted from [164], [165].

Name	Scenes /Rooms	Objects /Cat	UI	Physics Engine	Task	Observation	Action	Agent	Description	Related
Gibson [166]	572/-	–	–	Pybullet	Navi	RGB, D, N, S	–	–	Navi only	–
iGibson [165], [167], [168]	15/108	152/5	Mouse, VR	Pybullet	Navi, Mani	RGB, D, S, N, Flow, LiDAR	Force	TurtleBot v2, LoCoBot, etc.	VR, Continuous Extended States. Versions: 0.5, 1.0, 2.0	Benchmarks: BEHAVIOR-100 [164], BEHAVIOR-1K [169]
SAPIEN [170]	–	2346/-	Code	PhysX	Navi, Mani	RGB, D, S	Force	Franka	Articulation, Ray Tracing	VoxPoser. Benchmark: SIMPLER [171]
AI2-THOR [172]	-/120	118/118	Mouse	Unity	Navi, Mani	RGB, D, S, A	Force, PD	ManipulaTHOR, LoCoBot, etc.	Object States, Task Planning. Versions: [173], [174]	Benchmarks: ALFRED, RoPOR [175]
VirtualHome [176]	7/-	-/509	Lang	Unity	Navi, Mani	RGB, D, S	Force, PD	Human	Object States, Task Planning	LID, Translated⟨LM⟩, ProgPrompt
TDW [177]	15/120	112/50	VR	Unity, Flex	Navi, Mani	RGB, D, S, A	Force	Fetch, Sawyer, Baxter	Audio, Fluids	–
RLBench [178]	1/-	28/28	Code	Bullet	Mani	RGB, D, S	Force	Franka	Tiered Task Difficulty	Hiveformer, PerAct
Meta-World [179]	1/-	80/7	Code	MuJoCo	Mani	Pose	Force	Sawyer	Meta-RL	R3M, VC-1, Vi-PRoM, EmbodiedGPT
CALVIN [180]	4/-	7/5	–	Pybullet	Mani	RGB, D	Force	Franka	Long-horizon Lang-cond tasks	GR-1, HULC, RoboFlamingo
Franka Kitchen [181]	1/-	10/6	VR	MuJoCo	Mani	Pose	Force	Franka	Extended by R3M with RGB	R3M, Voltron, Vi-PRoM, Diffusion Policy, EmbodiedGPT
Habitat [182], [183]	(Matterport + Gibson)		Mouse	Bullet	Navi	RGB, D, S, A	Force	Fetch, Franka, AlienGO	Fast, Navi only. Versions: 1.0, 2.0, Rearrangement [184]	VC-1, PACT; Benchmark: OVMM [185]
ALFRED [186]	-/120	84/84	–	Unity	Navi, Mani	RGB, D, S	PD	Human	Diverse long-horizon tasks	$(SL)^3$, LLM-Planner
DMC [187]	1/-	4/4	Code	MuJoCo	Control	RGB, D	Force	–	Continuous RL	VC-1, SMART
OpenAI Gym [188]	1/-	4/4	Code	MuJoCo	Control	RGB	Force	–	Single agent RL environments	–
Genesis [189]	(Rigid, deformable, liquid, etc.)		Code	(Proprietary)	Navi, Mani	RGB, D, S, N	Force	Franka, Unitree, etc.	High-speed comprehensive physics simulation	–

requires considerable effort, often involving techniques such as 3D scanning and mesh editing. Despite these hurdles, simulated environments provide automated evaluation metrics that aid researchers in consistently evaluating robotic models. Many benchmarks are based on simulators because they can precisely reproduce the experimental setup and yield fair comparisons of different models. Another technique, known as real-to-sim, can improve simulation fidelity, recreate failure cases, or facilitate digital twins.

C. Automated Dataset Collection

Several approaches advocate for automated dataset collection. RoboGen [190] employs a generative simulation paradigm that proposes interesting skills, simulates corresponding environments, and selects optimal learning approaches to train policies for acquiring those skills. AutoRT [191] functions as a robot orchestrator driven by LLMs, generating tasks, filtering them by affordance, and utilizing either autonomous policies or human teleoperators to collect and evaluate data. DIAL [192] focuses on augmenting language instructions in existing datasets using VLMs. RoboPoint [92] generates scenes procedurally with randomized 3D layouts, objects, and camera viewpoints.

D. Human Datasets

An alternative strategy to address data scarcity in real-world settings is to leverage human data. Human behavior offers plentiful guidance for robot policies due to its dexterity and diversity [193]. However, this strategy also comes with inherent drawbacks. Capturing and transferring human hand/body motions to robot embodiments is inherently difficult. Moreover, the inconsistency in human data poses a hurdle, as some data may be egocentric while others are captured from third-person perspectives. Additionally, filtering human data to extract useful information can be labor-intensive. These obstacles underscore the complexities involved in incorporating human data into robot learning processes. UMI [194] proposes a method to mitigate these issues using hand-held grippers. For a more comprehensive comparison of human datasets, we refer interested readers to [195].

E. Task Planning Benchmarks

EgoPlan-Bench [196] focuses on benchmarking real-world task planning with human annotations. PlanBench [197], [198] comprehensively assesses various aspects of task planning ability, such as cost optimality, plan verification, and replanning. LoTa-Bench [199] directly evaluates task planning by executing the generated plans in simulators and calculating success rates. Embodied Agent Interface (EAI) [200] argues that this approach fails to pinpoint issues in LLMs. By formalizing the input-output of LLM-based modules for decision-making tasks, EAI enables more fine-grained metrics beyond success rates.

F. Embodied Question Answering Benchmarks

Embodied question answering (EQA) benchmarks, as summarized in Table VII, do not directly evaluate robotic tasks like manipulation and navigation, but they are aimed at other relevant abilities for embodied AI, such as spatial reasoning, physics understanding, as well as world knowledge. EQA is akin to previous visual question answering benchmarks, but differs in that the agent can actively explore the environment before providing an answer. EmbodiedQA [201] and IQUAD [202] were among the first works to introduce this type of benchmark. MT-EQA [203] focuses on complex questions involving multiple targets. MP3D-EQA [204] converts previous RGB inputs to point clouds, testing 3D perception capabilities.

Active exploration requires access to a simulator, limiting the types of data that can be used, such as real-world videos. EgoVQA [205] shifts the focus of VQA to egocentric videos. EgoTaskQA [206] emphasizes spatial, temporal, and causal relationship reasoning. EQA-MX [207] investigates multimodal expressions (MX), including regular verbal utterances and nonverbal gestures like eye gaze and pointing. OpenEQA [208] evaluates seven main categories, including functional reasoning and world knowledge.

VI. Challenges and Future Directions

a) Safety first: Safety is paramount in robotics, as robots interact directly with the physical world. Ensuring the safety

表VI：模拟器与模拟基准测试。对照组：连续控制任务。D、S、A、N：深度、分割、音频
正常力：末端执行器与物体之间的模拟接触力。PD：预定义值。表格改编自[164]。[165].

姓名	场景/房间	物体/猫	用户界面	物理引擎	任务	观察	行动	智能体	描述	相关
吉布森[166]	572/-	—	—	Pybullet	纳维	RGB, D, N, S	—	—	仅限 Navi	—
iGibson [165], [167],[168]	15/108	152/5	鼠标，Pybullet VR		纳维，马尼	RGB、D、S、N、Flow、LiDAR	强迫	TurtleBot v2、LoCoBot等	VR，连续扩展状态。版本：0.5、1.0、2.0	基准测试：behavior-100[164]、behavior-1K[169]
SAPIEN [170]	—	2346/-	PhysX代码导航，Mani RGB，D，S				弗兰克力量		关节运动，光线追踪	VoxPoser。基准测试：更简单[171]
AI2-THOR[172]-/120 118/118 小鼠统一导航系统，Mani RGB，D、S、A力场，PD ManipulaTHOR，对象状态，任务规划。基准测试：alfred，RoPOR								LoCoBot等版本：[173], [174]		[175]
VirtualHome [176]	7/-	-/509	长的	统一体	纳维，马尼	RGB, D, S	肌力，PD	人	对象状态、任务规划	LID，翻译〈LM〉，ProgPrompt
TDW [177]	15/120	112/50	虚拟现实	Unity，Flex	纳维，马尼	RGB, D, S, A	强迫	费奇、索耶、巴克斯特	音频、流体	—
RLBench [178]	1/-	28/28	编码	子弹	玛尼	RGB, D, S	强迫	弗兰卡	分级任务难度	Hiveformer, PerAct
元世界[179]1/- 80/7			MuJoCo Mani代码			姿势	力杆	元强化学习		R3M、VC-1、Vi-PRoM 具身GPT
卡尔文[180]	4/-	7/5	—	Pybullet	玛尼	RGB, D	强迫	弗兰卡	长视距朗-康德任务	GR-1, HULC, RoboFlamingo
Franka Kitchen [181]	1/-	10/6	VR	MuJoCo	Mani	姿势	弗兰克力量		由 R3M 通过 RGB R3M、Voltron、Vi-PRoM 进行扩展。	扩散策略，EmbodiedGPT
生境[182]，[183]	（Matterport + Gibson）鼠标子弹导航					RGB，D，S，A力获取，弗兰卡		AlienGO	仅限Fast和Navi版本。版本：1.0、2.0、重排版[184]	VC-1，PACT；基准：OVMM[185]
阿尔弗雷德[186]	-/120	84/84	—	统一体	纳维，马尼	RGB, D, S	PD	人	多种长期任务	(SL)³，LLM规划器
DMC [187]	1/-	4/4	编码	MuJoCo	统治	RGB, D	强迫	·	连续RL	VC-1，SMART
OpenAI Gym [188]	1/-	4/4	编码	MuJoCo	统治	RGB	强迫	—	单药强化学习环境	—
创世记[189]	（刚性、可变形、液态等）		编码	（专有）乙醇胺	纳维，马尼	RGB, D, S, N	强迫	Franka、Unitree等	高速综合物理模拟	—

这类模拟环境的构建需要投入大量精力，通常涉及三维扫描和网格编辑等技术手段。尽管存在这些挑战，模拟环境仍能提供自动化评估指标，帮助研究人员对机器人模型进行标准化评估。许多基准测试基于仿真器开展，因其能精准复现实验场景，确保不同模型间的公平对比。另一种被称为“真实到模拟”的技术，可提升仿真逼真度、重现故障案例或助力数字孪生技术应用。

C. 自动化数据集收集

目前有多种方法倡导自动化数据集收集。RoboGen[190]采用生成式模拟范式，通过提出有趣技能、模拟对应环境并选择最优学习方法来训练获取这些技能的策略。AutoRT[191]作为由大语言模型驱动的机器人协调器，负责生成任务、按可用性进行筛选，并利用自主策略或人类远程操作员来收集和评估数据。DIAL[192]专注于通过虚拟语言模型增强现有数据集中的语言指令。RoboPoint[92]则通过程序化方式生成场景，包含随机化的三维布局、物体配置和摄像机视角。

D. 人类数据集

在现实场景中应对数据稀缺问题的另一种策略是利用人类数据。人类行为因其灵活性和多样性为机器人策略提供了丰富的指导依据[193]。然而，该策略也存在固有缺陷：将人类手部/身体动作捕捉并转化为机器人模型存在本质困难；数据不一致性构成障碍——部分数据可能呈现自我中心视角，而另一些则来自第三人称视角；此外，筛选人类数据以提取有效信息可能耗费大量人力。这些挑战凸显了将人类数据融入机器人学习过程的复杂性。 UMI[194]提出了一种利用手持抓取器缓解这些问题的方法。如需获取更全面的人类数据集对比分析，感兴趣的读者可参考[195]。

E. 任务规划基准

EgoPlan-Bench[196]专注于通过人工标注对现实世界任务规划进行基准测试。PlanBench[197]，[198]该系统全面评估任务规划能力的多个维度，包括成本优化度、计划验证效果及重新规划能力。LoTa-Bench[199]通过在仿真环境中执行生成的规划方案并计算成功率，直接对任务规划效果进行量化评估。具身智能体接口（EAI）[200]指出该方法难以精准定位大语言模型（LLM）存在的问题。通过将基于LLM的决策模块输入输出关系进行形式化建模，EAI能够提供超越单纯成功率指标的精细化评估体系。

F. 基于实体的问答基准研究

如表VII所总结，具身问答（EQA）基准测试并未直接评估机械臂操作与导航等机器人任务，而是针对具身人工智能的其他关键能力展开评估，例如空间推理、物理理解及世界知识。 EQA 与早期视觉问答基准测试类似，但其核心区别在于智能体在提供答案前可主动探索环境。EmbodiedQA[201]与 IQUAD[202]是最早提出此类基准测试的研究成果。MT- EQA[203]专注于涉及多目标的复杂问题；MP3D- EQA[204]通过将传统RGB输入转换为点云数据来测试三维感知能力；主动探索功能需依赖模拟器支持，限制了可使用数据类型（如真实世界视频）；Ego-VQA[205]则将 VQA 重点转向以自我为中心的视频分析。EgoTaskQA[206]侧重于空间、时间及因果关系推理。EQA -MX[207]研究多模态表达（MX），包括常规言语表达及眼神接触、指向等非语言手势。OpenEQA[208]评估七大主要类别，涵盖功能推理与世界知识。

第六章.挑战与未来方向

*a) 安全第一：*机器人技术中安全至关重要，因为机器人会直接与物理世界进行交互。确保安全

Table VII: Embodied question answering benchmarks. Explore: active exploration.

Benchmark	QA Pairs	Scenes	Source	Answer Type	Collection	Explore	Metrics
EQA [201]	5K	750 envs	House3D simulator	Answer set (172 answers)	Template	✓	Accuracy
IQUAD [202]	75K	30 rooms	AI2-THOR	Multiple choice	Template	✓	Accuracy
MT-EQA [203]	19.3K	588 envs	House3D simulator	Binary answer	Template	✓	Accuracy
MP3D-EQA [204]	1,136	83 envs	MatterPort3D	Answer set (53 answers)	Template	✓	Accuracy
EgoVQA [205]	600	16 videos	IU Multi-view	Multiple choice (1 out of 5)	Human annotators	✗	Accuracy
EgoTaskQA [206]	40K	2K videos	LEMMA dataset	Open answer, binary verification	Human annotators & template	✗	Accuracy
EQA-MX [207]	8.2M	750K images	CAESAR simulator	Answer set	Question templates, answer set	✗	Accuracy
OpenEQA [208]	557 + 1079	180 envs	HM3D + ScanNet	Open answer	Human annotators	✗	LLM Score

of robotic systems requires the integration of real-world commonsense. This involves the incorporation of robust safety guardrails, risk assessment frameworks, and human-robot interaction protocols. RLHF and "evaluation without execution" can also significantly lower safety risks [19]. Interpretability and expandability of VLA decision-making processes are also crucial for enhancing robot safety through error diagnosis and troubleshooting.

b) Datasets & Benchmarks: In addition to the issues discussed in §V, comprehensive benchmarks that cover a wide range of skills, objects, embodiments, and environments remain to be developed. Moreover, metrics beyond the success rate are needed for a fine-grained diagnosis of issues in VLA models, as highlighted by EAI [200] for LLMs.

c) Foundation Models & Generalization: VLA foundation models or robotic foundation models (RFM) for embodied AI remain an open research topic primarily due to the diversity in embodiments, environments, and tasks. Many have made significant progress [111], [116], but still lack generalization capability on par with LLMs in NLP. Attaining such a level of generalization is very challenging, as it requires the development of many core AGI capabilities.

d) Multimodality: VLAs inherit many challenges associated with multimodal models, such as obtaining useful embeddings and aligning different modalities. Current approaches, like ImageBind [209] and LanguageBind [210], align different modalities to the image or language embedding space, respectively. Within a unified embedding space, MLLMs can accommodate diverse modalities and become more general. However, whether focusing on embeddings alone is sufficient remains under debate. Although beyond the scope of VLAs, other modalities—such as audio [43], haptics [211], and gaze [212]—have proven useful for certain embodied AI applications. For instance, modeling human gaze data using an additional gaze network [213] or incorporating it via an auxiliary loss [214] has been shown to enhance the performance of the primary policy network. These approaches demonstrate that RL policies can benefit from human-like visual attention, as human gaze often reveals the most salient locations in the environment [215]. While incorporating additional modalities is often advantageous, it inevitably introduces extra complexity into the model design.

e) Framework for Long-Horizon Tasks: The hierarchical framework is currently the most practical approach for long-horizon tasks. However, it increases system complexity and potential points of failure. Frequent task execution failures can trigger re-planning, which can cause significant latency. Moreover, monolithic task planners share similar architectures with LVLAs, so employing two large models can be redundant and may hinder scalability. Additionally, although modular task planners typically do not require training, they are not plug-and-play: language-based models may generate subtasks that control policies cannot execute, whereas code-based models require modules to be pre-wrapped in APIs manually. Therefore, developing a unified framework that directly translates long-horizon tasks into low-level control signals in an end-to-end fashion is worth exploring.

f) Real-Time Responsiveness: Unlike conversational AI, many robotic applications require real-time decision-making to respond to dynamic environments. If inference time cannot keep pace with environmental changes, the model may generate obsolete actions repeatedly. However, current VLA models—especially LVLAs and task planners—face a trade-off between speed and capacity. Novel mechanisms are thus needed to strike an optimal balance.

g) Multi-agent Systems: Cooperative multi-agent systems offer benefits such as distributed perception and collaborative fault recovery. However, they also face challenges, including effective communication, coordinated dispatching, and fleet heterogeneity. In certain scenarios, individual agents may have conflicting goals, which further increases complexity.

h) Ethical and Societal Implications: Robotics has always raised various ethical, societal, and legal concerns. These include risks related to privacy, job displacement, decision-making bias, and the impact on social norms and human relationships.

i) Applications: Most current VLAs focus on household or industrial settings, but a wider range of applications is possible, such as virtual assistants, autonomous vehicles, and agricultural robots. Various embodiments may also call for specialized VLAs, including dexterous hands, drones, quadruped robots, and humanoid robots. One particularly important field is healthcare, encompassing surgical robots [216], care robots [217]. Healthcare demands higher safety and privacy standards and may necessitate novel techniques such as human-in-the-loop (HITL) control and federated learning. Moreover, due to the significant domain gap, specialized vision models might be needed for medical images.

VII. Conclusion

Vision-language-action models hold immense promise for enabling embodied agents to interact with the physical world and fulfill users' instructions. This paper is the first survey to review large VLAs alongside generalized VLAs. Our taxonomy provides a high-level overview of three main lines of research: key components, control policies, and task planners. We meticulously analyze and compare their technical details,

表VII：具身化问答基准测试。探索：主动探索。

基准	QA对	场景	根源	答案类型	收集	探索	指标
EQA [201]	5K	750欧元	House3D模拟器	答案集（172个答案）	模板	✓	精度
IQUAD [202]	75K	30间客房	AI2-THOR	多选题	模板	✓	精度
MT-EQA [203]	19.3K	588个环境	House3D模拟器	二元答案	模板	✓	精度
MP3D-EQA [204]	1,136	83个环境	MatterPort3D	答案集（53个答案）	模板	✓	精度
EgoVQA [205]	600	16个视频	IU多视角	多项选择（5选1）	人工标注者	✗	精度
EgoTaskQA [206]	40K	2K 视频	词典数据集	开放答案，二元验证	人类标注员 & 模板	✗	精度
EQA-MX [207]	8.2M	75万张图像	凯撒模拟器	答案集	问题模板，答案集	✗	精度
OpenEQA [208]	557 + 1079	180欧元	HM3D + ScanNet	开放式答案	人工标注者	✗	LLM评分

机器人系统的开发需要融入现实世界的常识性知识，这包括建立完善的安全防护机制、风险评估框架以及人机交互协议。采用 RLHF（可验证逻辑架构）和“无需执行的评估”方法也能显著降低安全风险[19]。此外，VLA（虚拟逻辑架构）决策过程的可解释性和可扩展性对于通过错误诊断与故障排除来提升机器人安全性至关重要。

*b) 数据集与基准测试：*除第V节讨论的问题外，仍需开发涵盖广泛技能、对象、实施方式及环境的综合性基准测试。此外，如EAI[200]针对大语言模型（LLMs）所强调的，需要超越成功率的指标来对VLA模型中的问题进行细粒度诊断。

*c) 基础模型与泛化能力：*用于具身人工智能的VLA基础模型或机器人基础模型（RFM）仍是一个开放的研究课题，这主要源于其具身形式、运行环境及任务的多样性。该领域已取得显著进展[111]。[116]但其泛化能力仍无法与自然语言处理领域的大型语言模型（LLMs）相媲美。要达到这样的泛化水平极具挑战性，因为这需要开发多种核心的通用人工智能（AGI）能力。

*d) 多模态：*VLA继承了多模态模型面临的诸多挑战，例如获取有效的嵌入表示以及实现不同模态间的对齐。当前的方法（如ImageBind[209]和LanguageBind [210]）分别将不同模态对齐到图像或语言嵌入空间。在统一的嵌入空间内，MLLM能够兼容多种模态并更具通用性。然而，仅聚焦于嵌入表示是否足够仍存在争议。尽管超出VLA的研究范畴，其他模态——如音频[43]、触觉[211]及注视方向——同样值得关注。
[212]—已被证明对某些具身人工智能应用具有实用价值。例如，通过附加注视网络[213]对人类注视数据进行建模，或通过辅助损失函数[214]将其整合到模型中，已被证实能显著提升主策略网络的性能表现。这些研究结果表明，强化学习策略可从类人视觉注意力机制中获益，因为人类注视行为往往能揭示环境中最具显著性的位置特征[215]。尽管引入额外模态通常具有优势，但不可避免地会增加模型设计的复杂度。

*e) 长周期任务框架：*目前，分层架构仍是处理长周期任务最实用的方法。但该架构会增加系统复杂度和潜在故障点。频繁的任务执行失败可能触发重新规划，从而导致显著延迟。此外，单体式任务规划器与低层级逻辑架构（LVLAs）采用相似的架构设计，使用两个大型模型存在冗余性，可能制约系统扩展能力。虽然模块化任务规划器通常无需训练即可运行，但其并非即插即用方案：基于语言的模型可能生成控制策略无法执行的子任务，而基于代码的模型则需要手动将模块封装到API接口中。因此，开发能以端到端方式将长周期任务直接转化为低级控制信号的统一框架具有重要研究价值。

*f) 实时响应能力：*与对话式人工智能不同，许多机器人应用需要实时决策以应对动态环境。若推理时间无法跟上环境变化速度，模型可能会反复生成过时的动作。然而当前虚拟实验室自动化（VLA）模型——尤其是实验室虚拟实验室（LVLAs）和任务规划器——在速度与容量之间存在权衡关系。因此需要创新机制来实现最佳平衡。

*g) 多智能体系统：*协同式多智能体系统具有分布式感知与协同故障恢复等优势，但也面临有效通信、协调调度及车队异构性等挑战。在特定场景下，各智能体可能目标冲突，这会进一步增加系统复杂度。

*h) 伦理与社会影响：*机器人技术始终引发诸多伦理、社会及法律层面的争议。这些问题涉及隐私风险、就业岗位流失、决策偏见，以及对社会规范与人际关系的潜在冲击。

*i) 应用场景：*当前大多数虚拟现实增强设备（VLA）主要应用于家庭或工业场景，但其应用场景可拓展至虚拟助手、自动驾驶汽车及农业机器人等领域。不同应用场景可能需要定制化VLA设备，例如灵巧机械手、无人机、四足机器人及人形机器人。其中医疗健康领域尤为重要，涵盖手术机器人等应用。
[216]护理机器人[217]。医疗领域对安全性和隐私保护标准要求更高，可能需要采用人机协同（HITL）控制和联邦学习等创新技术。此外，由于存在显著的领域差异，医学图像可能需要专门的视觉模型。

VII.结论

视觉-语言-动作模型在实现具身智能体与物理世界交互及执行用户指令方面具有巨大潜力。本文首次对大规模视觉语言动作系统（VLAs）与通用型视觉语言动作系统进行系统综述。我们的分类体系从关键组件、控制策略和任务规划器三大研究方向提供高层次概述，并对各系统的技术细节进行了细致分析与对比。

including model architectures, training strategies, and individual modules. Additionally, we highlight essential resources for training and evaluating VLAs, such as datasets, simulators, and benchmarks. We hope this survey captures the rapidly evolving landscape of embodied AI and inspires future research.

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涵盖模型架构、训练策略及各模块设计等内容。此外，我们重点介绍了虚拟实验室自动化（VLAs）训练与评估所需的核心资源，包括数据集、仿真器及基准测试工具。本综述旨在呈现具身人工智能领域快速发展的现状，并为未来研究提供启发。

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Appendix

A. Background

1) Unimodal Models: Vision-language-action models integrate three modalities, often relying on existing unimodal models. The transition from convolutional neural networks (e.g., ResNet [218]) to visual Transformers (e.g., ViT [219], SAM [220]) in computer vision has facilitated the development of vision foundation models (VFMs). In natural language processing, the evolution from recurrent neural networks (e.g., LSTM [221], GRU [222]) to Transformers [3] initially led to the pretrain-finetune paradigm (e.g., BERT [26], ChatGPT [223]), followed by the recent success of prompt tuning driven by large language models. Reinforcement learning (e.g., DQN [4], AlphaGo [224], PPO [225], Dactyl [226]) has also witnessed a shift towards employing Transformers to model the Markov Decision Process as autoregressive sequential data. DPO [227] directly trains LLMs on human preferences, simplifying RLHF.

2) Vision-Language Models: Vision-language tasks, encompassing image captioning [228], visual question answering [229], visual grounding [230], require the fusion of computer vision and natural language processing models. Early efforts, such as Show and Tell [231], leveraged the success of early CNNs and RNNs. The advent of high-capacity language models, including BERT [26] and GPT [232], ushered in a new era of Transformer-based VLMs. One of the pioneering self-supervised pretraining methods is ViLBERT [233], while CLIP [22] popularized contrastive pretraining approaches for multimodal alignment. The recent emergence of large language models has driven the development of multimodal LLMs (MLLMs) or large multimodal models (LMMs), which achieve state-of-the-art performance on multimodal instruction-following tasks. Representative MLLMs include Flamingo [234], BLIP-2 [6], and LLaVA [7]. VLMs share a close relationship with VLAs, as the multimodal architectures of VLMs can be readily adopted for VLAs. Additionally, VLMs can function as high-level task planners if they possess sufficient reasoning capabilities.

B. Background (Extended Version): Unimodal Models

Vision-language-action models involve three modalities, and consequently, many VLAs depend on existing unimodal models for processing inputs from different modalities. Therefore, it is crucial to summarize representative developments in unimodal models, as they often serve as integral components in VLAs. Specifically, for the vision modality, we collect models designed for image classification, object detection, and image segmentation, as these tasks are particularly relevant for robotic learning. Natural language processing models play a crucial role in enabling VLAs to understand language instructions or generate language responses. Reinforcement learning is a foundational component for obtaining optimal policies, facilitating the generation of appropriate actions in a given environment and condition. A brief timeline of the development of unimodal models is depicted in Figure 7. Additionally, Figure 8 highlights the progressive increase in model size within these fields.

1) Computer Vision: Computer vision witnessed the inception of modern neural networks. In robotics, object classification models can be used to inform a policy about which objects are of interest, and models for object detection or image segmentation can help precisely locate objects. Therefore, we mainly summarize approaches for these tasks, but numerous excellent surveys on visual models, ranging from convolutional neural networks (CNNs) [235] to Transformers [236], offer more detailed insights. Interested readers are directed to these surveys for a more comprehensive introduction. Here, we will briefly touch upon some of the key developments in the field of computer vision.

a) Convolutional neural network: Early developments in computer vision (CV) were primarily focused on the image classification task. LeNet [237] was among the first convolutional neural networks, designed for identifying handwritten digits in zip codes. In 2012, AlexNet [2] emerged as a breakthrough by winning the ImageNet challenge, showcasing the potential of neural networks. VGG [238] demonstrated the benefits of increasing the depth of CNNs. GoogLeNet [239], also known as Inception-V1, introduced the concept of blocks. ResNet [218] introduced skip connections or residual connections. Inception-ResNet [240], as the name suggests, combines residual connects and inception blocks. ResNeXt [241] explored the concept of split, transform, and merge. SENet [242] introduces the squeeze-and-excitation blocks, utilizing a type of attention mechanism. EfficientNet [243] studied the width, depth, and resolution of CNN models with "compound scaling," highlighting the trade-off between efficiency and performance.

Alongside image classification, object detection became an integral component in many applications. Building upon the success of image classification backbone networks, a series of works optimized region-based methods: R-CNN [244], Fast R-CNN [245], Faster R-CNN [246], and Mask R-CNN [247]. Grid-based methods like YOLO [248] are also widely adopted. Bottom-up, top-down is also a popular strategy, employed by FPN [249], RetinaNet [250], BUTD [251], etc. In scenarios requiring more detailed and precise object detection, image segmentation aims to determine the exact outline of objects. Many popular models adopt an "encoder-decoder" architecture, where the encoder understands both the global and local context of the image, and the decoder produces a segmentation map based on this context information. Representative works following this idea include FCN [252], SegNet [253], Mask R-CNN [247], and U-Net [254].

b) Vision Transformer: Convolutional neural networks (CNNs) have historically been the foundation of computer vision models. However, the landscape shifted with the introduction of the Transformer architecture in the seminal work by [3]. This paradigm shift was initiated by ViT [219]. It revolutionizes image processing by breaking down images into 16-by-16 pixel patches, treating each as a token akin to those in NLP; leveraging a BERT-like model, ViT encodes these patches and has exhibited superior performance over many traditional CNN models in image classification tasks.

The transformative power of the Transformer extends beyond classification. DETR [255] employs an encoder-decoder

附录

A. 背景

1) 单模态模型: 视觉-语言-动作模型整合了三种模态，通常依赖于现有的单模态模型。计算机视觉领域从卷积神经网络（如ResNet[218]）向视觉Transformer（如ViT[219]、SAM[220]）的演进促进了视觉基础模型（VFMs）的发展。在自然语言处理领域，从循环神经网络（如LSTM[221]、GRU[222]）向Transformer[3]的演变最初催生了预训练-微调范式（如BERT[26]、ChatGPT[223]），随后大型语言模型驱动的提示调优取得了近期成功。强化学习（如 DQN[4]、AlphaGo[224]、PPO[225]、Dactyl[226]）也呈现出向采用Transformer建模马尔可夫决策过程作为自回归序列数据的转变。 DPO[227]直接基于人类偏好对LLMs进行训练，简化了 RLHF 。

2) 视觉-语言模型: 视觉-语言任务涵盖图像描述[228]、视觉问答[229]、视觉定位[230]等领域，需要计算机视觉与自然语言处理模型的融合。早期尝试如Show and Tell[231]借鉴了早期CNN和RNN的成功经验。随着BERT[26]和GPT[232]等大容量语言模型的出现，基于Transformer的视觉语言模型新时代应运而生。开创性自监督预训练方法ViLBERT[233]与CLIP[22]推动了对比预训练方法在多模态对齐中的普及。近期大型语言模型的涌现推动了多模态大语言模型（MLLMs）或大型多模态模型（LMMs）的发展，这些模型在多模态指令遵循任务中实现了顶尖性能。代表性MLLMs包括Flamingo[234]、BLIP-2[6]和LLaVA[7]。视觉语言模型与视觉语言架构（VLA）具有密切关联性，因其多模态架构可直接适配VLA。此外，若具备足够的推理能力，虚拟语言模型（VLMs）可作为高级任务规划器发挥作用。

B. 背景（扩展版）：单峰模型

视觉-语言-动作模型涉及三种模态，因此许多视觉语言动作系统（VLA）需要依赖现有的单模态模型来处理不同模态的输入。因此，总结单模态模型的代表性发展成果至关重要，因为这些模型往往是VLA的核心组成部分。具体而言，针对视觉模态，我们收集了专为图像分类、目标检测和图像分割设计的模型，这些任务对机器人学习具有重要应用价值。自然语言处理模型在使VLA理解语言指令或生成语言响应方面发挥着关键作用。强化学习作为获取最优策略的基础组件，能够帮助系统在特定环境和条件下生成恰当动作。单模态模型的发展脉络简要展示于图7，而图8则突显了该领域模型规模的持续增长趋势。

1) 计算机视觉: 计算机视觉见证了现代神经网络的诞生。在机器人技术中，物体分类模型可用于制定关于哪些物体值得关注的策略，而物体检测或图像分割模型则有助于精确定位物体。因此，本文主要总结这些任务的相关方法，但关于视觉模型的众多优秀综述——从卷积神经网络（CNN）[235]到Transformer模型[236]——提供了更深入的见解。感兴趣的读者可参考这些综述以获得更全面的介绍。在此，我们将简要探讨计算机视觉领域的一些关键进展。

a) 卷积神经网络: 计算机视觉（CV）早期发展主要聚焦于图像分类任务。LeNet[237]是最早用于识别邮政编码中手写数字的卷积神经网络之一。2012年，AlexNet[2]通过赢得ImageNet挑战赛成为突破性进展，展现了神经网络的潜力。 VGG[238]证明了增加CNN深度的优势。GoogLeNet[239]（又称Inception-V1）引入了模块化设计概念。ResNet[218]提出了跳跃连接或残差连接机制。Inception-ResNet[240]正如其名，融合了残差连接与Inception模块。ResNeXt[241]探索了分割-变换-合并算法框架。SENet[242]采用基于注意力机制的压缩-激励模块。EfficientNet[243]通过“复合缩放”技术研究CNN模型的宽度、深度与分辨率，揭示了效率与性能之间的权衡关系。

与图像分类并行发展的是，目标检测已成为众多应用中的核心组件。基于图像分类骨干网络的成功经验，一系列研究对区域方法进行了优化：R-CNN[244]、Fast R-CNN[245]、Faster R-CNN[246]以及Mask R-CNN[247]。基于网格的方法如YOLO[248]也得到广泛应用。自下而上与自上而下的策略同样流行，被 FPN[249]、RetinaNet[250]、BUTD[251]等模型采用。在需要更精细精准目标检测的场景中，图像分割旨在确定物体的精确轮廓。多数主流模型采用“编码器-解码器”架构，其中编码器同时理解图像的全局与局部上下文，解码器则基于这些上下文信息生成分割图。遵循该思路的代表性研究包括 FCN[252]、SegNet[253]、Mask R-CNN[247]以及U-Net[254]。

b) 视觉Transformer: 卷积神经网络（CNN）历来是计算机视觉模型的基础架构。然而随着[3]开创性研究中Transformer架构的提出，这一格局发生了根本性转变。ViT模型[219]率先推动了这一范式革新：通过将图像分解为16×16像素块（类似自然语言处理中的标记处理方式），并采用类似BERT的模型对这些像素块进行编码，ViT在图像分类任务中展现出超越传统CNN模型的卓越性能，彻底革新了图像处理领域。

Transformer的变革性能力超越了分类任务。DETR[255]采用了一种编码器-解码器架构。

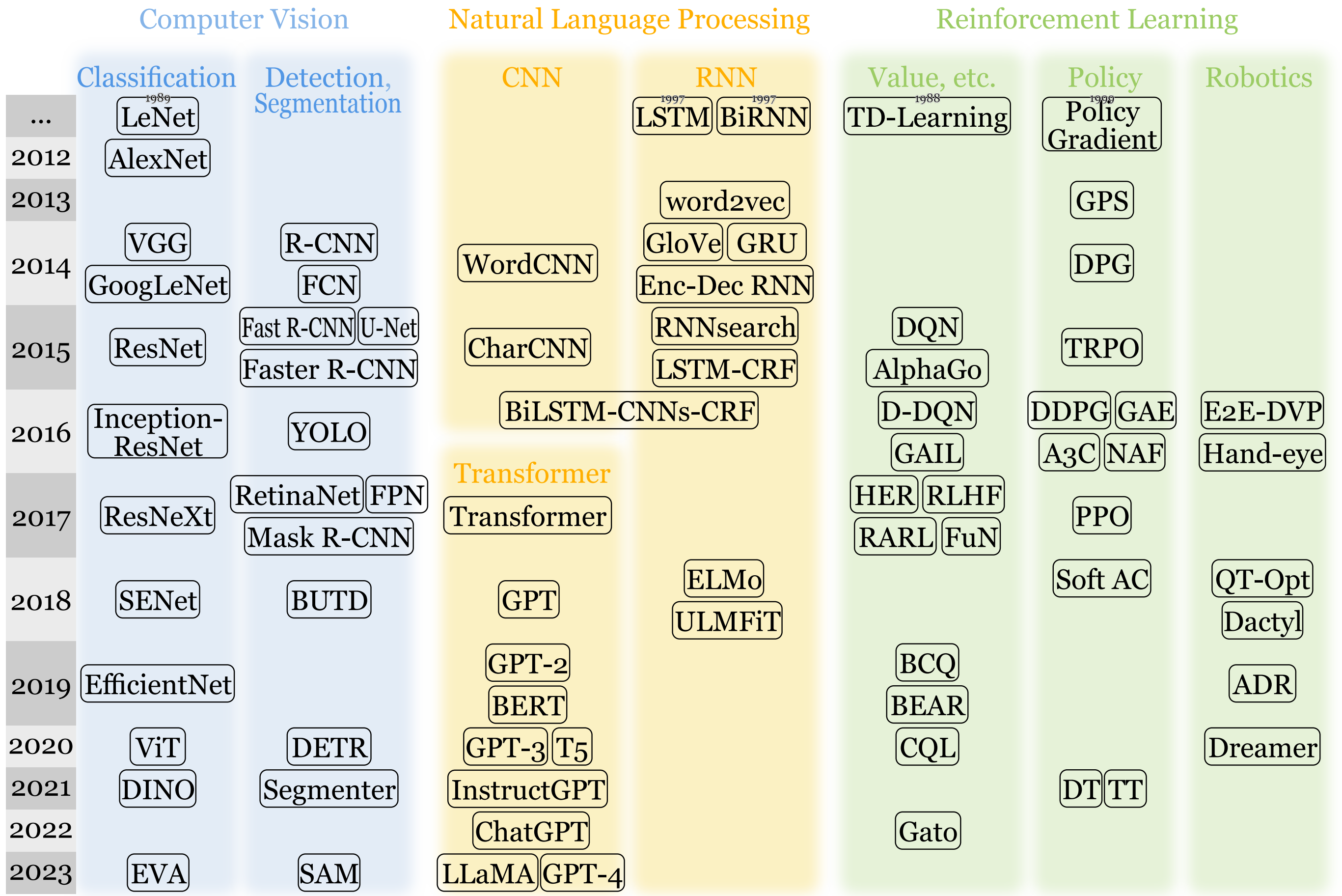

Figure 7: A brief timeline of pivotal unimodal models leading to the development of vision-language-action models, organized by their publication years. Details can be found in Appendix:B.

Transformer architecture to tackle object detection. The encoder processes the input image, and its output embeddings are fed into the decoder through cross-attention. Notably, DETR introduces learnable object queries to the decoder, facilitating the extraction of crucial object-wise information from the encoder's output. Venturing into image segmentation, Segmenter [256] was the first to utilize Transformer on this task. The Segment Anything model (SAM) [220] achieves remarkable milestones in promptable segmentation, zero-shot performance, and versatile architecture, further underlining the transformative impact of Vision Transformer models in various computer vision domains.

c) Vision in 3D: Aside from the most common RGB data, other types of visual inputs are widely used [257], [258]. In robotics, depth maps are useful since they provide essential 3D information that is not explicitly stored in RGB images. Depth maps can be captured with Microsoft Kinect [1] or Intel RealSense [2] or recovered from pure RGB images. Point clouds [259] are also popular visual input types due to the widespread adoption of LiDARs and 3D scanners; depth maps can be easily converted to point clouds. Volumetric data [260], such as voxels or octrees, is usually more information-rich than depth maps and is suitable for representing rigid objects. Despite the widespread use of 3D meshes as the default data format in computer graphics, their irregular nature poses challenges for neural networks [261].

2) Natural Language Processing: Natural Language Processing (NLP) plays a pivotal role in VLA, serving as a vital component for understanding user instructions or even generating appropriate textual responses. The recent surge in NLP owes much to the success of Transformer models [3]. In the landscape of contemporary NLP, there is a noticeable shift towards implicit learning of language syntax and semantics, a departure from the previous paradigms. To provide context, this subsection will commence with a concise overview of fundamental yet enduring concepts before delving into the noteworthy advancements in contemporary NLP. For an in-depth exploration of the progress in the NLP domain, readers are directed to comprehensive surveys by [262] and [263], which meticulously review the trajectory of advancements in NLP.

a) Early developments: The field of NLP, which was more frequently referred to as Computational Linguistics (CL) in the early days, tries to solve various tasks regarding to natural language. In CL, natural languages used to be processed in a hierarchical way: word, syntax, and semantics. Firstly, on the

[1] https://azure.microsoft.com/en-us/products/kinect-dk/
[2] https://www.intelrealsense.com

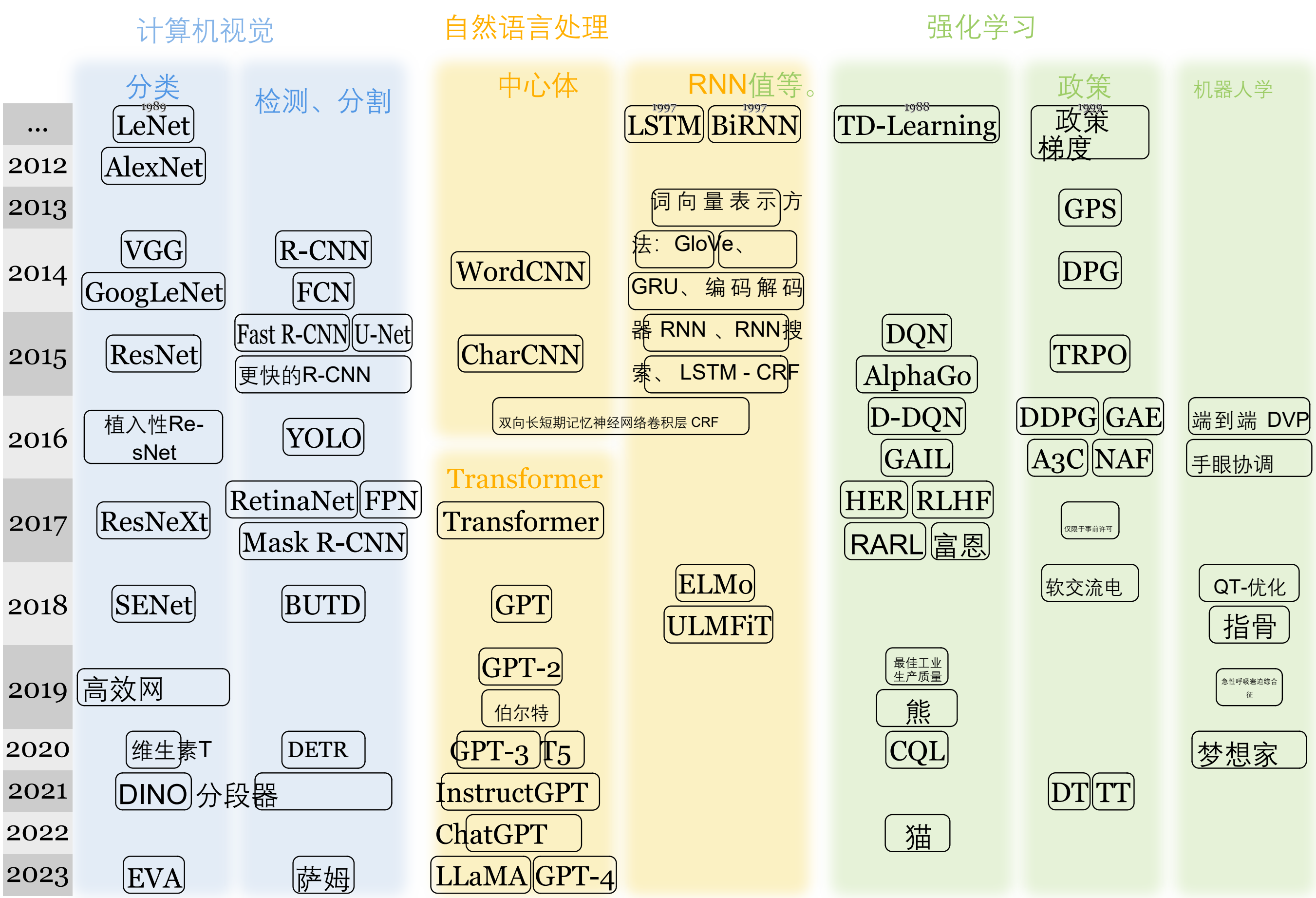

图7：按发表年份排序的关键单峰模型发展为视觉-语言-动作模型的时间线简表。详细信息参见附录B。

Transformer架构在目标检测领域的应用。编码器对输入图像进行处理，其生成的嵌入向量通过交叉注意力机制输入解码器。值得注意的是，DETR模型在解码器中引入了可学习的目标查询机制，有效促进了从编码器输出中提取关键目标信息的过程。在图像分割领域，Segmenter[256]是首个将Transformer架构应用于该任务的模型。Segment Anything模型（SAM）[220]在提示式分割、零样本性能及通用架构设计方面取得显著突破，进一步彰显了Vision Transformer模型在计算机视觉各领域中的变革性影响。

*c) 三维视觉：*除最常见的RGB数据外，其他类型的视觉输入也被广泛使用[257]。[258]在机器人技术领域，深度图具有重要应用价值，因其能提供RGB图像中无法直接获取的关键三维信息。深度图可通过微软Kinect [1]或英特尔RealSense [2]设备采集，也可从纯RGB图像中提取。随着激光雷达和三维扫描仪的广泛应用，点云[259]已成为主流视觉输入类型，深度图可轻松转换为点云数据。体素数据[260]（如体素或八叉树）通常比深度图包含更丰富的信息量。

[1]https://azure.microsoft.com/en-us/products/kinect-dk/

[2]https://www.intelrealsense.com

地图数据适合用于表示刚性物体。尽管三维网格（3D meshes）作为计算机图形学中的默认数据格式被广泛使用，但其不规则特性给神经网络带来了挑战[261]。

*2) 自然语言处理：*自然语言处理（NLP）在VLA中扮演着关键角色，是理解用户指令甚至生成恰当文本回应的重要组成部分。近年来NLP的迅猛发展很大程度上归功于Transformer模型的成功[3]。在当代NLP领域，语言句法与语义的隐式学习正呈现出显著趋势，这与以往的研究范式有所不同。为提供背景，本小节将首先简要概述一些基础且持久的概念，随后深入探讨当代NLP领域的重大进展。关于NLP领域进展的深入研究，读者可参考[262]与[263]的全面综述，这些文献详尽梳理了NLP技术的发展历程。

*a) 早期发展：*自然语言处理（NLP）领域在早期常被称为计算语言学（CL），致力于解决与自然语言相关的各类问题。在计算语言学中，自然语言处理通常采用分层化方法：从词汇层面、句法层面到语义层面依次进行分析。首先，

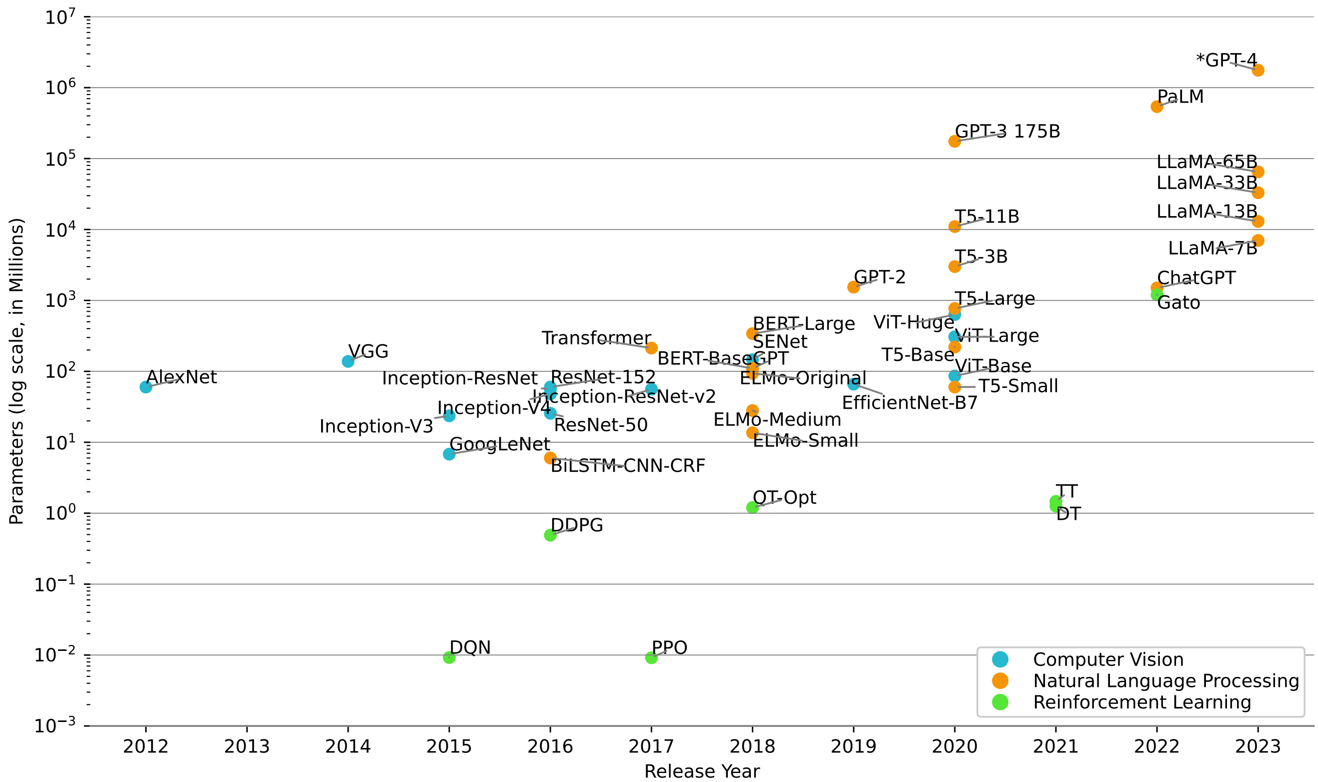

Figure 8: The growing scale of unimodal models over the years. ∗GPT-4's model size is estimated since it has not been officially disclosed.

word level, many aspects need to be accounted for, including morphology, lexicology, phonology, etc. This leads to problems like tokenization, lemmatization, stemming, semantic relations, word sense disambiguation, and Zipf's Law problem. Then, in terms of syntax, natural language, in contrast to formal language, has much less restricted grammar and thus is more challenging to parse: in the Chomsky hierarchy, natural language is generally considered to follow context-sensitive grammar, while programming languages are covered under context-free grammar. Syntactic parsing includes tasks such as part-of-speech tagging, constituency parsing, dependency parsing, and named entity recognition. Finally, to understand the semantics of a sentence in written language or an utterance in spoken language, the following tasks were studied: semantic role labeling, frame semantic parsing, abstract meaning representation, logical form parsing, etc.

b) Recurrent neural network & convolutional neural network: In the initial stages of NLP, rudimentary models relied on simple feed-forward neural networks to tackle various tasks [264]. After the introduction of word embeddings like word2vec [265], [266] and GloVe [267], NLP techniques embraced recurrent neural networks (RNNs) [268], such as LSTM [221], GRU [222], RNNsearch [269], LSTM-CRF [270], etc. Examples of representative RNNs in NLP include ELMo [271] and ULMFiT [272]. While RNNs played a significant role, alternative models utilizing convolutional neural networks also emerged. WordCNN [273] employed CNNs at the word level, with word2vec features serving as input to the CNNs. Another approach, CharCNN [274], focused on modeling language at the character level with CNN. Subsequent research [275] highlighted that character-level CNNs excel at capturing word morphology, and their combination with a word-level LSTM backbone can significantly enhance performance.

c) Transformer & large language model: The groundbreaking Transformer model, introduced by [3], revolutionized natural language processing through the introduction of the self-attention mechanism, inspiring a cascade of subsequent works. BERT [26] leverages the Transformer encoder stack, excelling in natural language understanding. On the other hand, the GPT family [232], [276]–[278] is built upon Transformer decoder blocks, showcasing prowess in natural language generation tasks. A line of work strives to refine the original BERT, including RoBERTa [279], ALBERT [280], ELECTRA [281]. Simultaneously, a parallel line of research following the GPT paradigm has given rise to models like XLNet [282], OPT [283]. BART [284], an encoder-decoder Transformer, distinguishes itself through pretraining using the denoising sequence-to-sequence task. Meanwhile, T5 [285] introduces modifications to the original Transformer, maintaining the encoder-decoder architecture. T5 unifies various NLP tasks through a shared text-to-text format, exhibiting enhanced performance in transfer learning. These diverse models collectively showcase the versatility and ongoing evolution within the NLP landscape.

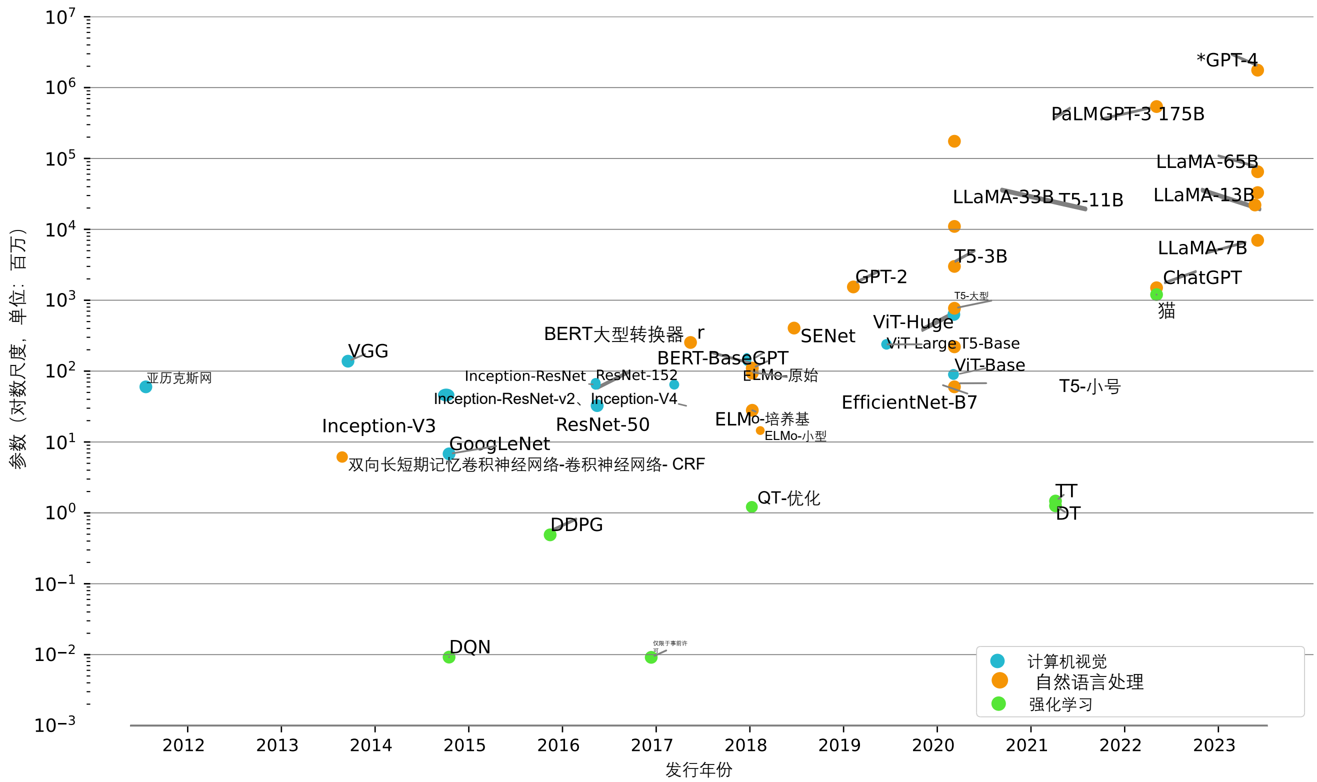

图8：多年来单峰模型规模的持续增长。 GPT-4的模型规模是根据其未公开的官方数据估算得出的。

在词层面，需要综合考虑形态学、词典学、音系学等诸多因素，这会导致诸如分词处理、词形还原、词干提取、语义关联分析、词义消歧以及齐普夫定律问题等技术难题。从句法层面来看，与形式语言相比，自然语言的语法约束较少，因此解析难度更大：在乔姆斯基层级结构中，自然语言通常遵循上下文敏感语法，而编程语言则适用上下文无关语法。句法解析包含词性标注、成分分析、依存关系解析及命名实体识别等任务。最后，为理解书面语句或口语表达的语义内涵，研究者重点探讨了语义角色标注、框架语义解析、抽象意义表征、逻辑形式解析等关键技术。

b) 循环神经网络与卷积神经网络工作原理: 在自然语言处理（NLP）初期阶段，基础模型依赖简单的前馈神经网络来处理各类任务[264]。随着word2vec等词嵌入技术的引入[265]，[266] 以及GloVe[267]，自然语言处理技术采用了循环神经网络（RNNs）[268]，例如 LSTM[221]、GRU[222]、RNNsearch[269]、 LSTM - CRF[270]等。自然语言处理中具有代表性的RNNs包括ELMo[271]和ULMFiT[272]。虽然RNNs发挥了重要作用，但利用卷积神经网络的替代模型也应运而生。WordCNN[273]在Word层级采用了CNN，以word2vec特征作为CNN的输入。另一种方法CharCNN[274]则专注于通过CNN在字符层级对语言进行建模。后续研究[275]指出，字符层级CNN在捕捉Word形态学方面表现优异，将其与词层级 LSTM 骨干网络结合可显著提升性能。

c) Transformer与大型语言模型: 由[3]提出的突破性Transformer模型通过引入自注意力机制彻底改变了自然语言处理领域，激发了后续一系列研究工作的涌现。BERT [26]基于Transformer编码器堆栈架构，在自然语言理解方面表现卓越。另一方面，GPT系列模型[232][276]–[278]基于Transformer解码器模块构建，在自然语言生成任务中展现出卓越能力。一系列研究致力于改进原始BERT模型，包括 RoBERTa[279]、albert[280]、electra[281]。与此同时，遵循GPT范式的平行研究路线催生了XLNet[282]、OPT[283]等模型。采用编码器-解码器架构的BART[284]通过使用去噪序列到序列任务进行预训练而脱颖而出。T5 [285]对原始Transformer模型进行改进，保持编码器-解码器架构。T5通过共享文本到文本格式统一多种NLP任务，在迁移学习中表现出更强的性能。这些多样化模型共同展现了NLP领域中的多功能性与持续演进态势。

Over the past few years, there has been a remarkable expansion in the size of language models, driven by the scalability of the Transformer architecture. This trend has given rise to a series of large language models (LLMs) that have demonstrated breakthrough performance and capabilities not achievable with smaller models. A landmark model in this evolution is ChatGPT [223], which has sparked considerable interest and inspired a series of works in this domain, such as GPT-4 [278], PaLM [286], PaLM-2 [287], LLaMA [288], LLaMA 2 [289], ERNIE 3.5 [290]. Notably, LLaMA stands out as one of the few open-source LLMs, fostering interesting developments. The introduction of "instruction-tuning" has allowed efficient fine-tuning of a pretrained LLM to become an instruction-following model. This technique is popularized by InstructGPT [291] and FLAN [292], [293]. Recent advancements in instruction-following models include Alpaca [294], employing self-instruction, and Vicuna [295], leveraging conversations from ShareGPT. As LLMs grow in scale and power, there is a shift away from the need for fine-tuning on downstream tasks. With appropriate prompts, LLMs can produce accurate outputs without task-specific training, a paradigm known as *prompt engineering*. This approach differs from the traditional *pretrain-finetune* paradigm. DPO [227] introduces a method to train LLMs directly on human preferences without requiring explicit reward modeling or reinforcement learning, simplifying upon previous RLHF methods.

3) Reinforcement Learning: Reinforcement learning (RL) seeks to acquire a policy capable of taking optimal actions based on observations of the environment. Numerous vision-language-action models are constructed based on paradigms such as imitation learning or Temporal Difference (TD) learning within RL. Many challenges faced in the development of robotic policies can be effectively addressed through insights gained from the field of RL. Consequently, delving into RL methods presents a valuable avenue for enhancing robotic learning. For a deeper exploration of RL methods, readers can refer to more comprehensive reviews provided by [296], [297], and [298].

a) Deep reinforcement learning: The advent of deep reinforcement learning can be attributed to the success of pioneering models, Deep Q-Network [4] and AlphaGo [224]. Deep learning, with its ability to provide low-dimensional representations, proved instrumental in overcoming traditional computational and memory complexity challenges in reinforcement learning. In recent years, a multitude of value-function-based approaches have surfaced. Double DQN (D-DQN) [299] addresses the action value overestimation issue of DQN. Hindsight experience replay (HER) [300] focuses on the sparse reward issue. Batch-Constrained deep Q-learning (BCQ) [301] presents an approach aimed at enhancing off-policy learning by constraining the action space. BEAR [302] endeavors to alleviate instability arising from bootstrapping errors in off-policy RL. [303] introduces conservative Q-learning (CQL) to address the overestimation of values by standard off-policy RL methods.

Another paradigm within reinforcement learning is policy search, encompassing methods such as policy gradient and actor-critic techniques. These approaches aim to combat persistent challenges, including instability, slow convergence, and data inefficiency. Guided policy search (GPS) [304] learns a policy with importance sampling guided by another controller toward a local optimum. Deterministic policy gradient (DPG) [305], deep deterministic policy gradient (DDPG) [306], and asynchronous advantage actor-critic (A3C) [307] improve efficiency without compromising stability. Normalized advantage functions (NAF) [308] are a continuous variant of Q-learning, allowing for Q-learning with experience replay. Soft actor-critic (Soft AC) [309] takes advantage of the maximum entropy RL framework to lower sample complexity and improve convergence properties. Trust region policy optimization (TRPO) [310] and proximal policy optimization (PPO) [225] utilize trust region methods to stabilize policy gradients, with PPO additionally incorporating a truncated generalized advantage estimation (GAE) [311].

Beyond these, various other RL methodologies exist, such as imitation learning and hierarchical reinforcement learning. Generative adversarial imitation learning (GAIL) [312] is an imitation learning method that uses a generative adversarial framework to discriminate expert trajectories against generated trajectories. Robust adversarial reinforcement learning (RARL) [313] incorporates adversarial agents to enhance generalization. RLHF [314] utilizes human preferences without access to the reward function. FeUdal Networks (FuN) [315] introduce a hierarchical reinforcement learning architecture featuring a Manager module and a Worker module.

b) Robotics: In the field of RL, robotics stands out as one of the most prevalent and impactful applications. A noteworthy contribution to this field is E2E-DVP [316]. This pioneering model represents one of the first end-to-end solutions for robotic control. Its neural network is designed to take raw images as input and generate motor torques as output. [5] curated a substantial real-world dataset and developed a CNN that predicts grasps based on monocular input images. Building upon these foundations, QT-Opt [317] further expands the dataset and model scale, introducing closed-loop control capabilities to enhance robotic control systems. Then Dreamer [51] addresses long-horizon tasks. OpenAI also developed a dexterous robot hand that can solve the Rubik's cube [226], [318].

4) Graph: Graph is ubiquitous in many scenarios, such as social networks, molecule structures, 3D object meshes, etc. Even images and text can be modeled as a 2D grid and a linear graph (path graph), respectively. To process graph-structured data, recurrent graph neural networks [319] were first introduced, which were later optimized by convolutional graph neural networks. The review of graph neural networks (GNN) [320] can be referred to for more in-depth details.

Convolutional graph neural networks can be generally divided into two categories: spectral-based and spatial-based. Spectral-based convolutional GNNs draw inspiration from graph signal processing, which provides theoretical support for the design of the networks. However, spatial-based convolutional GNNs have an advantage in terms of their efficiency and flexibility. Spectral CNN [321] is one of the first convolutional GNNs, but it is not robust to changes in graph structure and

过去几年间，得益于Transformer架构的可扩展性，语言模型的规模实现了显著扩张。这一趋势催生了一系列大型语言模型（LLMs），它们展现出小型模型无法企及的突破性性能与能力。这一演进历程中的里程碑式模型是ChatGPT[223]，它引发了广泛关注，并激励了该领域的多项研究工作，例如GPT-4[278]、PaLM[286]、PaLM-2[287]、LLAMA[288]、LLAMA 2[289]以及ernie 3.5[290]。值得注意的是，LLAMA作为少数开源LLM之一，推动了诸多有趣的发展。“指令调优”技术的引入使得预训练LLM能够高效地微调为遵循指令的模型。该技术通过InstructGPT [291]和FLAN[292]得到了广泛应用。[293]近期指令遵循模型的进展包括采用自我指令的Alpaca[294]，以及利用ShareGPT对话数据的Vicuna[295]。随着大语言模型规模和性能的不断提升，下游任务不再需要精细调参。通过适当的提示，大语言模型无需特定任务训练即可生成准确输出，这一范式被称为*提示工程*，与传统的*预训练-微调*方法截然不同。 DPO[227]提出了一种直接基于人类偏好训练大语言模型的方法，无需显式奖励建模或强化学习，从而简化了以往基于 RLHF 的方法。

*3) 强化学习：*强化学习（RL）旨在通过环境观察获取能够执行最优动作的策略。基于模仿学习或时差（TD）学习等范式，研究者构建了多种视觉-语言-动作模型。机器人策略开发过程中遇到的诸多挑战，均可通过强化学习领域的研究成果得到有效解决。因此，深入研究强化学习方法为提升机器人学习能力提供了重要途径。读者若想深入了解强化学习方法，可参考[296]提供的系统性综述文献。[297]，以及[298]。

*a) 深度强化学习：*深度强化学习的诞生归功于开创性模型的成功，即Deep Q-Network[4]和AlphaGo[224]。深度学习凭借其生成低维表示的能力，对克服强化学习中传统的计算与内存复杂性挑战起到了关键作用。近年来，多种基于价值函数的方法相继涌现：Double DQN（DDQN）[299]解决了 DQN 中的动作价值高估问题；Hindsight Experience Replay（HER）[300]专注于解决奖励稀疏性问题；Batch-Constrained Deep Q-Learning（BCQ）[301]通过约束动作空间来提升非策略学习效果；BEAR[302]则致力于缓解非策略强化学习中由引导误差引发的不稳定性。[303] 引入保守型Q学习（CQL）以解决标准离线强化学习方法对价值的高估问题。

强化学习领域的另一重要范式是策略搜索，涵盖策略梯度法和演员-评论家技术等方法。这些方法旨在解决系统中存在的长期挑战，包括稳定性不足、收敛速度缓慢以及数据利用率低下等问题。引导式策略搜索（GPS）[304]通过由其他控制器引导的重要性采样技术，逐步学习出接近局部最优解的策略。确定性策略梯度法（DPG）[305]深度确定性策略梯度（DDPG）[306]与异步优势演员-评论家（A3C）[307]在提升效率的同时不牺牲稳定性。归一化优势函数（NAF）[308]是Q学习的连续版本，支持经验回放缓放的Q学习。软演员-评论家（Soft AC）[309]利用最大熵强化学习框架来降低样本复杂度并改善收敛性能。信任域策略优化（TRPO）[310]与近端策略优化（PPO）[225]采用信任域方法来稳定策略梯度，其中PPO还结合了截断广义优势估计（GAE）[311]。

除此之外，还存在多种强化学习方法，例如模仿学习和分层强化学习。生成对抗性模仿学习（GAIL）[312]是一种利用生成对抗框架区分专家轨迹与生成轨迹的模仿学习方法。鲁棒对抗性强化学习（RARL）[313]通过引入对抗智能体来提升泛化能力。 RLHF[314]在无需接触奖励函数的情况下利用人类偏好进行学习。联邦网络（FuN）[315]提出了一种包含管理模块和工作模块的分层强化学习架构。

*b) 机器人技术：*在强化学习领域，机器人技术堪称最具影响力的应用之一。其中E2E- DVP[316]堪称该领域的里程碑式贡献。这款开创性模型是机器人控制领域首批端到端解决方案之一，其神经网络设计可直接接收原始图像作为输入，并输出电机扭矩信号。[5]研究人员构建了一个规模庞大的真实世界数据集，并开发出一种基于单目输入图像预测抓取动作的卷积神经网络（CNN）。在此基础上，QT-Opt[317]进一步扩展了数据集和模型规模，引入闭环控制功能以提升机器人控制系统性能；而Dreamer[51]则专注于长期任务处理。OpenAI还研发出一款灵巧的机器人手，能够完成魔方解谜任务[226]、[318]。

*4) 图结构：*图结构在社交网络、分子结构、三维物体网格等多种场景中无处不在。即便是图像和文本，也可以分别建模为二维网格和线性图（路径图）。为处理图结构数据，研究者最初提出了循环图神经网络[319]，随后通过卷积图神经网络对其进行了优化。关于图神经网络（GNN）的详细研究可参考相关综述文献[320]。

卷积图神经网络通常可分为两大类：基于谱的卷积图神经网络和基于空间的卷积图神经网络。基于谱的卷积图神经网络借鉴了图信号处理理论，为网络设计提供了理论支撑；而基于空间的卷积图神经网络则在计算效率和灵活性方面更具优势。谱卷积神经网络[321]作为最早提出的卷积图神经网络之一，但其对图结构变化的适应性存在局限性。

has a high computational cost. ChebNet [322] and GCN [323] significantly reduced the cost: ChebNet used an approximation based on Chebyshev polynomials, and GCN is its first-order approximation. Neural Network for Graphs (NN4G) [324] is the first spatial network. MPNN [325] introduced a general framework of spatial-based networks under which most existing GNNs can be covered. But the drawback of MPNN is that it does not embed graph structure information, which is later solved by GIN [326]. GraphSage [327] improves efficiency by sampling a fixed number of neighbors. Graph Attention Network (GAT) [328] incorporates the attention mechanism.

Besides recurrent and convolutional graph neural networks, there are graph autoencoders [329], [330] and spatial-temporal graph neural networks [331]. Equivariant message passing networks are recently introduced to handle 3D graphs, including E(n)- and SE(3)- Equivariant GNN [332], [333]. Graph Transformer models make use of the power of transformers to process graph data. There are already over 20 such graph Transformer models, such as GROVER [334] and SE(3)-Transformers [335].

a) Graph and vision: Graph structures also exist in some computer vision tasks. Scene graph [336] can be used to express object relationships in most visual inputs. In addition to detecting objects in an image, scene graph generation necessitates the understanding of the relationships between detected objects. For example, a model needs to detect a person and a cup, and then understand that the person is holding the cup, which is the relationship between the two objects. Knowledge graphs often contain visual illustrations, such as WikiData. Those graphs can be helpful in downstream computer vision tasks.

b) Graph and language: Graph structures are ubiquitous in language data [337]. Word-level graphs include dependency graphs, constituency graphs, AMR graphs, etc. Word-level means each node of those graphs corresponds to a word in the original text. These graphs can be used to explicitly represent the syntax or semantics of the raw sentence. Sentence-level graphs can be useful in dialog tracking [338], fact checking [339], etc. Document-level graphs include knowledge graphs [340], citation graphs [341], etc. They can be used in document-level tasks, such as document retrieval, document clustering, etc. Different types of language graphs are often processed using the aforementioned GNNs to facilitate downstream tasks.

C. Background (Extended Version): Vision-Language Models

Comprehensive surveys on VLMs exist, covering early BERT-based VLMs [386], [387] (Section C1), as well as more recent VLMs with contrastive pretraining [388], [389] (Section C2). Given the rapid evolution of this field and the emergence of new VLMs based on large language models, commonly known as large multi-modal models (LMMs), we also compile the latest developments of LMMs (Section C3). To compare the most representative VLMs, we include their specifications in Table VIII.

1) Self-supervised pretraining: The evolution of the Transformer architecture to accommodate various modalities has given rise to robust multi-modal Transformer models. Initial VLMs based on BERT can be broadly categorized into two types: single-stream and multi-stream [386]. Single-stream models employ a single stack of Transformer blocks to process both visual and linguistic inputs, whereas multi-stream models utilize a separate Transformer stack for each modality, with Transformer cross-attention layers exchanging multi-modal information. To enhance alignment among modalities, these models incorporate various pretraining tasks aimed at absorbing knowledge from out-of-domain data. ViLBERT [233] stands as the pioneer in this line of work, featuring a multi-stream Transformer architecture. Text input undergoes standard processing in the language Transformer; image input is first processed using Faster R-CNN, and the output embeddings of all objects are then passed into the vision stream. The two Transformer outputs—language embeddings and vision embeddings—are combined using a novel co-attention transformer layer. VL-BERT [344] adopts a single-stream multi-modal Transformer, simply concatenating vision and language tokens into a single input sequence. VideoBERT [390] adapts mulitmodal Transformer models to video inputs. UNITER [345] proposes the word-region alignment loss to explicitly align word with image regions. ViLT [346] uses ViT-style [219] image patch projection to embed images, deviating from previous region or grid features.

SimVLM [347] opts for a streamlined approach, relying solely on a single prefix language modeling objective to reduce training costs. VLMo [391] and BEiT-3 [351] both introduce mixture-of-modality-experts Transformers to effectively handle multi-modal inputs.

2) Contrastive pretraining: Vision-language pretraining in the initial series of BERT-based VLMs has evolved, with refinements such as curating larger-scale pretraining datasets, leveraging multi-modal contrastive learning, and exploring specialized multi-modal architectures. CLIP [22] is one of the earliest attempts in vision-language contrastive pretraining. By contrastive pretraining on a large-scale image-text pair dataset, CLIP exhibits the capability to be transferred to downstream tasks in a zero-shot fashion. In the same line of work, other few-/zero-shot learners have emerged. FILIP [352] concentrates on finer-grained multi-modal interactions with a token-wise contrastive objective. ALIGN [353] focuses on learning from noisy datasets collected without filtering and post-processing. "Locked-image Tuning" (LiT) [392] posits that only training the text model while freezing the image model yields the best results on new tasks. In Frozen [393], the pretrained language model is frozen and a vision encoder is trained to produce image embeddings as a part of language model prompts, exemplifying an instance of prompt tuning.

Unlike the two-tower frameworks of CLIP, FILIP, and ALIGN, which solely train unimodal encoders (image encoder and text encoder), ALBEF [355] additionally incorporates training a multimodal encoder on top of the unimodal encoders, with FLAVA [356] sharing a similar idea. In contrast to contrastive pretraining methods, CoCa [394] seeks to amalgamate the strengths of CLIP's contrastive learning and SimVLM's generative objective. Florence [350] generalizes representations from coarse, scene-level to fine, object-level,

其计算成本较高。ChebNet[322]和 GCN[323]显著降低了计算成本：ChebNet采用了基于切比雪夫多项式的近似方法，而 GCN 则是其一阶近似。Graphs Neural Network（NN4G）[324]是首个空间网络模型。 MPNN[325]提出了一个通用的空间网络框架，可涵盖大多数现有的图神经网络（GNN）。但 MPNN 的缺点在于未嵌入图结构信息，这一缺陷后来由GIN[326]解决。GraphSage[327]通过采样固定数量的邻域节点提升了效率；Graph Attention Network（GAT）[328]则引入了注意力机制。

除了循环图神经网络和卷积图神经网络之外，还有图自编码器[329]。[330] 以及时空图神经网络[331]。等变消息传递网络最近被引入以处理三维图结构，包括E(n)和SE(3)等变图神经网络[332]，[333] 图Transformer模型利用Transformer的强大能力来处理图数据。目前已存在超过20种此类图 Transformer 模型，例如 Grover[334] 和 SE(3)-Transformer[335]。

*a) 图结构与计算机视觉：*图结构在某些计算机视觉任务中同样具有重要应用价值。场景图[336]能够有效表征大多数视觉输入中的物体关系。除了实现图像中的物体检测功能外，场景图生成还需要理解被检测物体之间的关联关系。例如，模型在识别出人物与杯子后，还需理解“人物持杯”这一物体间的关系。知识图谱通常包含维基数据等可视化呈现形式，这类图谱结构对下游计算机视觉任务具有重要辅助作用。

*b) 图与语言：*图结构在语言数据中无处不在[337]。词级图包括依存图、构式图、AMR图等。词级意味着这些图中的每个节点都对应原始文本中的一个Word。这些图可用于显式表示原始句子的语法或语义。句级图在对话追踪[338]、事实核查[339]等领域具有应用价值。文档级图包括知识图谱[340]、引文图[341]等，可用于文档检索、文档聚类等文档级任务。不同类型的语言图通常采用上述图神经网络（GNN）进行处理，以支持下游任务。

C. 背景（扩展版）：视觉-语言模型

关于虚拟语言模型（VLMs）的综合调查研究已有文献，涵盖早期基于BERT的VLMs[386]。[387] （第C1节），以及采用对比预训练的最新视觉语言模型[388]，[389]（第C2节）。鉴于该领域快速发展及基于大型语言模型（通常称为大型多模态模型，LMMs）的新VLMs的涌现，我们还汇总了LMMs的最新进展（第C3节）。为比较最具代表性的VLMs，我们在表VIII中列出了其技术参数。

*1) 自监督预训练：*Transformer架构为适应多种模态而不断演进，从而催生出稳健的多模态Transformer模型。基于BERT的早期视觉语言模型（VLM）大致可分为两类：单流模型和多流模型[386]。单流模型采用单一Transformer模块堆栈同时处理视觉和语言输入，而多流模型则为每种模态使用独立的Transformer堆栈，并通过Transformer交叉注意力层交换多模态信息。为增强模态间对齐效果，这些模型融入了多种预训练任务以吸收领域外数据的知识。ViLBERT[233]是该领域的先驱者，采用多流Transformer架构：文本输入先在语言Transformer中进行标准处理；图像输入首先通过Faster R-CNN处理，随后将所有对象的输出嵌入传递至视觉流；语言嵌入与视觉嵌入这两个Transformer输出通过创新的协同注意力Transformer层进行融合。VL-BERT[344]采用单流多模态Transformer架构，通过简单拼接视觉与语言标记形成单一输入序列。VideoBERT[390]将多模态Transformer模型适配视频输入场景。Uniter[345]提出词-区域对齐损失函数，实现词义与图像区域的显式对齐。ViLT[346]采用ViTstyle[219]图像块投影技术进行图像嵌入，突破了传统区域或网格特征的局限。

SimVLM[347]采用精简方案，仅依赖单一前缀语言建模目标以降低训练成本。VLMo[391]与BEiT-3[351]均引入模态混合专家Transformer架构，有效处理多模态输入。

*2) 对比预训练：*基于BERT的视觉语言模型（VLM）初期系列研究中，视觉-语言预训练技术持续演进，主要体现在三大改进方向：构建更大规模预训练数据集、运用多模态对比学习方法以及探索专用多模态架构。CLIP[22]是视觉-语言对比预训练领域的早期尝试之一，通过在大规模图像-文本数据集上进行对比预训练，该模型展现出零样本迁移至下游任务的能力。同期研究中还涌现出其他零样本/少量样本学习器： FILIP[352]专注于通过标记级对比目标实现细粒度多模态交互；align[353]则聚焦于未经过滤和后处理的噪声数据集学习；而“锁定图像调优”（LiT）[392]理论认为，仅训练文本模型并冻结图像模型才能在新任务中获得最佳效果。在《Frozen[393]》中，预训练语言模型被冻结，并训练视觉编码器生成图像嵌入作为语言模型提示的一部分，这体现了提示调优的一个实例。

与CLIP、 FILIP 和align这两种仅训练单模态编码器（图像编码器和文本编码器）的双塔框架不同， ALBEF[355]在单模态编码器基础上额外引入了多模态编码器训练机制，flava[356]也采用了类似思路。相较于对比预训练方法，CoCa[394]致力于融合CLIP对比学习的优势与SimVLM生成目标的特性。Florence[350]实现了从粗粒度场景级表征到细粒度物体级表征的泛化能力，

Table VIII: Vision-language models. In the "Objective" column: "MLM": masked language modeling. "MVM": masked vision modeling, reconstructing masked image regions. "VLM": binary classification of whether vision and language inputs are a match. "LM": autoregressive language modeling. "VLCL": vision-language contrastive learning. We only include representative multi-modal datasets due to limited space. "MM" includes multi-modal tasks such as visual question answering, image captioning, and vision-language retrieval. "Vision" represents computer vision tasks, like image classification. "Language" represents natural language processing tasks.

		Vision Encoder		Language Encoder					
	Model	Name	Params	Name	Params	VL-Fusion	Objectives	Datasets	Tasks
Self-supervised	ViLBERT [233]	Faster R-CNN [246], [251]	44M	Dual-stream BERT-base [26]	221M	Dual-stream	MLM, MVM, VLM	COCO, VG	MM
	LXMERT [342]	Faster R-CNN [246], [251]	44M	Dual-stream BERT-base [26]	183M	Dual-stream	MLM, MVM, VLM, VQA	COCO, VG, VQA, GQA, VGQA	MM
	VisualBERT [343]	Faster R-CNN [246], [251]	60M	BERT-base [26]	110M	Single-stream	MLM, VLM	COCO	MM
	VL-BERT [344]	Faster R-CNN [246], [251]	44M	BERT-base [26]	110M	Single-stream	MLM, MVM	CC	MM
	UNITER [345]	Faster R-CNN [246], [251]	44M	BERT-base/ BERT-large [26]	86M/ 303M	Single-stream	MLM, VLM, MVM, WRA	COCO, VG, SBU, CC	MM
	ViLT [346]	Linear projection [219]	2.4M	BERT-base [26]	85M	Single-stream	MLM, VLM	COCO, VG, SBU, CC	MM
	SimVLM [347]	ViT/CoAtNet-huge (shared encoder) [348]	632M	Shared encoder	632M	Single-stream	PrefixLM	ALIGN dataset	MM
	GIT [349]	Florence [350]	637M	Transformer [3]	60M	Single-stream	LM	COCO, VG, SBU, CC, etc.	MM
	BEiT-3 [351]	V-FFN (+Shared Attn)	692M (+317M)	L-FFN (+Shared Attn)	692M (+317M)	Modality experts	MLM, VLM	COCO, VG, SBU, CC	MM, Vision
Contrastive	CLIP [22]	ViT [219]	428M	GPT-2 [276]	63M	Two-tower	VLCL	WTI	Vision
	FILIP [352]	ViT-L/14 [219]	428M	GPT [276]	117M	Two-tower	VLCL	FILIP300M (Self-collect)	MM, Vision
	ALIGN [353]	EfficientNet-L2 [354]	480M	BERT-large [26]	336M	Two-tower	VLCL	ALIGN dataset (Self-collect)	MM, Vision
	ALBEF [355]	ViT-B/16 [219]	87M	BERT-base [26]	85M	Dual-stream	MLM, VLM, VLCL	COCO, VG, CC, SBU	MM
	FLAVA [356]	ViT-B/16 [219]	87M	RoBERTa-base [279]	125M	Dual-stream	MLM, MVM, VLM, VLCL	COCO, VG, CC, SBU, etc. (PMD)	MM, Vision, Language
	Florence [350]	Hierarchical Vision Transformers [357], [358]	637M	RoBERTa [279]	125M	Two-tower	VLCL	FLD-900M (Self-collect)	Vision
Large Multi-modal Model	Flamingo [234]	NFNet-F6 [359]	438M	Chinchilla [360]	70B	Dual-stream	LM	M3W, ALIGN dataset, LTIP, VTP	MM
	BLIP-2 [6]	CLIP ViT-L/14 [22], EVA ViT-G/14 [361] + Q-Former	428M, 1B	OPT [283] Flan-T5 [293]	6.7B 3B/11B	Single-stream	BLIP, LM	COCO, VG, CC, SBU, LAION	MM
	PaLI [362]	ViT-e [362]	4B	mT5-XXL [363]	13B	Single-stream	Mixed	WebLI, etc	MM
	PaLI-X [364]	ViT-22B [365]	22B	UL2 [366]	32B	Single-stream	Mixed	WebLI, etc	MM
	LLaMA-Adapter [367]	CLIP ViT-B/16 [22]	87M	LLaMA [288]	7B	Single-stream	LM	Self-instruct	Instruction-following
	LLaMA-Adapter-V2 [368]	CLIP ViT-L/14 [22]	428M	LLaMA [288]	7B	Single-stream	LM	GPT-4-LLM, COCO, ShareGPT	Instruction-following
	Kosmos-1 [369], Kosmos-2 [370]	CLIP ViT-L/14 [22]	428M	Magneto [371]	1.3B	Single-stream	LM	LAION, COYO, CC; Unnatural Instructions, FLANv2	Instruction-following (Kosmos-2 w/ grounding, referring)
	InstructBLIP [372]	EVA ViT-G/14 [361]	1B	Flan-T5 [293] Vicuna [295]	3B/11B 7B/13B	Single-stream	BLIP, LM	COCO, VQA, LLaVA-Instruct-150K, etc. (26 datasets)	Instruction-following
	LLaVA [7]	CLIP ViT-L/14 [22]	428M	LLaMA [288]	13B	Single-stream	LM	CC, (FT: GPT-assisted Visual Instruction Data Generation)	Instruction-following
	MiniGPT-4 [373]	EVA ViT-G/14 [361] + Q-Former [6]	1B	Vicuna [295] LLaMA2 [289]	7B/13B 7B	Single-stream	LM	CC, SBU, LAION (FT: SC)	Instruction-following
	Video-LLaMA [374]	EVA ViT-G/14 [361] + Q-Former [6]	1B	LLaMA [288]	7B/13B	Single-stream	BLIP, LM	CC595k	Instruction-following
	PandaGPT [375]	ImageBind ViT-H [209]	632M	Vicuna [295]	13B	Single-stream	LM	(FT: LLaVA data, MiniGPT-4 data)	Instruction-following
	VideoChat [376]	EVA ViT-G/14 [361] + Q-Former [6]	1B	StableVicuna [377]	13B	Single-stream	LM	COCO, VG, CC, SBU (FT: SC, MiniGPT-4, LLaVA data)	Instruction-following
	ChatSpot [378]	CLIP ViT-L/14 [22]	428M	Vicuna [295]	7B	Single-stream	LM	MGVLID, RegionChat	Instruction-following, Vision
	mPLUG-Owl [379], mPLUG-Owl2 [380]	CLIP ViT-L/14 [22] + Visual Abstractor	428M	LLaMA [288]	7B	Single-stream	LM	LAION, COYO, CC, COCO (FT: Alpaca, Vicuna, Baize [381] data)	Instruction-following
	Visual ChatGPT [382]	(22 different models)	-	ChatGPT [223]	-	Prompt Manager	-	Add image understanding and generation to ChatGPT	
	X-LLM [383]	ViT-G [384] + Q-Former [6] + Adapter	1.8B	ChatGLM [385]	6B	Single-stream	Three-stage training	MiniGPT-4 data, AISHELL-2, ActivityNet, VSDial-CN (SC)	Instruction-following

表VIII：视觉-语言模型。在「目标」列中：「 MLM 」：掩码语言建模。「 MVM 」：掩码视觉建模（重建被遮挡图像区域）。「 VLM 」：判断视觉输入与语言输入是否匹配的二分类任务。「LM」：自回归语言建模。「 VLCL 」：视觉-语言对比学习。由于篇幅限制，仅选取代表性多模态数据集。「MM」涵盖视觉问答、图像描述生成及视觉-语言检索等多模态任务。「Vision」代表计算机视觉任务（如图像分类）。「Language」代表自然语言处理任务。

	模型	视觉编码器 姓名	参数	语言编码器名称	参数	VL-Fusion	目的	数据集	任务
自监督的	ViLBERT [233]	更快的R-CNN[246]，44M [251]		双流BERT基础模型 [26]	221M	双流	MLM, MVM, VLM	COCO，VG	MM
	LXMERT [342]	更快的R-CNN[246]，[251]	44M	双流BERT基础模型 [26]	183M	双流	MLM, MVM, VLM, VQA	COCO、VG、VQA，GQA，VGQA	MM
	VisualBERT [343]	更快的 R-CNN[246]、[251]	60M	BERT-base [26]	110M	单流	MLM, VLM	COCO	MM
	VL-BERT [344]	更快的 R-CNN[246]、[251]	44M	BERT-base [26]	110M	单流	MLM, MVM	CC	MM
	Uniter[345]	更快的R-CNN[246]，[251]	44M	BERT-base/ BERT-large [26]	86M/ 303M	单流	MLM, VLM, MVM, WRA	COCO、VG、SBU，CC	MM
	ViLT [346]	线性投影[219]	2.4M	BERT-base [26]	85M	单流	MLM, VLM	COCO、VG、SBU，CC	MM
	SimVLM [347]	ViT/CoAtNet-huge（共享编码器）[348]	632M	共享编码器	632M	单流	前缀LM	对齐数据集	MM
	GIT [349]	佛罗伦萨[350]	637M	变压器[3]	60M	单流	LM	COCO、VG、SBU、CC等	MM
	BEiT-3 [351]	V-FFN（共同收件人）	692M (+317M)	L-FFN（共同收件人）	692M (+317M)	模态专家	MLM, VLM	COCO、VG、SBU，CC	MM，愿景
对比的	CLIP [22]	ViT [219]	428M	GPT-2 [276]	63M	双塔式	VLCL	西得克萨斯轻油	视觉
	FILIP [352]	ViT-L/14 [219]	428M	GPT [276]	117M	双塔式	VLCL	FILIP300M（自采样）	MM，愿景
	对齐[353]	EfficientNet-L2 [354]	480M	BERT-large [26]	336M	双塔式	VLCL	对齐数据集（自行收集）	MM，愿景
	ALBEF [355]	ViT-B/16 [219]	87M	BERT-base [26]	85M	双流	MLM, VLM, VLCL	COCO、VG、CC、SBU	MM
	黄酮[356]	ViT-B/16 [219]	87M	罗伯塔-碱基 [279]	125M	双流	MLM, MVM, VLM, VLCL	COCO、VG、CC、SBU 等 胸大肌缺损	MM，视力 语言
	佛罗伦萨[350]	层次视觉变换器[357][358]	637M	Roberta[279]125M		双塔式	VLCL	FLD-900M（自采样）	视觉
大型多模态模型	火烈鸟[234]	NFNet-F6 [359]	438M	鼠兔[360]	70B	双流	LM	M3W、对齐数据集、LTIP、VTP	MM
	BLIP-2 [6]	CLIP ViT-L/14[22] EVA ViT-G/14 [361] + Q-Former	428M,1B	OPT [283] Flan-T5 [293]	6.7B 3B/ 11B	单流	BLIP，LM	COCO、VG、CC、SBU，LAION	MM
	PaLI [362]	ViT-e [362]	4B	mT5-XXL [363]	13B	单流	混合型	WebLI等	MM
	PaLI-X [364]	ViT-22B [365]	22B	UL2 [366]	32B	单流	混合型	WebLI等	MM
	LLaMA适配器 [367]	CLIP ViT-B/16 [22]	87M	LLaMA [288]	7B	单流	LM	自我指导	指令 接下来的
	LLAMA-Adapter-V2 [368]	CLIP ViT-L/14 [22]	428M	LLaMA [288]	7B	单流	LM	GPT-4-LLM、COCO、ShareGPT	指令 接下来的
	Kosmos-1[369]，Kosmos-2 [370]	CLIP ViT-L/14 [22]	428M	马格南托[371]	1.3B	单流	LM	LAION、COYO、CC；非自然指令，FLANv2	指令遵循（Kosmos-2卫星接地状态，参照）
	InstructBLIP [372]	EVA ViT-G/14 [361]	1B	Flan-T5 [293] 维库尼亚[295]	3B/11B 7B/13B	单流	BLIP，LM	COCO、VQA、LLaVA-Instruct-150K等（26个数据集）	指令 接下来的
	LLaVA [7]	CLIP ViT-L/14 [22]	428M	LLaMA [288]	13B	单流	LM	CC（FT：基于GPT辅助的视觉指令数据生成）	指令 接下来的
	MiniGPT-4 [373]	EVA ViT-G/14 [361] + Q-Former [6]	1B	Vicuna [295] LLaMA2 [289]	7B/13B 7B	单流	LM	CC, SBU, LAION (FT: SC)	指令 接下来的
	Video-LLaMA [374]	EVA ViT-G/14 [361] + Q-Former [6]	1B	LLaMA [288]	7B/13B	单流	BLIP，LM	CC595k	指令 接下来的
	PandaGPT [375]	ImageBind ViT-H [209]	632M	Vicuna [295]	13B	单流	LM	（来源：LLaVA 数据、MiniGPT-4 数据）	指令 接下来的
	VideoChat [376]	EVA ViT-G/14 [361] + Q-Former [6]	1B	StableVicuna [377]	13B	单流	LM	COCO、VG、CC、SBU（FT：SC、MiniGPT-4、LLaVA数据集）	指令 接下来的
	ChatSpot [378]	CLIP ViT-L/14 [22]	428M	维库尼亚[295]7B		单流 LM		MGVLID, RegionChat	指令 随访，视觉
	mPLUG-Owl[379]，mPLUG-Owl2 [380]	CLIP ViT-L/14 [22] + 可视化摘要器	428M	LLaMA [288]	7B	单流	LM	LAION、COYO、CC、COCO（FT：羊驼、骆马、巴伊泽[381]数据）	指令 接下来的
	Visual ChatGPT [382]	（22种不同型号）	-	ChatGPT [223]	-	提示管理器	-	在ChatGPT中添加图像理解与生成功能	
	X-LLM [383]	ViT-G[384]+ Q-Former[6]+ 适配器	1.8B	ChatGLM [385]	6B	单流	三阶段培训	MiniGPT-4 数据、aishell-2、ActivityNet、VSDial-CN（SC）	指令 接下来的

expands from images to videos, and encompasses modalities beyond RGB channels. OFA [395] draws inspiration from T5 [285] and proposes unifying diverse unimodal and multi-modal tasks under a sequence-to-sequence learning framework.

3) Large multi-modal model: Large language models (LLMs) encapsulate extensive knowledge, and efforts have been made to transfer this knowledge to multimodal tasks. Given the resource-intensive nature of fine-tuning entire LLMs on multi-modal tasks due to their large size, various techniques have been explored to effectively connect frozen LLMs with vision encoders, enabling the combined model to acquire multi-modal capabilities. Flamingo [234] connects a pretrained vision encoder, NFNet [359], and a large language model, Chinchilla [360], by inserting trainable gated cross-attention layers while keeping the rest of the model frozen. BLIP-2 [6] introduces Q-Former, bootstrapping vision-language representation learning first from a frozen CLIP ViT [22] and then from a frozen LLM, OPT [283] or Flan-T5 [293]. PaLI [362] and PaLI-X [364] investigate the advantages of jointly scaling up the vision and language components using large-scale multilingual image-text data.

Similar to developments in NLP, instruction-following has become a crucial aspect of VLMs, prompting the exploration of various multimodal instruction-tuning methods. LLaMA-Adapter [367], [368] employs a parameter-efficient finetuning (PEFT) technique, enabling LLaMA [288] to process visual inputs. Kosmos-1 [369] introduces a less restrictive input format that accommodates interleaved image and text. Its Magneto LLM [371] serves as a "general-purpose interface" for docking with perception modules [22]. Kosmos-2 [370] adds additional grounding and referring capabilities. InstructBLIP [372] achieves instruction-following using an instruction-aware Q-Former based on BLIP-2's Q-Former [6]. Comparable to Kosmos-2, ChatSpot [378] excels at following precise referring instructions, utilizing CLIP ViT [22] and Vicuna [295]. X-LLM [383] converts multi-modality data into LLM inputs using X2L interfaces and treats them as foreign languages, where the X2L interface is inspired by the Q-Former from BLIP-2 [6]. mPLUG-Owl [379], [380] introduces a two-stage training paradigm that establishes a connection between a pretrained LLM with a visual encoder and a visual abstractor, thereby endowing LLMs with multi-modality abilities. Visual ChatGPT [382] proposes a prompt manager that manages the interaction between ChatGPT and 22 visual foundation models, with the goal of equipping ChatGPT with the capability to understand and generate images.

Rather than employing intricate mechanisms to connect components for different modalities, both LLaVA [7] and MiniGPT-4 [373] propose connecting vision encoders with LLMs through a single linear layer. LLaVA adopts a two-stage instruction-tuning approach, pretraining the CLIP ViT vision encoder [22] in the first stage and finetuning the linear layer and the LLaMA LLM [288] in the second stage. In contrast, MiniGPT-4 freezes both the vision encoder (BLIP-2's ViT + Q-Former [6]) and the Vicuna LLM [295], only training the linear layer. Following MiniGPT-4, Video-LLaMA [374] handles videos by incorporating two branches for video and audio, each comprising a video/audio encoder and a BLIP-2-style Q-Former [6]. PandaGPT [375] leverages ImageBind [209] to encode vision/text/audio/depth/thermal/IMU data, feeding them to the Vicuna model [295] also through a linear layer. PandaGPT diverges from MiniGPT-4 by using LoRA [396] to train Vicuna alongside the linear layer.

D. Supplementary Related Work

Due to page limitations, we were unable to provide citations for all the mentioned models in the main text. Here is a list of supplementary related works:

Computer Vision:

- Mask R-CNN [247]
- YOLOv8 [248]
- EfficientNet [243]
- UNet [254]
- VQ-GAN [397]
- MoVQ [398]
- SigLIP [399]
- ViLD [400]
- MDETR [401]
- HLSM object detector [402]
- ResNet [218]
- ViT, ViT-B, ViT-L [25], [219]
- MAE [403]
- ViT-22B [365]
- EVA ViT-G/14 [361]
- OWL-ViT [404]
- ConvNeXt [405]
- SAM [220]
- Imagen video [406]
- C-ViViT [407]
- OSRT [408]
- PointNet [409]
- PointNet++ [410]
- 3D-CLR [411]
- ConceptFusion [412]

Natural Language Processing, LLM, VLM:

- Long Short-Term Memory (LSTM) [221]
- Transformer [3]
- Sentence-BERT / Sent.-BERT [413]
- DistilBERT [414]
- T5-small, T5-base, T5-XXL [285]
- LLaMA [288]
- LLaMA 2 [289]
- Vicuna-V1.5 13B [295]
- MPT [415]
- FLAN [292]
- Flamingo [234], [416]
- Pythia [417]
- Phi-2 [418]
- GPT-NeoX [419]
- Codex [420]
- InstructGPT [375]
- ChatGPT [223]
- GPT-2 [276]

该方法从图像扩展到视频，并涵盖超越RGB通道的多种模态。OFA[395]借鉴了T5[285]的思路，提出将各类单模态与多模态任务统一纳入序列到序列的学习框架之中。

*3) 大型多模态模型：*大型语言模型（LLMs）蕴含了丰富的知识，学界已致力于将这些知识迁移至多模态任务。鉴于大型LLMs在多模态任务上微调时资源消耗巨大，研究人员探索了多种技术，将冻结的LLMs与视觉编码器有效连接，使组合模型能够获得多模态能力。Flamingo[234]通过插入可训练的门控交叉注意力层，将预训练的视觉编码器NFNet[359]与大型语言模型Chinchilla[360]相连接，同时保持模型其余部分冻结。BLIP-2[6]提出了Q-Former方法，首先从冻结的CLIP ViT[22]开始进行视觉-语言表征学习，随后分别从冻结的LLM OPT[283]或Flan-T5[293]学习。PaLI[362]和PaLI-X[364]则研究了利用大规模多语言图像-文本数据联合扩展视觉与语言组件的优势。

与自然语言处理（NLP）领域的进展类似，指令遵循已成为虚拟语言模型（VLM）的关键组成部分，这推动了多种多模态指令调优方法的探索。LLaMAAdapter[367]便是其中一例。[368] 采用参数高效微调（PEFT）技术，使LLaMA[288]能够处理视觉输入。Kosmos-1[369]引入了更宽松的输入格式，支持图像与文本的交错输入。其Magneto LLM[371]作为“通用接口”用于对接感知模块[22]。Kosmos-2[370]增加了额外的定位与指代能力。InstructBLIP[372]基于BLIP-2的Q-Former[6]开发了指令感知型Q-Former，实现指令遵循功能。与Kosmos-2类似，ChatSpot [378]擅长遵循精确指代指令，采用CLIP ViT[22]和Vicuna [295]技术。X-LLM[383]通过X2L接口将多模态数据转换为LLM输入并视为外语处理，其中X2L接口灵感源自BLIP-2的QFormer[6]。mPLUG-Owl[379][380] 该方法提出了一种两阶段训练范式，通过建立预训练大语言模型（LLM）与视觉编码器及视觉抽象器之间的连接，从而赋予LLMs多模态处理能力。Visual ChatGPT[382]设计了一种提示管理器，用于协调ChatGPT与22个视觉基础模型之间的交互，旨在使ChatGPT具备理解和生成图像的能力。

与采用复杂机制连接不同模态组件的方法不同，LLaVA[7]和MiniGPT-4[373]均提出通过单个线性层将视觉编码器与大语言模型连接。LLaVA采用两阶段指令调优策略：第一阶段预训练CLIP ViT视觉编码器[22]，第二阶段微调线性层与LLaMA大语言模型[288]。相比之下，MiniGPT-4同时冻结视觉编码器（BLIP-2的ViT+Q-Former [6]）和Vicuna大语言模型[295]，仅训练线性层。继MiniGPT-4之后，Video-LLaMA[374]通过整合视频与音频两个分支处理视频，每个分支包含视频/音频编码器及BLIP-2式Q-Former[6]。PandaGPT[375]利用ImageBind[209]对视觉/文本/音频/深度/热成像/IMU数据进行编码，并通过线性层将这些数据输入Vicuna模型[295]。PandaGPT与MiniGPT-4的区别在于，它采用LoRA[396]在训练线性层的同时同步训练Vicuna模型。

D. 补充相关工作

由于篇幅限制，我们未能在正文中为所有提及的模型提供参考文献。以下是相关的补充研究列表：

计算机视觉：
- Mask R-CNN [247]
- YOLOv8 [248]
- EfficientNet [243]
- UNet [254]
- VQ-GAN [397]
- MoVQ [398]
- SigLIP [399]
- ViLD [400]
- MDETR [401]
- HLSM 目标检测器[402]
- ResNet [218]
- ViT, ViT-B, ViT-L [25], [219]
- MAE [403]
- ViT-22B [365]
- EVA ViT-G/14 [361]
- OWL-ViT [404]
- ConvNeXt [405]
- SAM [220]
- 视频图像[406]
- C-ViViT [407]
- OSRT [408]
- PointNet [409]
- PointNet++ [410]
- 3D-CLR [411]
- ConceptFusion [412]

自然语言处理、大语言模型（LLM）、 VLM：
- 长短期记忆（LSTM）[221]
- 变压器[3]
- Sentence-BERT / Sent.-BERT [413]
- DistilBERT [414]
- T5-small、T5-base、T5- XXL[285]
- LLaMA [288]
- LLaMA 2 [289]
- Vicuna-V1.5 13B [295]
- MPT [415]
- FLAN [292]
- 火烈鸟[234]，[416]
- 皮提娅[417]
- Phi-2 [418]
- GPT-NeoX [419]
- 法典[420]
- InstructGPT [375]
- ChatGPT [223]
- GPT-2 [276]

- GPT-3 [277]
- GPT-4 [278]
- PaLI-X [364]
- Prismatic-7B VLM [421]
- PaliGemma [422]
- PaliGemma 2 [423]
- Chameleon [424]
- Emu3 [425]
- LWM-Chat-1M [426]
- LLaMA-Adapter [367]

Action Modules:

- LingUNet [427]
- Latent Motor Plans (LMP) [428]
- STORM [429]
- PerceiverIO [430]
- DD-PPO [431]
- ScaleDP [432]
- Symbol-tuning [433]
- DiT [434]

Others:

- DDPM [435]
- DDIM [436]
- Ravens [437]
- RLBench [178]
- Something V2: Something-something V2 [438]

Abbreviations:

- TFM: transformer
- MCIL: multicontext imitation learning
- MLE: maximum likelihood estimation
- MPC: model predictive control
- CVAE: conditional variational autoencoder
- EDR: Everyday Robots

E. Additional VLA-Related Work

We include additional VLA-related work that could not be included in the main text due to the page limit.

1) Components of VLA: Pretraining: Value-Implicit Pretraining (VIP) [24] capitalizes on the temporal sequences present in videos, distinguishing itself from R3M [23] by attracting both the initial and target frames while simultaneously repelling successive frames. This objective aims to capture long-range temporal relationships and uphold local smoothness. While their own experiments demonstrate superior performance compared to R3M in specific tasks, subsequent research presents conflicting findings through more comprehensive evaluations [31].

SpawnNet [439] utilizes a two-stream architecture incorporating adapter layers to fuse features from both a pretrained vision encoder and features learned from scratch. This innovative approach eliminates the necessity of training the pretrained vision encoders while surpassing the performance of parameter-efficient finetuning (PEFT) methods, as evidenced by experimental results in robot manipulation tasks.

Holo-Dex [440] proposes learning low-dimensional representations for visual policies using self-supervised learning (SSL). In Dobb·E [441], SSL is also explored for 3D visual inputs, through a method termed “Home Pretrained Representations,” using data collected from 22 homes in NYC. The authors extend SSL-pretrained representations beyond visual inputs. AuRL [43] introduces learning dynamic behaviors from audio, an often-overlooked yet important information source. Additionally, visual inputs were found to be insufficient for multi-fingered robotic hands, leading to T-Dex [211], a tactile-based dexterous policy.

2) Components of VLA: World Models: Masked World Model (MWM) [442] innovates by modifying the vision encoder of DreamerV2 to a hybrid composition of convolutional neural network and vision Transformer. In training this novel Convolutional-ViT vision encoder, MWM draws inspiration from the approach proposed in MAE [403]. Through the incorporation of an auxiliary reward prediction loss, the resulting learned latent dynamics model exhibits a notable performance improvement across various visual robotic tasks.

SWIM [443] advocates for the use of human videos in training a world model due to the availability of large-scale human-centric data. However, acknowledging the substantial gap between human and robot data, SWIM addresses this disparity by grounding actions in visual affordance maps. This approach involves inferring target poses based on the affordance maps, facilitating an effective transfer of knowledge from human data to enhance robot control.

Iso-Dream [444] introduces two key enhancements to the Dreamer framework. The first enhancement focuses on optimizing inverse dynamics by decoupling controllable and noncontrollable dynamics. The second enhancement involves optimizing agent behavior based on the decoupled latent imaginations. This approach is particularly advantageous, as the noncontrollable state transition branch can be rolled out independently of actions, yielding benefits for long-horizon decision-making processes.

Transformer-based World Model (TWM) [56] shares the same motivation as Dreamer but adopts a different approach by constructing a world model based on the Transformer-XL architecture [445]. This Transformer-based world model trains a model-free agent in latent imagination by generating new trajectories. Additionally, TWM suggests a modification to the balanced KL divergence loss from DreamerV2 and introduces a novel thresholded entropy loss tailored for the advantage actor-critic framework.

SceneScript [446] introduces a set of structured language commands that can define an entire scene, specifying the layout and objects. This language-based scene representation distinguishes itself from previous methods—such as meshes, voxel grids, point clouds, or radiance fields—by generating scenes in an autoregressive, token-based fashion. It achieves competitive results against state-of-the-art methods in layout estimation and object detection.

3) Components of VLA: Imitation Learning: BeT [447], an imitation learning variant of the Trajectory Transformer, addresses the noisy and multimodal nature of non-expert human demonstration data by using k-means-based action discretization coupled with continuous action correction. C-BeT [448] extends BeT by adding a target frame or demonstration

- GPT-3 [277]
- GPT-4 [278]
- PaLI-X [364]
- Prismatic-7B VLM [421]
- PaliGemma [422]
- PaliGemma 2 [423]
- 变色龙[424]
- Emu3 [425]
- LWM-Chat-1M [426]
- LLaMA适配器[367]

行动模块：
- LingUNet [427]
- 潜在运动计划（LMP）[428]
- 风暴[429]
- PerceiverIO [430]
- DD-PPO [431]
- ScaleDP [432]
- 符号调谐[433]
- DiT[434] 其

他：
- DDPM [435]
- DDIM [436]
- 乌鸦[437]
- RLBench [178]
- Something V2：Something-something V2[438]

缩写：
- TFM：变压器
- MCIL：多上下文模仿学习
- MLE：最大似然估计
- MPC：模型预测控制
- CVAE：条件变分自编码器
- EDR：日常机器人

E. 其他VLA相关研究

我们补充了因篇幅限制未能纳入正文的其他与甚大阵列望远镜（VLA）相关研究工作。

*1) 视觉语言模型组件：预训练：*价值隐式预训练（VIP）[24]充分利用视频中的时间序列特征，与R3M[23]的区别在于：它既能同时吸引初始帧和目标帧，又能排斥后续帧。该方法旨在捕捉长时序关系并保持局部平滑性。尽管其自身实验在特定任务中展现出优于R3M的性能，但后续研究通过更全面的评估结果却呈现了相互矛盾的结论[31]。

SpawnNet[439]采用双流架构设计，通过适配器层融合预训练视觉编码器与从头学习特征。该创新方案不仅省去了预训练视觉编码器的训练步骤，其性能表现更超越参数高效微调（PEFT）方法，机器人操作任务的实验结果充分印证了这一优势。

Holo-Dex[440]提出利用自监督学习（SSL）为视觉策略学习低维表示。在Dobb·E[441]中，研究者通过一种名为“家庭预训练表示”的方法，利用从纽约市22户家庭收集的数据，探索了SSL在三维视觉输入中的应用。作者将SSL预训练表示的应用范围扩展至视觉输入之外。AuRL[43]则介绍了从音频中学习动态行为的方法——音频是常被忽视但至关重要的信息源。此外，研究发现视觉输入对于多指机械手而言并不充分，由此催生了基于触觉的灵巧策略T-Dex[211]。

*2) 视觉语言模型组件：世界模型：*掩码世界模型（MWM）[442]通过将DreamerV2的视觉编码器改进为卷积神经网络与视觉Transformer的混合架构实现了创新突破。在训练这款新型卷积-ViT视觉编码器时， MWM 借鉴了MAE模型[403]提出的优化方案。通过引入辅助奖励预测损失机制，最终构建的潜在动力学模型在各类视觉机器人任务中展现出显著的性能提升。

SWIM[443]倡导在训练世界模型时使用人类视频，这得益于大规模以人类为中心的数据资源。然而，鉴于人类数据与机器人数据之间存在显著差距，SWIM 通过将动作基于视觉功能映射来弥合这一差异。该方法通过根据功能映射推断目标姿态，实现了从人类数据向机器人控制能力的有效知识迁移。

Iso-Dream[444]为 Dreamer 框架引入了两项关键改进。第一项改进专注于通过解耦可控与不可控动力学来优化逆向动力学；第二项改进则是基于解耦后的潜在想象来优化智能体行为。该方法具有显著优势：不可控状态转换分支可独立于动作进行展开，从而在长时程决策过程中发挥积极作用。

基于Transformer的世界模型（TWM）[56]与Dreamer具有相同的研究动机，但采用了基于Transformer-XL架构构建世界模型的不同方法[445]。该Transformer基础世界模型通过生成新轨迹，在潜在想象空间中训练无模型智能体。此外， TWM 对DreamerV2中的平衡KL散度损失函数进行了改进，并针对优势演员-评论家框架提出了新型阈值熵损失函数。

SceneScript[446]提出了一套结构化语言指令集，能够通过定义场景布局与物体来构建完整场景。这种基于语言的场景表征方法与传统技术（如网格模型、体素网格、点云或辐射场）形成鲜明对比，其创新之处在于采用自回归式、基于标记的生成机制。在布局估计与物体检测领域，该方法展现出与当前最先进方法相当的性能表现。

*3) VLA组件：模仿学习：*BeT[447]作为轨迹变换器的模仿学习变体，通过结合基于k均值的动作离散化与连续动作校正，有效解决了非专家人类演示数据的噪声性和多模态特性。C-BeT[448]在BeT基础上进一步扩展，增加了目标帧或演示内容。

as the goal specification. VQ-BeT [449] further improves BeT by incorporating vector quantization instead of k-means, enabling better modeling of long-range actions.

4) Subsequent Surveys on VLA: Since the initial release of this survey was made public, the field of embodied AI has experienced a proliferation of VLA models, leading to the emergence of several related surveys [450]–[460].

5) Latest Developments of VLA: LLaRA [461] introduces a data-centric method that uses a Vision-Language Model to automatically relabel existing robot demonstration videos with rich, descriptive language, significantly boosting the performance of downstream vision-language policies.

π_0 [116] undergoes further enhancements, including FAST [462] and $\pi_{0.5}$ [463]. FAST is a novel robot action tokenization method designed to prioritize dexterous skills. $\pi_{0.5}$ pushes the boundaries of in-the-wild generalization by incorporating co-training with diverse data sources.

Mobility VLA [464] proposes a hierarchical policy that integrates the commonsense reasoning capabilities of VLMs with a topological-graph-based low-level navigation policy.

While many VLAs focus primarily on robotic arms, recent advancements have extended their application to more complex embodiments, including humanoid and quadruped robots.

Humanoid Robot:

- GR00T N1 [133]
- Humanoid-VLA [465]

Quadruped Robot:

- QUAR-VLA [466]
- QUART-Online [467]
- MoRE [468]

Dexterous Hand:

- DexGraspVLA [469]
- DexGrasp-VLA [470]

Autonomous Vehicle:

- Alpamayo-R1 [471]
- CoVLA [472]
- DriveVLA-W0 [473]
- EMMA [474]

Additionally, some VLAs are designed to function as virtual assistants capable of taking actions within operating systems, video games, and other environments. Notable examples include:

- CombatVLA [475]
- JARVIS-VLA [476]

We utilize various charts to visualize key aspects of VLA developments from 2020 to 2025, including Figure 9, Figure 10, Figure 11, and Figure 12. To supplement the VLAs discussed in the main text, we employed a hybrid approach combining automated scripting and manual searching to retrieve VLA-related papers published between January 2020 and December 2025. We queried the keywords "VLA", "Vision-language-action", and "Vision language action", filtering false positives based on their relevance to "embodied AI" and "robotics". This pipeline yielded approximately 400 VLA-related papers. Acknowledging the potential for automated errors, we welcome feedback and requests for corrections regarding the included data.

Abbreviations of institutes appearing in Figure 9:

- **AI2**: Allen Institute for AI
- **AIBD**: Advanced Institute of Big Data
- **ANU**: Australian National University
- **AgiBot**: AgiBot Research
- **Alexandria Univ.**: Alexandria University
- **Alibaba**: Alibaba Group, DAMO Academy
- **Anyverse**: Anyverse Intelligence
- **BAAI**: Beijing Academy of Artificial Intelligence
- **BACFBC**: Innovation Center for Future Blockchain and Privacy Computing, Beijing
- **BIGAI**: State Key Lab of General Artificial Intelligence
- **BIT**: Beijing Institute of Technology
- **BJUT**: Beijing University of Technology
- **BKLSISMI**: Beijing Key Laboratory of Super Intelligent Security of Multi-Modal Information
- **BNU**: Beijing Normal University
- **BUAA**: Beihang University
- **BUPT**: Beijing University of Posts and Telecommunications
- **Bosch**: Bosch Center for Artificial Intelligence, Bosch Corporate Research, Bosch Mobility Solutions, Bosch Research
- **Bretagne**: Bretagne INP - ENIB
- **ByteDance**: ByteDance Research, ByteDance Seed
- **CAS**: Chinese Academy of Sciences
- **CAU**: Clark Atlanta University
- **CMU**: Carnegie Mellon University
- **CNRS**: CNRS IRL 2010 CROSSING
- **CSU**: Central South University
- **CUHK**: The Chinese University of Hong Kong
- **CUHK(SZ)**: The Chinese University of Hong Kong (Shenzhen)
- **CWRU**: Case Western Reserve University
- **Caltech**: California Institute of Technology
- **Cambridge**: Cambridge University
- **China Mobile**: China Mobile Information Technology Co., Ltd.,
- **China Telecom**: China Telecom
- **CityU**: City University of Hong Kong
- **Cognitive AI**: Cognitive AI Lab
- **Columbia**: Columbia University
- **DSO**: DSO National Laboratories
- **DUT**: Dalian University of Technology
- **Didi Chuxing**: Didi Chuxing
- **Drexel**: Drexel University
- **Duke**: Duke University
- **ECNU**: East China Normal University
- **EIT**: Eastern Institute of Technology
- **EKUT**: University of Tübingen
- **ENS**: École Normale Supérieure
- **ER**: Everyday Robots
- **ETH**: ETH Zurich

作为目标规范。 VQ -BeT[449]通过采用向量量化而非k均值聚类算法，进一步提升了BeT模型的性能，从而能够更精准地建模长距离动作。

*4) 后续关于VLA的调查：*自该调查首次公开发布以来，具身人工智能领域经历了VLA模型的激增，由此催生了多项相关调查[450]–[460]。

*5) VLA的最新进展：*LLaRA[461]提出了一种以数据为中心的方法，该方法利用视觉-语言模型自动为现有机器人演示视频添加丰富的描述性语言标签，显著提升了下游视觉-语言策略的性能。

π_0[116]经历了进一步改进，包括 FAST[462]和π_0。$_5$[463]。FAST是一种新型机器人动作分词方法，旨在优先提升灵巧技能。π_0。$_5$通过结合多种数据源的协同训练，拓展了实际场景中的泛化能力边界。

Mobility VLA[464]提出了一种分层策略，该策略将视觉语言模块（VLMs）的常识推理能力与基于拓扑图的低级导航策略相结合。

尽管许多视觉定位算法（VLA）主要针对机械臂设计，但近期技术进展已将其应用范围扩展至更复杂的系统形态，包括人形机器人和四足机器人。

人形机器人：

- GR00T N1 [133]
- 人形-VLA[465]四足机器

人：

- QUAR-VLA [466]
- quart-Online[467]
- MoRE[468] 精

巧之手：

- DexGraspVLA [469]
- DexGrasp-VLA[470] 自动

驾驶车辆：

- Alpamayo-R1 [471]
- CoVLA [472]
- DriveVLA-W0 [473]
- EMMA [474]

此外，部分虚拟生活助理（VLAs）被设计为虚拟助手功能，能够在操作系统、视频游戏及其他环境中执行操作。典型示例包括：

- CombatVLA [475]
- jarvis-VLA[476]

我们采用多种图表对2020年至2025年间VLA发展关键要素进行可视化呈现，包括图9、图10、图11及图12。为补充正文讨论的VLA研究，我们采用自动化脚本与人工检索相结合的混合方法，筛选出2020年1月至2025年12月期间发表的VLA相关论文。通过检索关键词“VLA”、“Visionlanguage-action”及“Vision language action”，并根据与“具身人工智能”和“机器人学”的相关性进行误报过滤，该流程共获取约400篇VLA相关论文。鉴于自动化流程可能存在误差，我们欢迎针对所含数据的反馈与更正请求。

图9中研究所的缩写：

- **AI2：**艾伦人工智能研究所
- **AIBD：**大数据高级研究院
- **ANU：**澳大利亚国立大学
- **AgiBot：**AgiBot研究
- **亚历山大大学：**亚历山大大学
- **阿里巴巴：**阿里巴巴集团， DAMO 学院
- **Anyverse：**Anyverse智能系统
- **BAAI：**北京人工智能研究院
- **BACFBC：**未来区块链与隐私计算创新中心，北京
- **BIGAI：**通用人工智能国家重点实验室
- **BIT：**北京理工大学
- **BJUT：**北京理工大学
- **bklsismi：**北京多模态信息超智能安全重点实验室
- **BNU：**北京师范大学
- **BUAA：**北京航空航天大学
- **BUPT：**北京邮电大学
- **博世：**博世人工智能中心、博世企业研究院、博世移动解决方案部门、博世研究院
- **布列塔尼：**布列塔尼INP - ENIB
- **字节跳动：**字节跳动研究院，字节跳动种子基金
- **CAS：**中国科学院
- **CAU：**克拉克亚特兰大大学
- **CMU：**卡内基梅隆大学
- **CNRS：**CNRS IRL 2010交叉点
- **CSU：**中南大学
- **CUHK：**香港中文大学
- **CUHK (SZ)：**香港中文大学（深圳）
- **CWRU：**凯斯西储大学
- **加州理工学院：**California Institute of Technology
- **剑桥：**剑桥大学
- **中国移动：**中国移动信息技术有限公司，
- **中国电信：**中国电信
- **CityU：**香港城市大学
- **认知人工智能：**认知人工智能实验室
- **哥伦比亚大学：**哥伦比亚大学
- **DSO：**DSO国家实验室
- **DUT：**大连理工大学
- **滴滴出行：**滴滴出行
- **德雷塞尔大学：**Drexel University
- **杜克大学：**杜克大学
- **ECNU：**华东师范大学
- **EIT：**东部理工学院
- **EKUT：**T仆ubingen 大学
- **ENS：**巴黎第一大学教育学院é里埃里埃
- **ER：**《日常机器人》
- **ETH：**ETH苏黎世

- **EdUHK**: The Education University of Hong Kong
- **Fudan**: Fudan University
- **GDUT**: Guangdong University of Technology
- **Gachon Univ.**: Gachon University
- **Galaxea**: Galaxea AI
- **Georgia Tech**: Georgia Institute of Technology
- **Google**: Google, Google DeepMind, Robotics at Google, Google Research, Brain Team, Gemini Robotics Team
- **HBUT**: Hebei University of Technology
- **HIT**: Harbin Institute of Technology
- **HIT(SZ)**: Harbin Institute of Technology (Shenzhen)
- **HKU**: The University of Hong Kong
- **HKUST**: Hong Kong University of Science and Technology
- **HKUST(GZ)**: Hong Kong University of Science and Technology (Guangzhou)
- **HUST**: Huazhong University of Science and Technology
- **Hanyang Univ.**: Hanyang University
- **Harvard**: Harvard University
- **Horizon**: Horizon Robotics
- **Huawei**: Huawei Technologies, Huawei Noah's Ark Lab
- **HuggingFace**: Hugging Face
- **Hyundai**: Hyundai Motor Company
- **IC**: Imperial College London
- **IEIT**: IEIT SYSTEMS Co., Ltd.
- **IIIT-H**: IIIT Hyderabad
- **IIT**: Istituto Italiano di Tecnologia
- **IMT**: IMT Atlantique
- **Infinigence**: Infinigence AI
- **JHU**: Johns Hopkins University
- **JLU**: Jilin University
- **KAIST**: KAIST AI
- **KIT**: Karlsruhe Institute of Technology
- **Kobe Univ.**: Kobe University
- **Korea Univ.**: Korea University
- **LZU**: Lanzhou University
- **Lehigh Univ.**: Lehigh University
- **LiAuto**: LiAuto Inc.
- **LimX**: LimX Dynamics
- **Logos**: Logos Robotics
- **Lumina**: Lumina Group
- **Lumos**: Lumos Robotics
- **MBZUAI**: Mohamed bin Zayed University of Artificial Intelligence
- **MEGVII**: MEGVII Technology
- **MIPT**: IAI MIPT
- **MIT**: Massachusetts Institute of Technology
- **MMLab**: MMLab @ CUHK, MMLab @ HKU
- **Macalester**: Macalester College
- **Meta**: Meta AI, FAIR-MetaAI, Meta Reality Labs
- **Microsoft**: Microsoft Research, Microsoft Research Asia, Microsoft Zurich
- **Midea**: Midea Group
- **Mila**: Mila — Quebec AI Institute
- **Monash Univ.**: Monash University
- **Moxin**: Moxin (Huzhou) Technology Co., Ltd.
- **NAVER**: NAVER AI Lab
- **NEU**: Northeastern University
- **NJU**: Nanjing University
- **NJUPT**: Nanjing University of Posts and Telecommunications
- **NKU**: Nankai University
- **NPU**: Northwestern Polytechnical University
- **NTU**: Nanyang Technological University
- **NUS**: National University of Singapore
- **NVIDIA**: NVIDIA Research
- **NWU**: Northwestern University
- **NYU**: New York University
- **NaUKMA**: National University of Kyiv-Mohyla Academy
- **Noematrix**: Noematrix Intelligence
- **OXE Team**: Open X-Embodiment Collaboration
- **OpenHelix**: OpenHelix Team
- **PCL**: Peng Cheng Laboratory
- **PI**: Physical Intelligence
- **PKU**: Peking University
- **PSL**: PSL Research University
- **Penn State**: Pennsylvania State University
- **Pitt**: University of Pittsburgh
- **PolyU**: The Hong Kong Polytechnic University
- **Princeton**: Princeton University
- **QMUL**: Queen Mary University of London
- **Qi Zhi**: Shanghai Qi Zhi Institute, Shanghai Qizhi Institute
- **Qualcomm**: Qualcomm AI Research
- **RU**: Rutgers University
- **RUC**: Renmin University of China
- **SBU**: Stony Brook University
- **SCUT**: South China University of Technology
- **SH AI Lab**: Shanghai AI Laboratory
- **SHU**: Shanghai University
- **SII**: Shanghai Innovation Institute
- **SJTU**: Shanghai Jiao Tong University
- **SKKU**: Sungkyunkwan University
- **SNU**: Seoul National University
- **STU**: ShanghaiTech University
- **SUST**: Southern University of Science and Technology
- **SUTD**: Singapore University of Technology and Design
- **SYSU**: Sun Yat-sen University
- **SZU**: Shenzhen University
- **Sea AI**: Sea AI Lab
- **SenseTime**: SenseTime Research
- **Siemens**: Siemens Corporation
- **Simplexity**: Simplexity Robotics
- **Sofia Univ.**: Sofia University "St. Kliment Ohridski"
- **Sony**: Sony Interactive Entertainment, Sony Research
- **Sorbonne Univ.**: Sorbonne University
- **Spirit**: Spirit AI
- **Stanford**: Stanford University
- **Syracuse**: Syracuse University
- **T-Stone**: T-Stone Robotics Institute
- **TJU**: Tianjin University
- **TUM**: Technical University of Munich
- **Taiwan Univ.**: National Taiwan University
- **Tencent**: Tencent Inc., Tencent Robotics X, Wechat AI
- **Tongji**: Tongji University

- **香港教育大学**：香港教育大学
- **复旦**：复旦大学
- **加川大学**：加川大学
- **银河**: 银河AI
- **佐治亚理工学院**：佐治亚理工学院
- **Google**：Google、Google DeepMind、Google机器人技术部门院、脑科学团队、Gemini机器人团队
- **HBUT**：河北工业大学
- **HIT**：哈尔滨工业大学
- **HIT (SZ)**：哈尔滨工业大学（深圳）
- **HKU**：香港大学
- **HKUST**：香港科技大学
- **HKUST（GZ）**：香港科技大学技术（广州）
- **HUST**：华中科技大学
- **汉阳大学**: Hanyang University
- **哈佛大学**：哈佛大学
- **Horizon**：Horizon Robotics
- **华为**：华为技术，华为诺亚方舟实验室
- **HuggingFace**：Hugging Face
- **现代汽车**：现代汽车公司
- **IC**：伦敦帝国理工学院
- **IEIT**： IEIT 系统有限公司
- **IIT**：意大利技术学院
- **IMT**：IMT Atlantique
- **Infinigence**: Infinigence AI
- **JHU**：约翰霍普金斯大学
- **JLU**：吉林大学
- **KAIST**: KAIST AI
- **KIT**：卡尔斯鲁厄理工学院
- **神户大学**：神户大学
- **韩国大学**：韩国大学
- **LZU**：兰州大学
- **利哈伊大学**：利哈伊大学
- **LiAuto**: LiAuto Inc.
- **LimX**：LimX动力学
- **Logos**：Logos机器人公司
- **Lumina**：Lumina集团
- **Lumos**：Lumos Robotics
- **MBZUAI**：穆罕默德·本·扎耶德人工智能大学智力
- **MEGVII**： MEGVII 技术
- **MIPT**: IAI MIPT
- **MIT**：麻省理工学院
- **MMLab**：MMLab@ CUHK， MMLab@ HKU
- **麦卡莱斯特**：麦卡莱斯特学院
- **Meta**：Meta AI、FAIR-MetaAI、Meta Reality Labs
- **微软**：微软研究院、微软亚洲研究院，微软苏黎世分公司
- **美的**：美的集团
- **米拉**：Mila——魁北克人工智能研究所
- **莫纳什大学**：莫纳什大学
- **莫信**：莫信（湖州）科技有限公司
- **NAVER**： NAVER 人工智能实验室
- **NEU**：东北大学

•**NJU**：南京大学
•**NJUPT**：南京邮电大学
•**GDUT**：广东理工大学 阳离子
NKU：南开大学
•**NPU**：西北工业大学
NTU：南阳理工大学
•**新加坡国立大学（NUS）**：谷歌研究
•**NVIDIA**： NVIDIA 研究
•**NWU**：西北大学
•**纽约大学**：纽约大学
•**NaUKMA**：基辅莫希拉国立大学学院
•**Noematrix**：Noematrix智能公司
•**OXE 团队**：Open X-Embodiment Collaboration
•**OpenHelix**：OpenHelix团队
•**PCL**：彭程实验室
•**PI**：物理智能
•**PKU**：北京大学
•**PSL**：PSL研究大学
宾夕法尼亚州立大学：Pennsylvania State University
•**匹兹堡大学**：Pittsburgh University
•**PolyU**：香港理工大学
普林斯顿：普林斯顿大学
•**QMUL**：伦敦玛丽女王大学
•**奇智**：上海奇智研究院，上海奇智研究院 瞳孔
高通：高通人工智能研究院
RU：罗格斯大学
•**RUC**：中国人民大学
•**SBU**：石溪大学
•**SCUT**：华南理工大学
•**SH AI实验室**：上海人工智能实验室
上海大学：
•**SII**：上海创新研究院
•**SJTU**：上海交通大学
•**SKKU**：成均馆大学
SNU：首尔国立大学
•**STU**：上海科技大学
•**SUST**：南方科技大学
SUTD：新加坡科技设计大学
•**SYSU**：中山大学
•**SZU**：深圳大学
•**Sea AI**：Sea AI实验室
SenseTime：SenseTime研究团队
西门子：西门子公司
•**单形性**：单形性机器人技术
•**索菲亚大学**：索菲亚大学“圣克莱门特·奥赫里德斯基”分校
索尼：索尼互动娱乐、索尼研究院
索邦大学：索邦大学
•**Spirit**：Spirit AI
斯坦福大学：斯坦福大学
雪城大学：雪城大学
•**T-Stone**：T-Stone机器人研究所
•**TJU**：天津大学
•**慕尼黑工业大学**：Technical University of Munich
•**台湾大学**：国立台湾大学
•**腾讯**：腾讯公司、腾讯机器人X、微信AI
同济大学：同济大学

- **Toyota**: Toyota Research Institute
- **Tsinghua**: Tsinghua University
- **Turing**: Turing Inc.
- **U of U**: University of Utah
- **UATX**: The University of Austin at Texas
- **UAlberta**: University of Alberta
- **UArizona**: University of Arizona
- **UArkansas**: University of Arkansas
- **UBC**: The University of British Columbia
- **UCAS**: University of Chinese Academy of Sciences
- **UCB**: University of California, Berkeley
- **UCL**: University College London
- **UCLA**: University of California, Los Angeles
- **UCSD**: University of California San Diego
- **UCSI**: UCSI University
- **UCSP**: Universidad Católica San Pablo
- **UChicago**: University of Chicago
- **UESTC**: University of Electronic Science and Technology of China
- **UIC**: University of Illinois Chicago
- **UIUC**: University of Illinois Urbana-Champaign
- **UM**: University of Macau
- **UMD**: University of Maryland, College Park
- **UMU**: Umeå University
- **UNSW**: University of New South Wales
- **UOsaka**: The University of Osaka
- **UPenn**: University of Pennsylvania
- **UQ**: University of Queensland
- **USC**: University of Southern California
- **USST**: University of Shanghai for Science and Technology
- **USTC**: University of Science and Technology of China
- **USYD**: The University of Sydney
- **UT Austin**: The University of Texas at Austin
- **UT Dallas**: University of Texas at Dallas
- **UTN**: University of Technology Nuremberg
- **UTS**: University of Technology Sydney
- **UTokyo**: The University of Tokyo
- **UW**: University of Washington
- **UWarsaw**: University of Warsaw
- **UW–Madison**: University of Wisconsin-Madison
- **UdeM**: Université de Montréal
- **Uni Freiburg**: University of Freiburg
- **UofT**: University of Toronto
- **VT**: Virginia Tech
- **VinUni**: VinUniversity
- **WHU**: Wuhan University
- **WMU**: Wenzhou Medical University
- **Waseda**: Waseda University
- **Westlake Univ.**: Westlake University
- **X-Era**: X-Era AI Lab
- **X-Humanoid**: Beijing Innovation Center of Humanoid Robotics
- **XDU**: Xidian University
- **XJTU**: Xi'an Jiaotong University
- **XMU**: Xiamen University Malaysia
- **XPeng**: XPeng Motors
- **Xiaomi**: Beijing Xiaomi Robot Technology Co., Ltd., Xiaomi EV, Xiaomi Robotics
- **Yale**: Yale University
- **Yinwang**: Yinwang Intelligent Technology Co., Ltd.
- **Yuandao**: Yuandao AI
- **ZGCA**: Zhongguancun Academy
- **ZGCI**: Zhongguancun Institute of Artificial Intelligence
- **ZJU**: Zhejiang University
- **ZZU**: Zhengzhou University
- **ZhiCheng**: ZhiCheng AI
- **Zhongke Huiling**: Beijing Zhongke Huiling Robot Technology Co
- **iFlyTek**: iFlyTek Research and Development Group
- **mimic**: mimic robotics
- **valeo**: valeo.ai

List of VLAs appearing in Figure 9:

- 3D Diffuser Actor [477]
- 3DS-VLA [478]
- 4D-VLA [479]
- A2C2 [480]
- A3VLM [481]
- ACG [482]
- ACT [98]
- Act3D [89]
- ActDistill [483]
- ActionFlow [484]
- AdaMoE [485]
- ADP [486]
- AFI [487]
- AgentWorld [488]
- AINA [489]
- Alpamayo-R1 [471]
- AMS [490]
- ANNIE [491]
- AsyncVLA [492]
- ATE [493]
- AutoDrive-R2 [494]
- AutoPrune [495]
- AutoVLA [496]
- AVA-VLA [497]
- Avi [498]
- BadVLA [499]
- BayesVLA [500]
- BC-Z [80]
- Being-H0 [501]
- Beyond Success [502]
- Bi-VLA [503]
- Bi-VLA 2025 [504]
- BridgeVLA [505]
- BYOVLA [506]
- CCoL [507]
- ChatVLA-2 [508]
- CLAW [509]
- CLIPort [28]
- CLIP-RT [510]
- ClutterDexGrasp [511]
- CoC-VLA [512]
- CogACT [120]

- **丰田**：丰田研究所
- **清华**大学：清华大学
- **图灵**：图灵公司。
- **犹他大学**：University of Utah
- **UATX**：德克萨斯州奥斯汀大学
- **阿尔伯塔大学**：阿尔伯塔大学
- **亚利桑那大学**：亚利桑那大学
- **阿肯色大学**：阿肯色大学
- **UBC**：不列颠哥伦比亚大学
- **UCAS**：中国科学院大学
- **UCB**：加利福尼亚大学伯克利分校
- **UCL**：伦敦大学学院
- **UCLA**：加利福尼亚大学洛杉矶分校
- **UCSD**：加利福尼亚大学圣地亚哥分校
- **UCSI**： UCSI 大学
- **UCSP**：卡塔赫纳大学6利卡·圣巴勃罗
- **UChicago**：芝加哥大学
- **UESTC**：中国电子科技大学
- **UIC**：伊利诺伊大学芝加哥分校
- **UIUC**：伊利诺伊大学厄巴纳-香槟分校
- **UM**：澳门大学
- **UMD**：马里兰大学帕克分校
- **UMU**: Umeå 大学
- **UNSW**：新南威尔士大学
- **大阪大学**：大阪大学
- **宾夕法尼亚大学**：University of Pennsylvania
- **UQ**：昆士兰大学
- **USC**：南加州大学
- **USST**：上海科技大学
- **中国科学技术大学**：中国科学技术大学
- **USYD**：悉尼大学
- **德克萨斯大学奥斯汀分校**：德克萨斯大学奥斯汀分校
- **达拉斯德克萨斯大学**：德克萨斯大学达拉斯分校
- **UTN**：纽伦堡工业大学
- **UTS**：悉尼科技大学
- **东京大学**：东京大学
- **UW**：华盛顿大学
- **华沙大学**：华沙大学
- **威斯康星大学麦迪逊分校**：威斯康星大学麦迪逊分校
- **UdeM**: Université de Montré阿尔
- **弗莱堡大学**：弗莱堡大学
- **多伦多大学**：University of Toronto
- **VT**：弗吉尼亚理工大学
- **VinUni**: VinUniversity
- **WHU**：武汉大学
- **WMU**：温州医科大学
- **早稻田大学**：早稻田大学
- **西湖大学**：Westlake University
- **X-Era**: X-Era AI实验室
- **X-人形机器人**：北京人形机器人创新中心
- **XDU**：西电大学
- **XJTU**：西安交通大学
- **XMU**：厦门大学马来西亚分校
- **XPeng**：XPeng汽车
- **小米**：北京小米机器人科技有限公司、小米电动汽车、小米机器人
- **耶鲁大学**：耶鲁大学
- **银网**：银网智能科技有限公司
- **元道**：元道AI
- **ZGCA**：中关村学院
- **ZGCI**：中关村人工智能研究院
- **ZJU**：浙江大学
- **ZZU**：郑州大学
- **智诚**：智诚人工智能
- **中科惠灵**：北京中科惠灵机器人技术有限公司
- **iFlyTek**: iFlyTek研发团队
- **模拟**：模拟机器人技术
- **valeo**: valeo.ai

图9中出现的VLA列表：

- 3D扩散器演员[477]
- 3DS-VLA [478]
- 4D-VLA [479]
- A2C2 [480]
- A3VLM [481]
- ACG [482]
- ACT [98]
- Act3D [89]
- ActDistill [483]
- ActionFlow [484]
- AdaMoE [485]
- ADP [486]
- AFI [487]
- AgentWorld [488]
- AINA [489]
- Alpamayo-R1 [471]
- AMS [490]
- 安妮[491]
- AsyncVLA [492]
- ATE [493]
- AutoDrive-R2 [494]
- AutoPrune [495]
- AutoVLA [496]
- AVA-VLA [497]
- Avi [498]
- BadVLA [499]
- BayesVLA [500]
- BC-Z [80]
- 存在状态H0[501]
- 超越成功[502]
- Bi-VLA [503]
- Bi-VLA 2025 [504]
- BridgeVLA [505]
- BYOVLA [506]
- CCoL [507]
- ChatVLA-2 [508]
- CLAW [509]
- CLIPort [28]
- CLIP-RT [510]
- ClutterDexGrasp [511]
- CoC-VLA [512]
- CogACT [120]

- CogVLA [513]
- ColaVLA [514]
- CombatVLA [475]
- ConRFT [515]
- ControlVLA [516]
- Control Your Robot [517]
- CoReVLA [518]
- CoT4AD [519]
- CoT-VLA [77]
- Counterfactual VLA [520]
- CoVLA [472]
- CRISP [521]
- CycleManip [522]
- DEAS [523]
- DeeAD [524]
- DeepThinkVLA [525]
- DeeR-VLA [102]
- Dejavu [526]
- DepthVLA [527]
- Dexbotic [528]
- DexGrasp-VLA [470]
- DexGraspVLA [469]
- DexVLA [121]
- Diffusion Policy [529]
- Diffusion-VLA [530]
- DiffVLA [531]
- DIPOLE [532]
- DLR [533]
- Don't Blind Your VLA [534]
- DP3 [106]
- DP-VLA [535]
- DreamTacVLA [536]
- DreamVLA [537]
- Dream-VLA [538]
- DriveVLA-W0 [473]
- DualVLA [539]
- DuoCore-FS [540]
- dVLA [541]
- EBT-Policy [542]
- ECoT [543]
- ECoT-Lite [544]
- EfficientVLA [545]
- Ego-PM [546]
- EmbodiedCoder [547]
- Embodied-SlotSSM [548]
- EMMA [549]
- Emma-X [550]
- EndoVLA [551]
- EO-1 [552]
- ERIQ [553]
- ERMV [554]
- ET-VLA [555]
- EveryDayVLA [556]
- Evo-1 [557]
- ExpReS-VLA [558]
- EyeVLA [559]
- F1 [560]
- FailSafe [561]
- FAST [462]
- FastDriveVLA [562]
- Fast-in-Slow [563]
- FLARE [564]
- FLOWER [565]
- FlowVLA [566]
- ForceVLA [567]
- FORGE-Tree [568]
- FPC-VLA [569]
- FreezeVLA [570]
- FTM, FLA [571]
- G0 [572]
- Gato [573]
- GEN-0 [574]
- Genie Envisioner [135]
- GeoAware-VLA [575]
- GeRM [576]
- GEVRM [577]
- GF-VLA [578]
- GLaD [579]
- GLUESTICK [580]
- GR00T N1 [133]
- GR 1.5 [581]
- GRAPE [582]
- GraphCoT-VLA [583]
- GraspVLA [584]
- GraSP-VLA [585]
- GR-Dexter [586]
- GR-RL [587]
- HAMSTER [588]
- Helix [589]
- HiF-VLA [590]
- HiMoE-VLA [591]
- Hi-ORS [592]
- Hi Robot [593]
- HiRT [594]
- Hiveformer [87]
- HULC [82]
- HULC++ [83]
- Humanoid-VLA [465]
- HumanVLA [595]
- HybridVLA [122]
- iFlyBot-VLA [596]
- ImaginationPolicy [597]
- Impromptu VLA [598]
- INSIGHT [599]
- Instruct2Act [103]
- InstructVLA [600]
- IntentionVLA [601]
- InteractGen [602]
- Interactive Language [86]
- InternVLA-M1 [603]
- iRe-VLA [604]
- IRL-VLA [605]
- JARVIS-VLA [476]
- KV-Efficient VLA [606]
- Language costs [85]
- LAPA [123]

- CogVLA [513]
- ColaVLA [514]
- CombatVLA [475]
- ConRFT [515]
- ControlVLA [516]
- 控制你的机器人[517]
- CoReVLA [518]
- CoT4AD [519]
- CoT-VLA [77]
- 反事实VLA[520]
- CoVLA [472]
- 脆性[521]
- CycleManip [522]
- DEAS [523]
- DeeAD [524]
- DeepThinkVLA [525]
- DeeR-VLA [102]
- 似曾相识[526]
- DepthVLA [527]
- Dexbotic [528]
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- DexVLA [121]
- 扩散策略[529]
- 扩散-VLA[530]
- DiffVLA [531]
- 偶极子[532]
- DLR [533]
- 不要盲目使用VLA[534]
- DP3 [106]
- DP-VLA [535]
- DreamTacVLA [536]
- DreamVLA [537]
- Dream-VLA [538]
- DriveVLA-W0 [473]
- DualVLA [539]
- DuoCore-FS [540]
- dVLA [541]
- EBT 政策[542]
- ECoT [543]
- ECoT-Lite [544]
- EfficientVLA [545]
- Ego-PM [546]
- EmbodiedCoder [547]
- 具身-槽位SSM[548]
- EMMA [549]
- Emma-X [550]
- EndoVLA [551]
- EO-1 [552]
- ERIQ [553]
- ERMV [554]
- ET-VLA [555]
- EveryDayVLA [556]
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- EyeVLA [559]
- F1 [560]
- FailSafe [561]
- FAST [462]
- FastDriveVLA [562]
- 快慢交替[563]
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- FPC-VLA [569]
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- GEN-0 [574]
- Genie Envisioner [135]
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- GeRM [576]
- GEVRM [577]
- GF-VLA [578]
- GLaD [579]
- GLUESTICK [580]
- GR00T N1 [133]
- GR 1.5 [581]
- 葡萄[582]
- GraphCoT-VLA [583]
- GraspVLA [584]
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- GR-Dexter [586]
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- HiF-VLA [590]
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- 高ORS[592]
- Hi Robot [593]
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- Hiveformer [87]
- HULC [82]
- HULC++ [83]
- 人形-VLA[465]
- HumanVLA [595]
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- iFlyBot-VLA [596]
- ImaginationPolicy [597]
- 即兴VLA[598]
- 意见[599]
- Instruct2Act [103]
- InstructVLA [600]
- IntentionVLA [601]
- InteractGen [602]
- 交互式语言[86]
- InternVLA-M1 [603]
- iRe-VLA [604]
- IRL-VLA [605]
- jarvis-VLA[476]
- KV高效VLA[606]
- 语言成本[85]
- LAPA [123]

- LatBot [607]
- LAWM [608]
- LCDrive [609]
- LITEN [610]
- Lite VLA [611]
- LLaRA [461]
- LoHoVLA [612]
- LoLA [613]
- Long-VLA [614]
- LVP [615]
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- Mantis [617]
- ManualVLA [618]
- MAP-VLA [619]
- $\mathcal{E}_0$ [620]
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- MemER [622]
- MemoryVLA [623]
- MetaVLA [624]
- MG-Select [625]
- MimicDreamer [626]
- MindDrive [627]
- Mind to Hand [628]
- MiVLA [629]
- MLA [630]
- MM-ACT [631]
- Mobility VLA [464]
- MoH [632]
- MoIRA [633]
- MoMaGen [634]
- MoManipVLA [635]
- MonoDream [636]
- MOO [637]
- MoRE [638]
- MotionTrans [639]
- MoTo [640]
- MotoVLA [641]
- Motus [642]
- Moxin-VLA [643]
- MT-ACT [99]
- MUVLA [644]
- NaVILA [645]
- NICE [646]
- NitroGen [647]
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- OccLLaMA [650]
- OccVLA [651]
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- OC-VLA [652]
- OmniJARVIS [653]
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- PerAct [88]
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- PIVOT [132]
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- Point-VLA [666]
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- QDepth-VLA [670]
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- ReflectDrive [676]
- REGENT [677]
- ReKep [678]
- RETAIN [679]
- RetoVLA [680]
- ReVLA [681]
- RICL [682]
- RoboCat [683]
- RoboChemist [684]
- RoboDual [685]
- RoboFlamingo [686]
- RoboMamba [117]
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- RoboMonkey [79]
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- RoboOmni [690]
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- RoboPoint [692]
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- Robotic Assistant [693]
- RoboUniView [101]
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- RT-1 [95]
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- RT-H [701]
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- RT-Trajectory [703]
- RT-X [704]

- LatBot [607]
- LAWM [608]
- LCDrive [609]
- LITEN [610]
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- LLaRA [461]
- LoHoVLA [612]
- LoLa[613]
- 长VLA[614]
- LVP [615]
- ManiAgent [616]
- 螳螂[617]
- ManualVLA [618]
- MAP-VLA [619]
- $\mathcal{E}_0$ [620]
- MCIL [82]
- MDT [621]
- MEMER[622]
- MemoryVLA [623]
- MetaVLA [624]
- MG-Select [625]
- MimicDreamer [626]
- MindDrive [627]
- Mind to Hand [628]
- MiVLA [629]
- MLA [630]
- MM-ACT [631]
- 移动性VLA[464]
- MoH [632]
- MOIRA[633]
- MoMaGen [634]
- MoManipVLA [635]
- MonoDream [636]
- MOO [637]
- MoRE [638]
- MotionTrans [639]
- MoTo [640]
- MotoVLA [641]
- Motus [642]
- Moxin-VLA [643]
- MT-ACT [99]
- MUVLA [644]
- NaVILA [645]
- NICE [646]
- 氮[647]
- NORA-1.5 [134]
- Oat-VLA [648]
- 服从-VLA[649]
- OccLLaMA [650]
- OccVLA [651]
- Octo [108]
- OC-VLA [652]
- OmniJARVIS [653]
- OmniReason [654]
- OmniSAT [655]
- OmniVLA [656]
- OneTwoVLA [657]
- OpenHA [658]
- OpenVLA [659]
- OpenVLA-OFT [114]
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- PhysBrain [661]
- PhysiAgent [662]
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- $\pi_{0.5}$ [463]
- $\pi^*_{0.6}$ [18]
- π_{RL} [663]
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- QDepth-VLA [670]
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- QUAR-VLA [466]
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- RoboMamba [117]
- RoboMatrix [687]
- RoboMonkey [79]
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- USIM & U0 [751]
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- V-GPS [78]
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- VLA-R [768]
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- VLA-RAIL [770]
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- WorldVLA [124]
- WristWorld [782]
- X-Humanoid [783]
- XR-1 [784]
- X-VLA [785]

6) Beyond VLA: Recent research has expanded beyond standard VLA architectures in several key directions:

- **VLA + World Model:** Architectures that unify VLAs with world models.
  - WorldVLA [124]
  - UniVLA [125]
  - NORA-1.5 [134]
  - RynnVLA-002 [707]
  - Motus [642]
- **PLA, MLA:** Models incorporating perception modalities beyond vision.
  - Perception-Language-Action framework (PLA) [665]
  - Multisensory Language–Action model (MLA) [630]
- **VAM:** Video-Action Models that jointly capture semantics and dynamics during video pretraining, enabling more efficient post-training for low-level robot control.
  - mimic-video (VAM) [753]

- RVT [90]
- RVT-2 [705]
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- RynnVLA-002 [707]
- SafeVLA [708]
- Sara-RT[709]
- ScaleDP [710]
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- SeeDo [713]
- SeqVLA [714]
- ShowUI [715]
- 西格玛[716]
- SimpleVLA-RL [717]
- 情景喜剧[718]
- SmolVLA [719]
- SOLAMI [720]
- 空间强迫效应[721]
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- Spec-VLA [723]
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- VINE [756]
- ViVLA [757]
- VLA-0 [758]
- VLA-4D [759]
- VLA适配器[760]
- VLA-AN [761]
- VLA缓存[762]
- VLA-Fool [763]
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- VLM2VLA [775]
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- VoxPoser [777]
- VST [778]
- WAM-Flow [779]
- WholeBodyVLA [780]
- WMPO [781]
- WorldVLA [124]
- WristWorld [782]
- X-类人机器人[783]
- XR-1 [784]
- X-VLA [785]

6) 超越VLA：近期研究已突破标准VLA架构框架，在多个关键方向上取得进展：

- **VLA + 世界模型**：将 VLA 与世界模型统一起来的架构。
  - WorldVLA [124]
  - UniVLA [125]
  - NORA-1.5 [134]
  - RynnVLA-002 [707]
  - Motus [642]
- **PLA，MLA**：融合视觉之外感知模态的模型。
  - 感知-语言-行动框架（PLA）[665]
  - 多感官语言-动作模型（MLA）[630]
- **VAM**：视频动作模型能够在视频预训练过程中同时捕捉语义与动态特征，从而实现更高效的后训练过程，以优化低级机器人控制性能。
  - 拟态视频（VAM）[753]

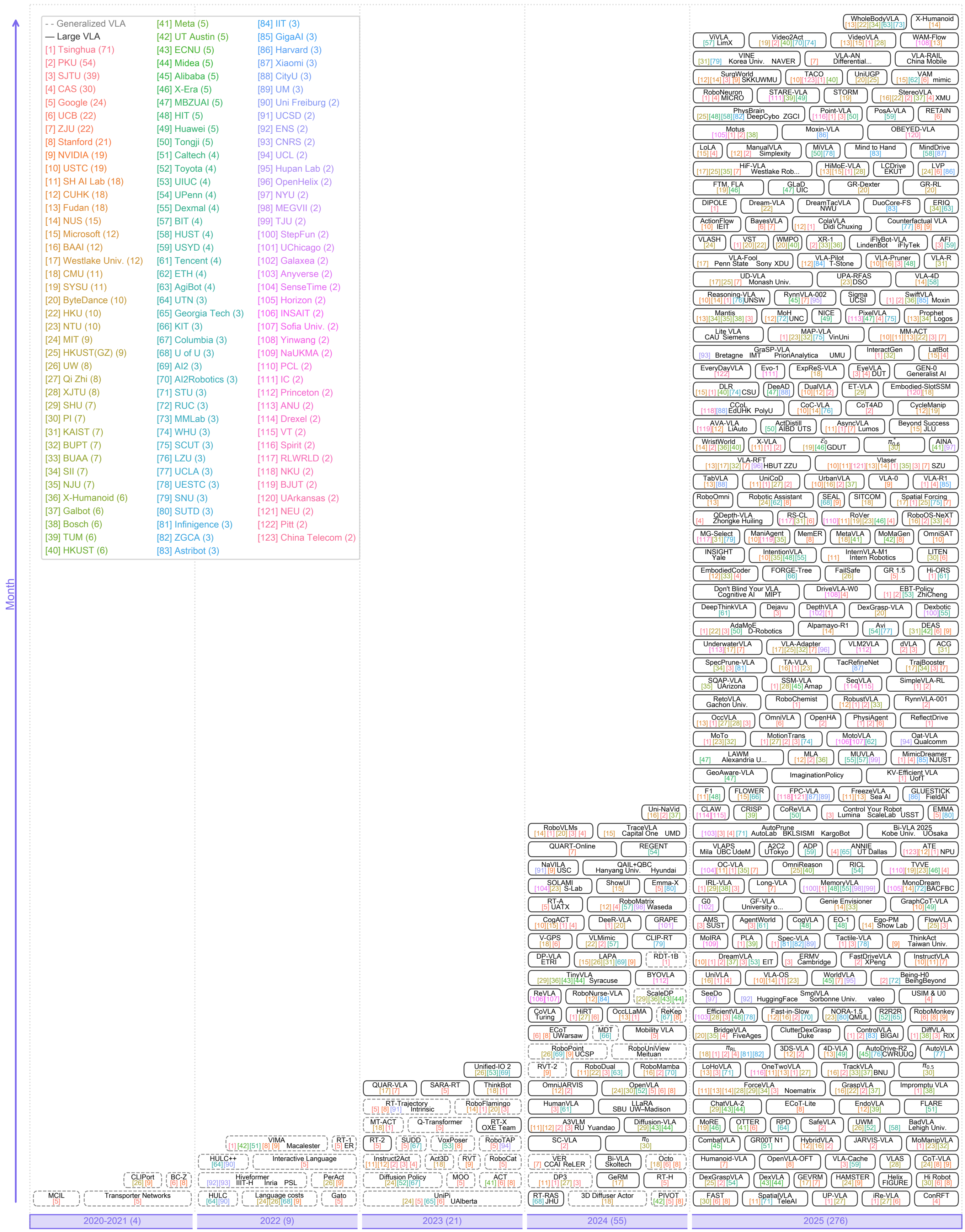


Figure 9: Timeline of Vision-Language-Action models from 2020 to 2025. Bracketed numbers indicate the publication count for the corresponding year or institute.

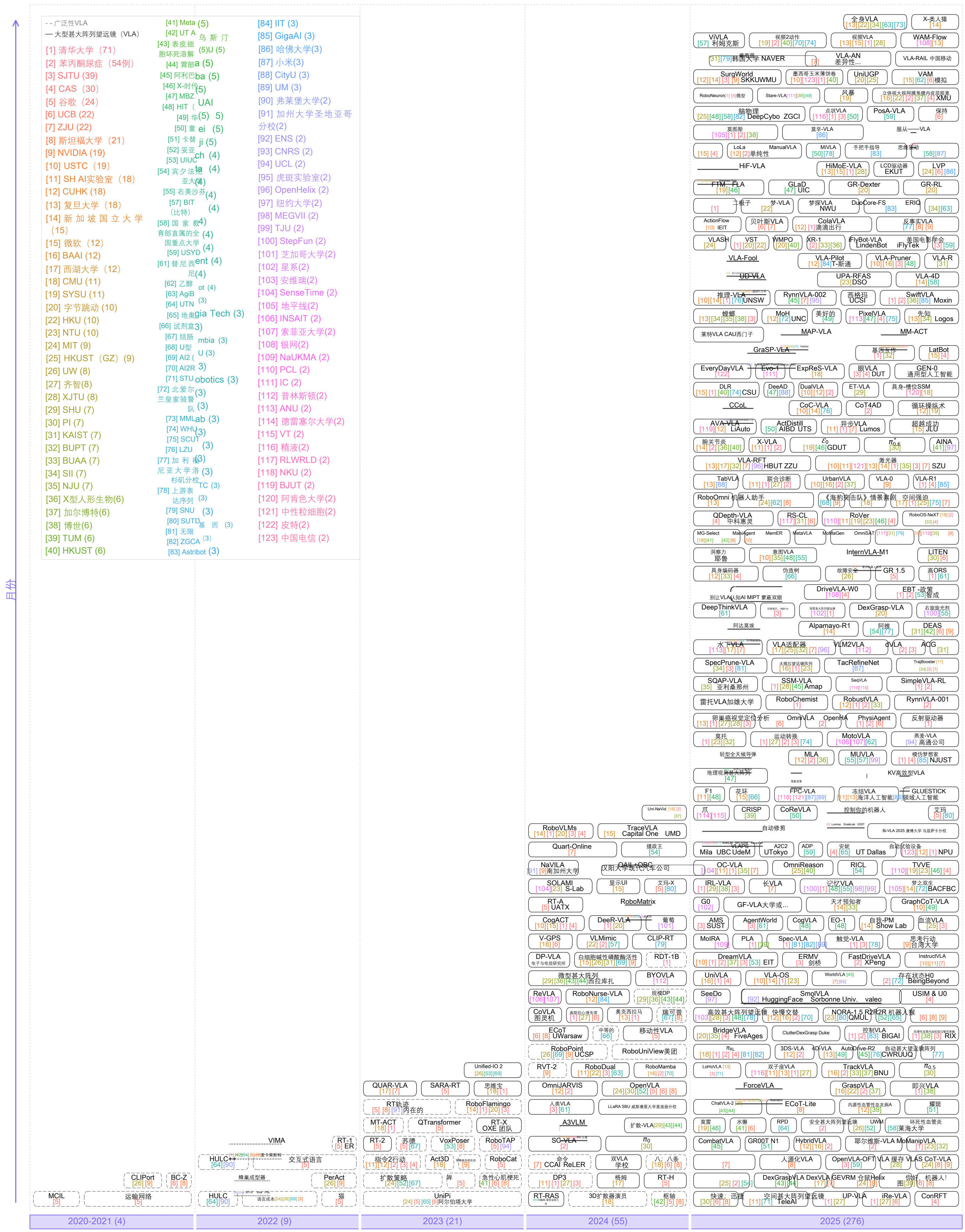


图9：2020年至2025年视觉-语言-动作模型时间线。括号内数字表示对应年份或机构的发表数量。

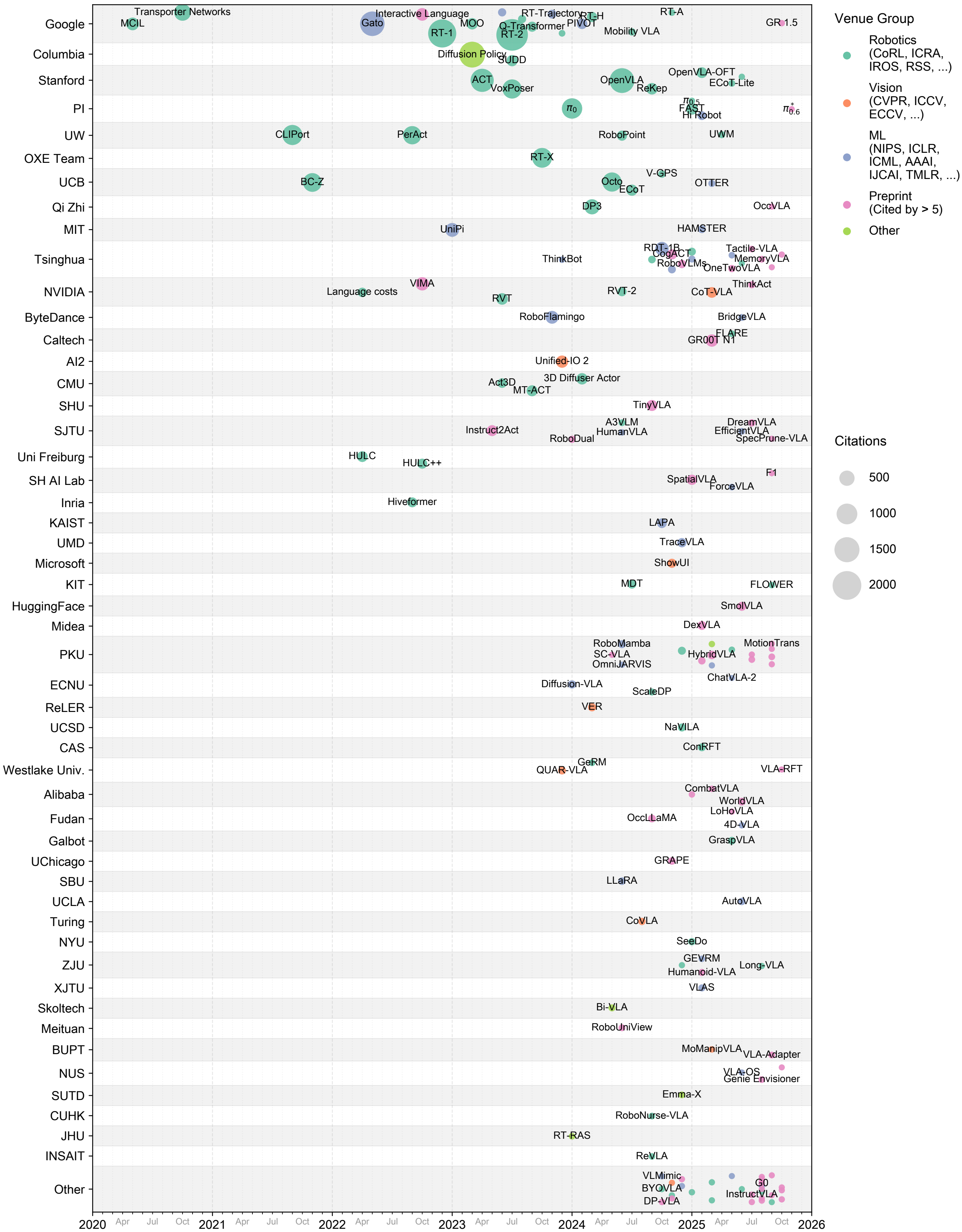


Figure 10: Visualization of the VLA research landscape.

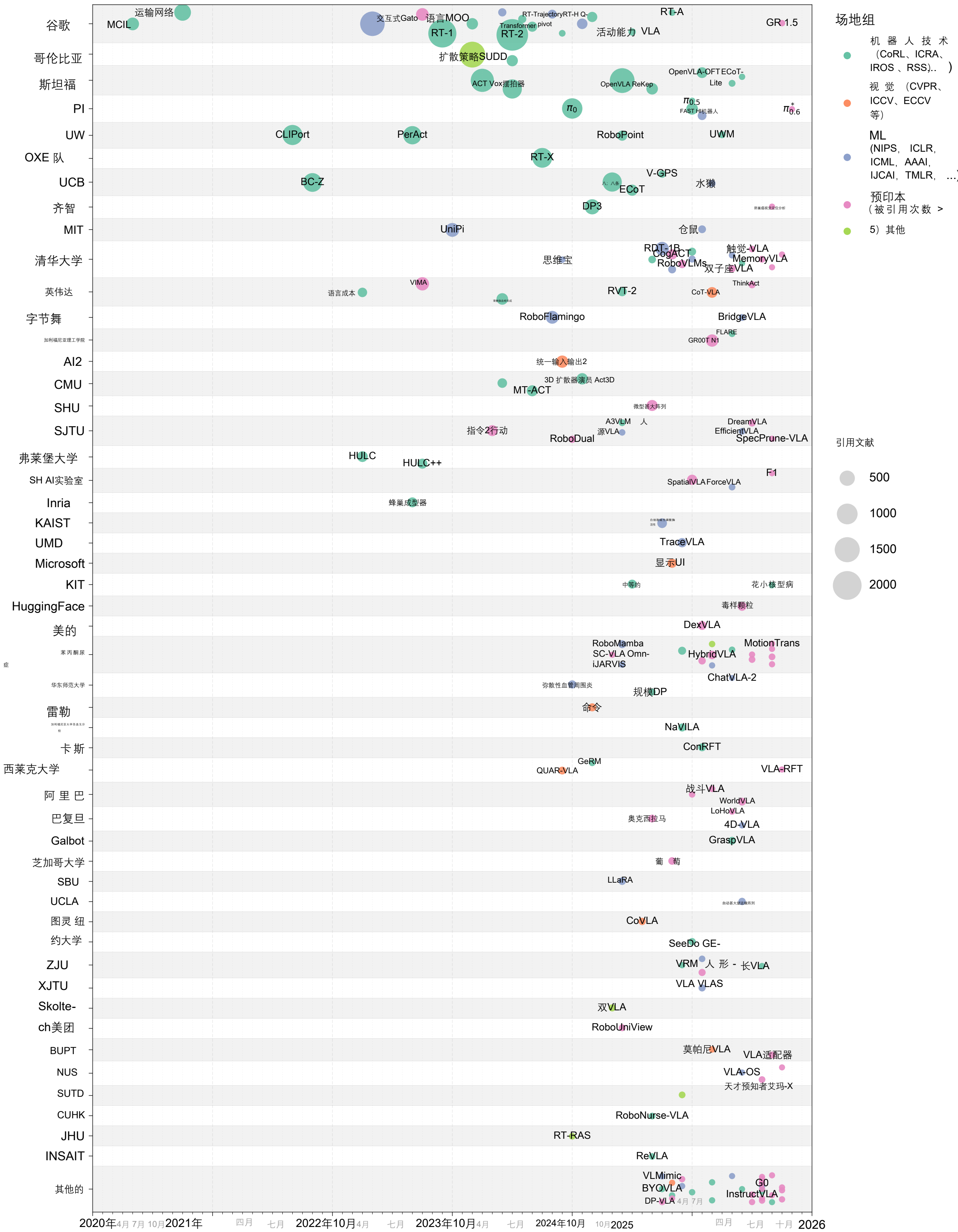
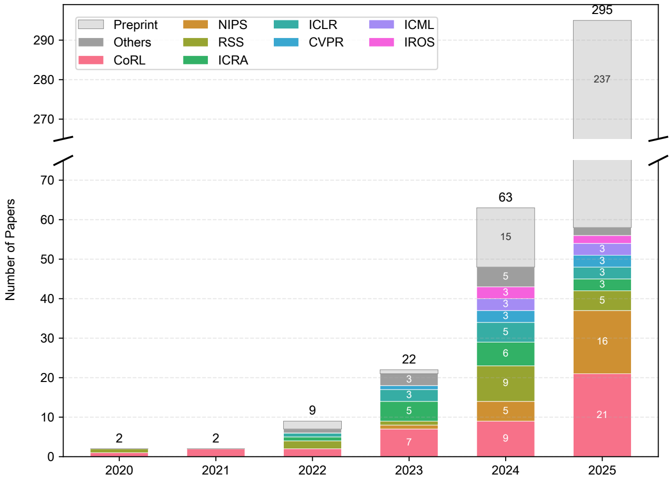
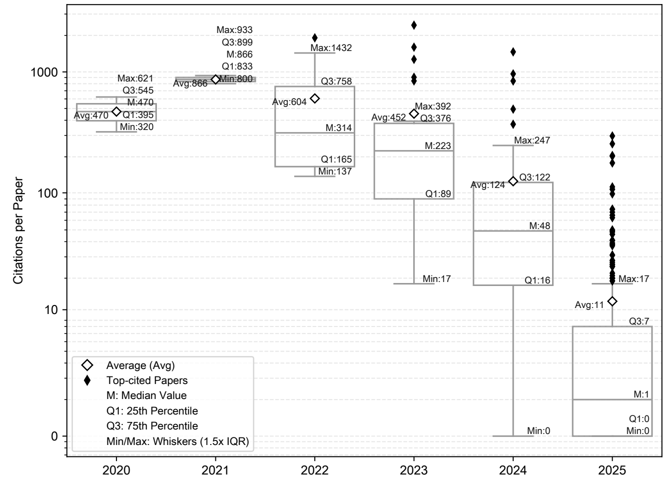


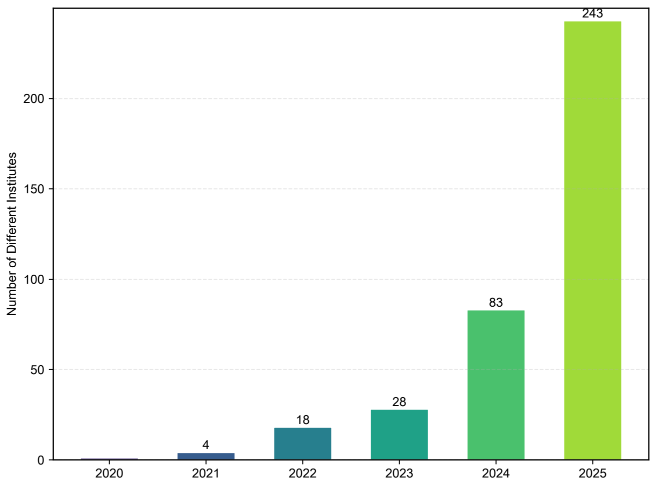
图10：VLA研究领域可视化展示



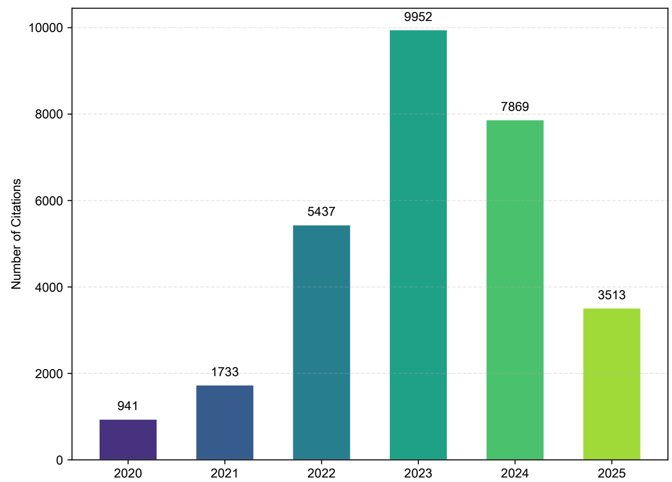
(a) Papers per Year



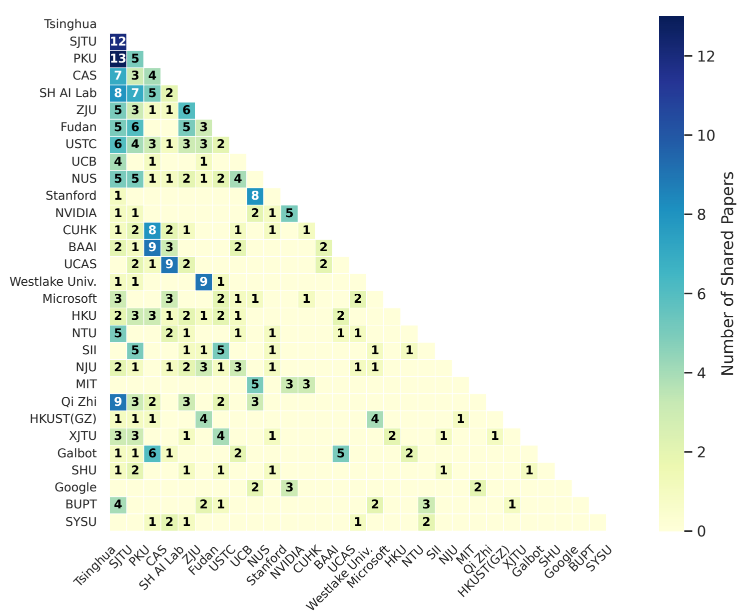
(b) Individual Paper Citations per Year



(c) Institutes per Year

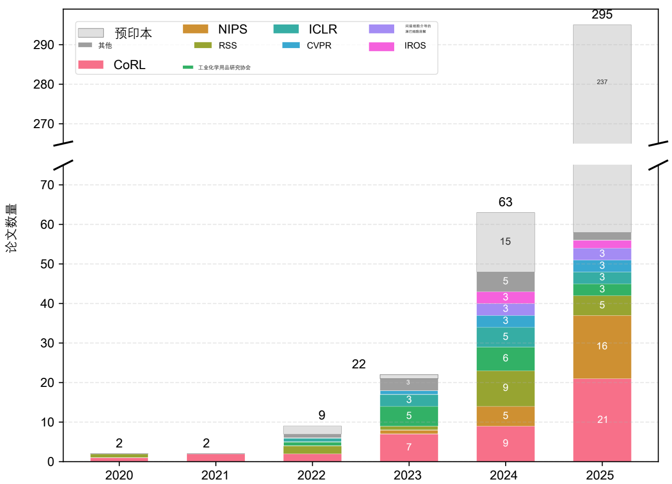


(d) Total Citations per Year

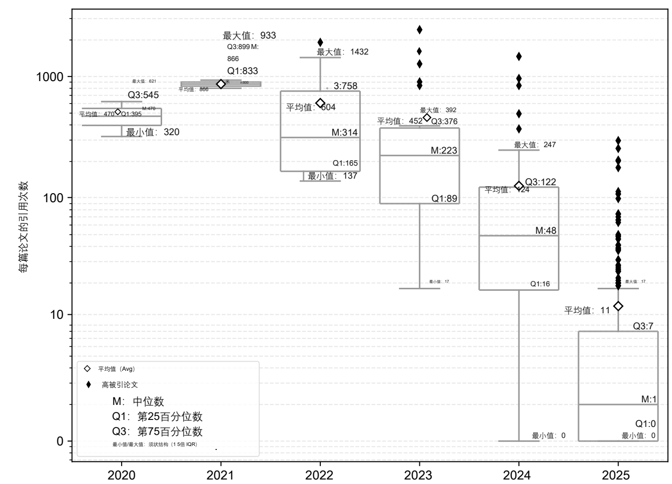


(e) Collaboration Heatmap

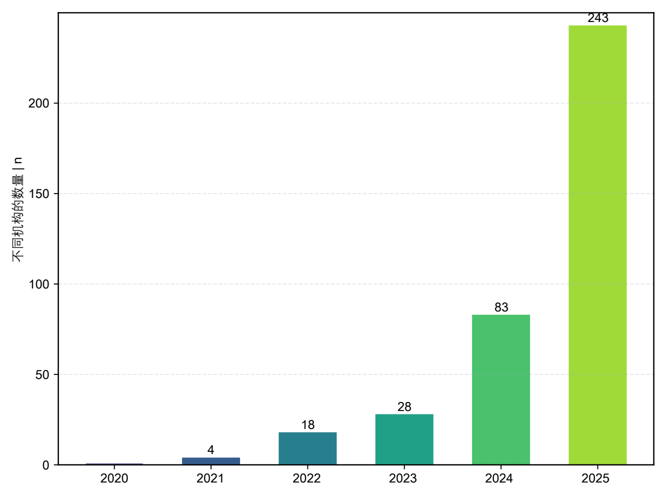
Figure 11: Quantitative analysis of VLA development trends.



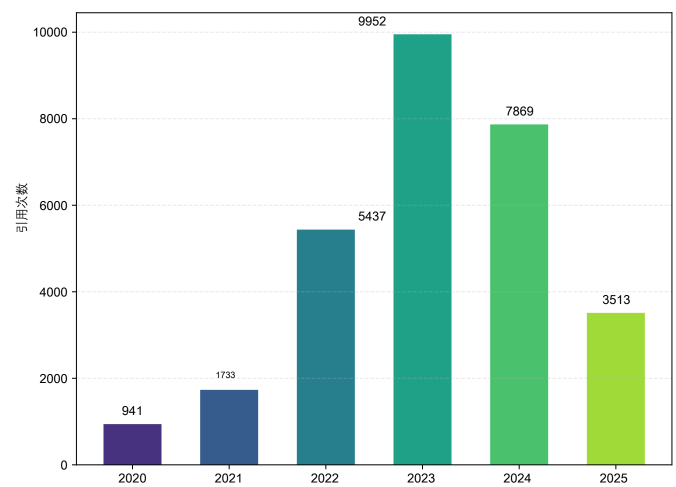
(a) 年论文数



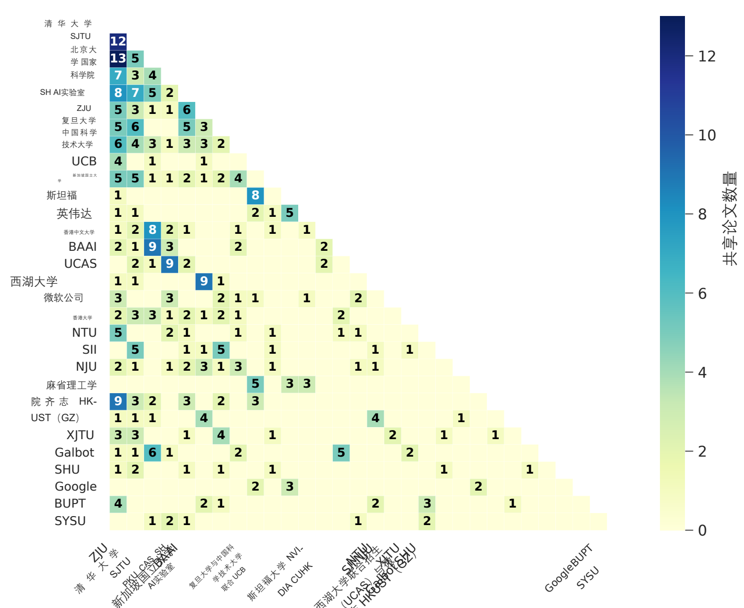
(b) 每年个人论文引用次数



(c) 研究所每年

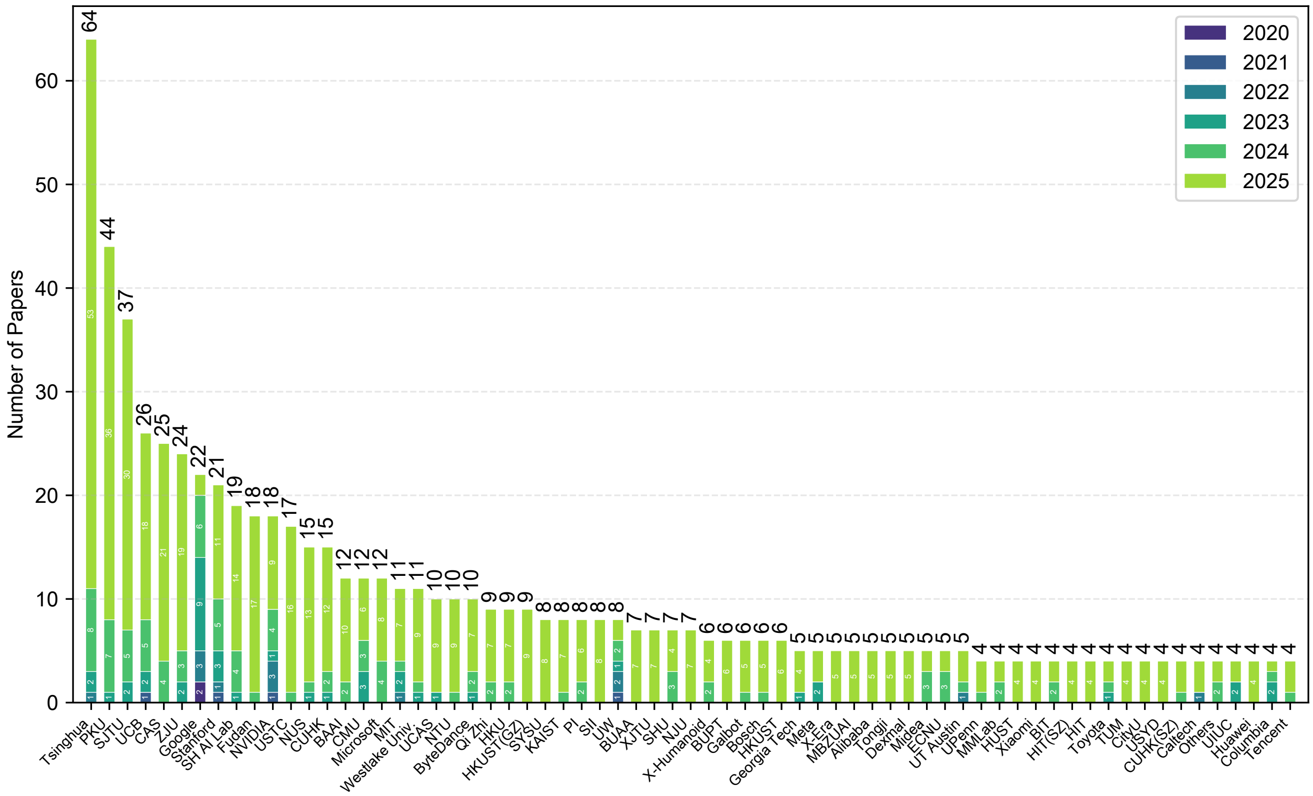


(d) 年总引用量

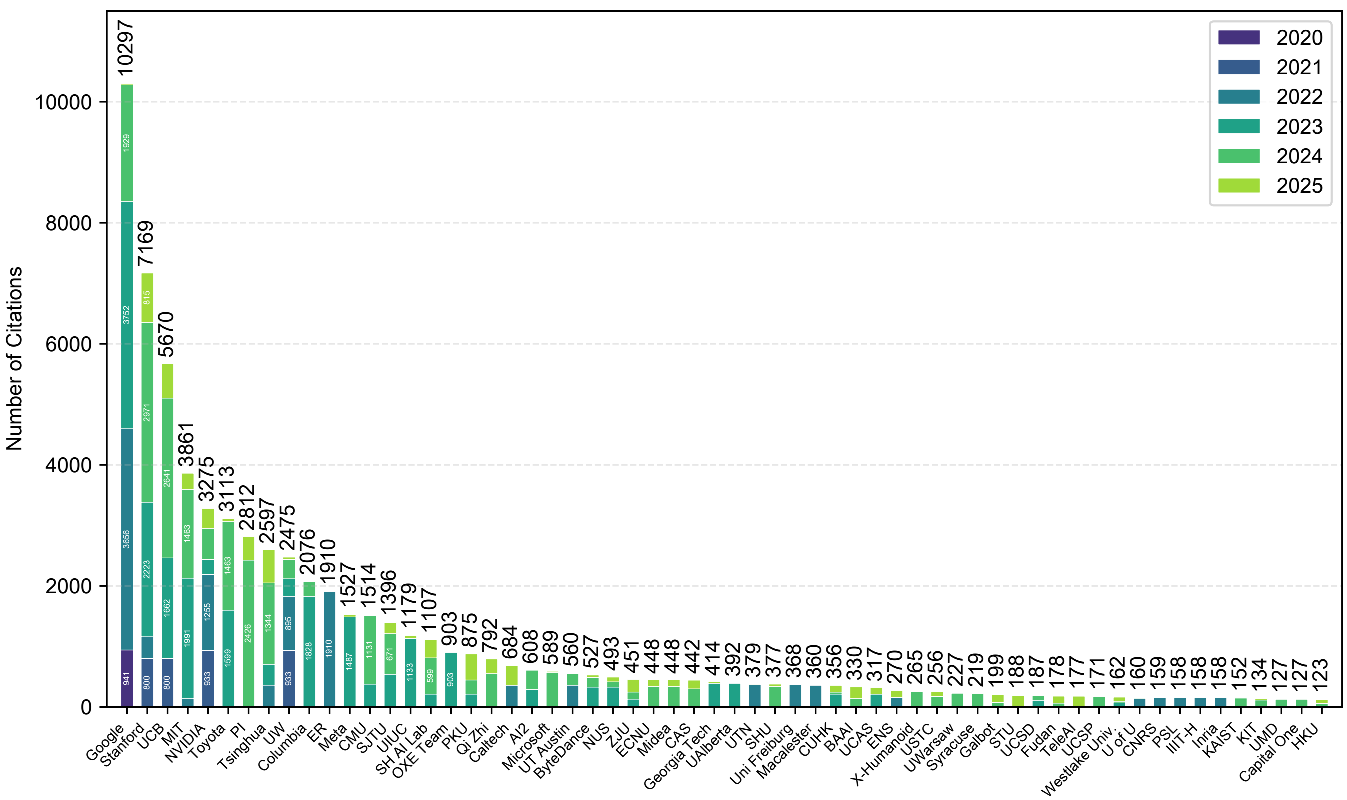


(e) 协作热力图

图11：VLA发展趋势的定量分析

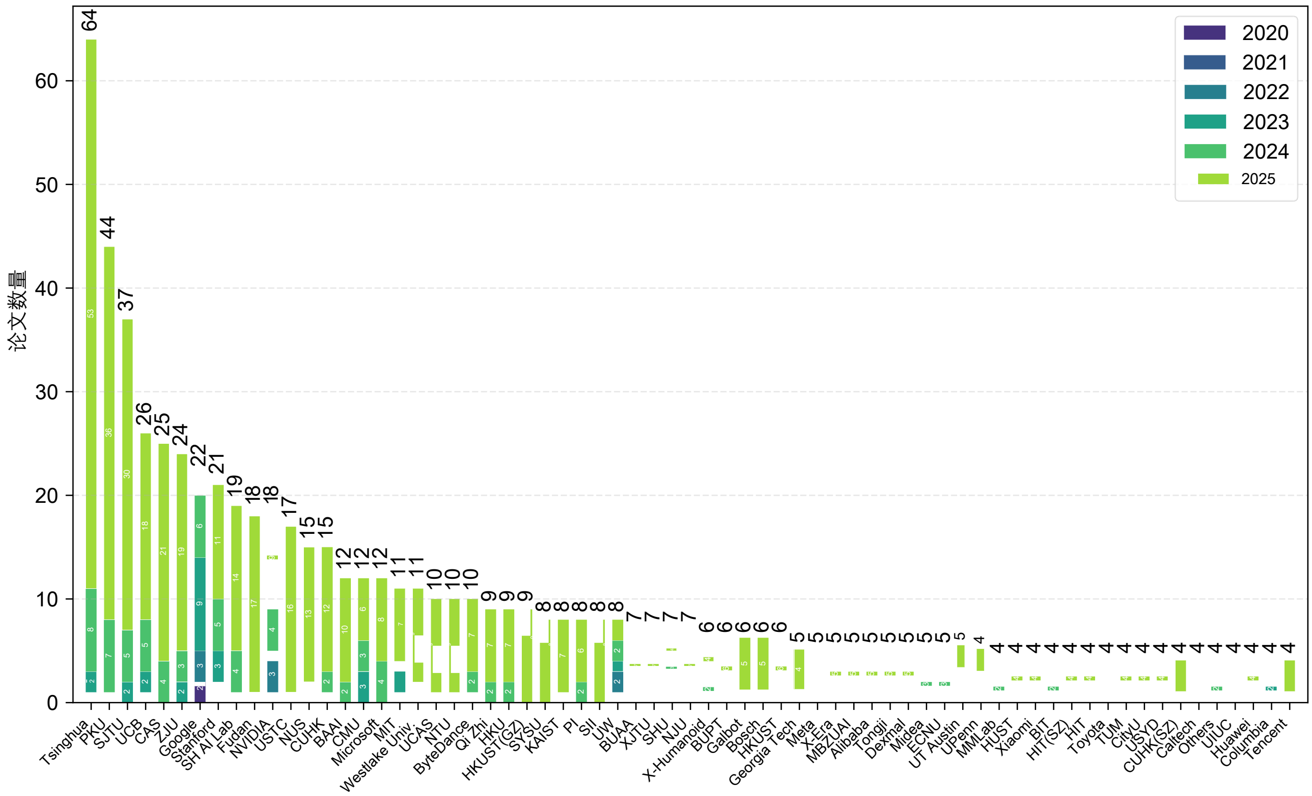


(a) Number of Papers per Institute

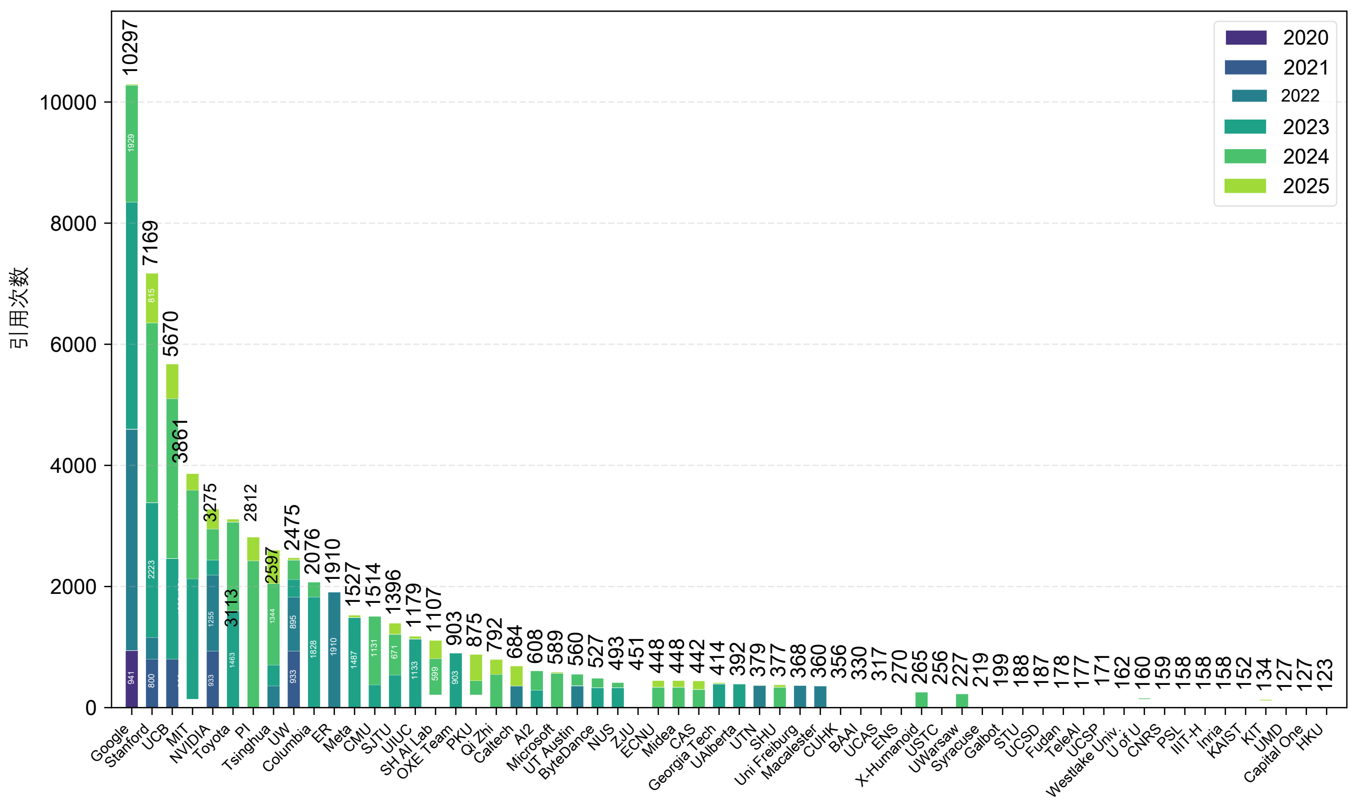


(b) Number of Citations per Institute

Figure 12: VLA research output and impact by institution.



(a) 各研究所发表论文的数量



(b) 各研究所的引用次数

图12：按机构划分的VLA研究成果及影响。